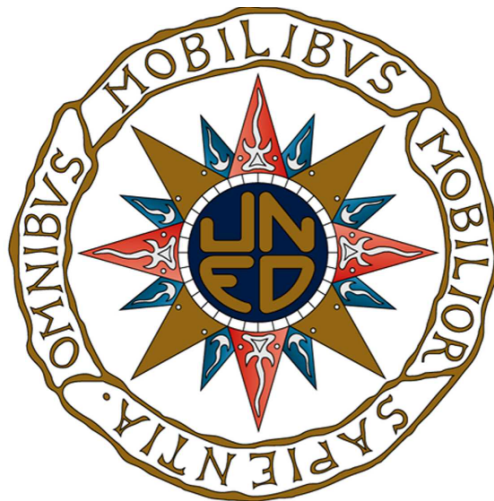


UNIVERSIDAD NACIONAL DE EDUCACIÓN A DISTANCIA

FACULTAD DE CIENCIAS



Máster en Matemáticas Avanzadas

TRABAJO FIN DE MÁSTER

Classical and Rough Volatility Models

Autor:

Héctor Zalama Alonso

Tutor:

Dr. Carlos Escudero Liébana

Madrid, April 2026

Aus dem Paradies, das Cantor uns geschaffen hat, soll uns niemand vertreiben können.

We shall never be expelled from the paradise Cantor created for us.

David Hilbert

Preface

This document constitutes the Master's Thesis of Héctor Zalama Alonso, prepared as the final requirement for the Master's Degree in Advanced Mathematics at UNED, under the supervision of Carlos Escudero Liébana, Full Professor at UNED.

The aim of this work is to present a natural progression from the classical foundations of stochastic calculus to the study and application of modern tools such as Rough Path Theory, with the ultimate goal of employing these tools in the field of financial modelling — particularly in the calibration of advanced volatility models. The intention is to illustrate how techniques originally developed in purely analytical contexts can find natural and effective applications in real-world problems in quantitative finance.

Contents

I	Foundations and Classical Models	1
0.1	A worthy note on context	3
0.2	Objective and Scope	4
1	Stochastic and Financial Foundations	5
1.1	Probabilistic Preliminaries	5
1.1.1	Fundamental Distributions	10
1.1.2	Markov Processes	13
1.1.3	Martingales, semimartingales and local martingales	14
1.1.4	Brownian Motion	17
1.1.5	Path regularity	20
1.2	Stochastic Calculus	23
1.2.1	The Itô Integral	23
1.2.2	Itô's Chain Rule	27
1.2.3	Stochastic Differential Equations	29
1.2.4	The multidimensional case	33
1.2.5	Probability measure changes	35
1.2.6	Itô Diffusions	39
1.2.7	References to Section 1.2	43
1.3	Financial Derivatives	44
1.3.1	The Concept of a Derivative	44
1.3.2	Derivative Pricing	46
1.4	The Black-Scholes Framework	48
1.4.1	The Model	49
1.4.2	Smile and Skewness	53
1.5	Other Important Models	53

1.5.1	Local Volatility Models	54
1.5.2	Stochastic Volatility Models	56
1.5.3	Path-Dependence and American Options	58
1.5.4	References to Sections 1.3 to 1.5	61

II Modern Approaches 63

2 Rough Paths 65

2.1	Introduction	65
2.2	Rough Paths	65
2.2.1	Tensor Spaces	65
2.2.2	Multiplicative Property	65
2.2.3	Types of Rough Paths	65
2.3	Fundamental Results	65
2.3.1	Integration	65
2.3.2	Differential Equations	65
2.3.3	Existence	65
2.3.4	Uniqueness and Continuity	65
2.4	Signature Transformation	65

3 Applications 67

3.1	Calibration of a Volatility Surface	67
3.1.1	The Model	67
3.2	Joint Calibration (?)	67
3.2.1	The Challenge (?)	67
3.2.2	Calibration of a (???)	67
3.3	Implementation	67
3.4	Results	67

4 Conclusions 69

Bibliography 71

A Auxiliary Results 75

A.1	Complements on conditional expectation	75
-----	--	----

A.2	Characteristic functions	76
A.3	Moment indeterminacy of the log-normal law	76
A.4	Grönwall's inequality	77
A.5	Radon–Nikodym and the Jordan decomposition	77

B Deferred Proofs 79

B.1	Doob's submartingale inequality	79
B.2	Density of step processes in $L^2_{ad}([0, T] \times \Omega)$	79
B.3	Continuity of the sample paths of an Itô integral	80
B.4	The Itô integral as a limit of Riemann sums	80
B.5	Lévy's characterisation theorem	81
B.6	Girsanov's theorem	81

Notation

A recurring difficulty in a text of this kind is that probabilists and practitioners often write the same object in different ways, and occasionally write different objects in the same way. The following conventions are fixed once and used without exception throughout. Where a financial convention differs from the probabilistic one, the discrepancy is pointed out in a footnote at the place where it first matters.

Probability space and information

$(\Omega, \mathcal{F}, \mathbb{P})$	Complete probability space; $\mathbb{P}$ is always the <i>physical</i> measure.
$\mathbb{Q}$	Risk-neutral measure associated with the numeraire S^0 .
$\mathbb{Q}^T$	T -forward measure, associated with the numeraire $P(\cdot, T)$.
$(\mathcal{F}_t)_{t \in [0, T]}$	Filtration, always satisfying the usual conditions (Definition 1.1.6).
$(\mathcal{F}_t^W)_{t \in [0, T]}$	Augmented filtration generated by the Brownian motion W .
$\mathbb{E}[\cdot], \mathbb{E}^{\mathbb{Q}}[\cdot]$	Expectation under $\mathbb{P}$, and under a named measure.
$\mathbb{V}[\cdot]$	Variance. (Not to be confused with V , an option value.)
$\mathcal{B}(E)$	Borel σ -algebra of the topological space E .
$\mathbb{1}_A$	Indicator function of the event A .

Time

$[0, T]$	Time interval. Always anchored at 0; T is simultaneously the time horizon and the maturity of the contract under consideration, which is exactly why the same letter is used. ¹
τ	A stopping time; $\mathcal{T}_{[t, T]}$ is the set of stopping times valued in $[t, T]$.

Processes

W	Brownian motion. The letter B is never used for it.
$\mathbf{W}$	Multidimensional Brownian motion with independent components.
W^H	Fractional Brownian motion with Hurst parameter H .
μ	A <i>drift</i> , always. Abstract measures are denoted ν, λ .
σ	A <i>diffusion coefficient</i> ; $\Sigma = \sigma\sigma^\top$ is the diffusion matrix.
ρ	A <i>correlation</i> , always. Densities are denoted p or f .
$p_{s,x}(t, y)$	Transition density of a Markov process.
$[X], \langle X \rangle$	Realised (pathwise) and predictable quadratic variation of X ; the two coincide for the continuous processes considered here.
$\mathcal{A}, \mathcal{A}^*$	Infinitesimal generator of a Markov semigroup and its adjoint.
$\mathcal{E}_\alpha$	Stochastic (Doléans-Dade) exponential of $\int \alpha \, dW$.

Finance

S_t	Spot price of the underlying.
F_t or $F(t, T)$	Forward price for delivery at T .
$P(t, T)$	Price at t of a zero-coupon bond maturing at T .
S^0	Numeraire asset.
H_T, Φ	Terminal payoff (a random variable) and payoff function.
K	Strike price; $k = \log(K/F)$ is log-moneyness.
r, q	Short rate and continuous dividend (or repo) yield.
$\sigma_{\text{imp}}, \sigma_{\text{loc}}$	Implied and local volatility; $w = \sigma_{\text{imp}}^2 T$ is total implied variance.
V	The value of a derivative. (Not to be confused with $\mathbb{V}$, variance.)

¹Probability texts customarily work on a generic interval $[a, b]$. We deliberately fix $a = 0$: in a financial context the present instant is the natural origin, every quoted quantity is a function of time-to-maturity, and carrying an arbitrary left endpoint through the pricing formulas obscures rather than generalises. Nothing is lost, since any statement on $[a, b]$ transfers by translation.

Part I

Foundations and Classical Models

Introduction

0.1 A worthy note on context

The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a gift which we neither understand nor deserve.

Eugene Wigner expressed his admiration in his 1960 article in *Communications in Pure and Applied Mathematics*. It is a sentiment many would agree with: the effectiveness of mathematics in describing Nature is both remarkable and fascinating. Moreover, over the last century its branches have reached into areas where, not so long ago, mathematicians were considered too abstract to be of much use in business terms.

In 1827, Robert Brown, while studying the behaviour of pollen grains suspended in water, noticed something unexpected: the grains wandered along irregular, hard-to-describe paths. Physics was not prepared to tackle such a problem yet. What was needed belonged to what we now call *statistical mechanics*, the study of collective behaviour of particles and molecules, intimately tied to *thermodynamics*. The decisive step came later, in 1905, when Albert Einstein provided a theory of this motion — the same year he also explained the photoelectric effect, introduced the Special Relativity theory and provided the first appearance of the mass-energy equivalence formula, $E = mc^2$. What a year.

Regarding the pollen grains, a parallel story took place in finance. Louis Bachelier, a doctoral student of Henri Poincaré, was interested in modelling stock prices. His model, which we now recognise as essentially Gaussian, worked with some shaky concepts on probability theory (Kolmogorov and his peers were not born yet, even Lebesgue had not published his work on measure theory yet). Worse, it allowed negative prices — a point critics did not miss.

The common thread in both stories is *Brownian motion*. This stochastic process is a central object in probability: geometers find its fractal dimension fascinating, while probabilists value the rich family of processes built upon it. The idea of treating “random noise” as a genuine mathematical object took shape gradually, but it was Kiyoshi Itô’s work on *stochastic integration* that truly changed the picture.

Finance joined somewhat later, but with undeniable impact. Today mathematics is everywhere in banks, consultancy and audit firms, underpinning the valuation of derivatives, that is, assigning a *fair price* (according to market data) to contracts involving random future cash flows. This pricing orientation will be the focus of the present thesis, among many other applications in finance, such as risk management and portfolio optimization.

0.2 Objective and Scope

The primary objective of this Thesis is to provide a solid review of *Volatility*, aiming to fill the gap between those who are strictly Mathematics-oriented—who often lack expertise in Finance—and those who are Finance-oriented and do not fully appreciate the depth of the ideas they are working with. For this reconciliatory purpose, the Thesis embraces a clear and rigorous mathematical mindset, while maintaining continuous references to practical and financial interpretations.

Part I aims to provide a thorough introduction to the subject. The core mathematical ideas, rooted in Probability Theory and forming the basis of modern Financial Theory and basic Pricing Theory, are presented in Chapter 1. Particular attention is paid there to the regularity of Brownian trajectories, since it is precisely the delicate balance between roughness and tractability — non-vanishing quadratic variation on the one hand, unbounded total variation on the other — that both makes Itô’s theory necessary and marks the limits of its reach.

Part II is devoted to more recent developments in volatility modelling, including Rough Paths and Rough Volatility theories (Chapter 2). The latter is introduced from a general and mathematical perspective: although it finds direct applications in Finance, the theory of rough paths has far-reaching implications in other areas of mathematics as well. These chapters aim to highlight the necessity of more sophisticated tools to describe and operate in the market.

Finally, these theoretical constructs are brought to practice in Chapter 3, where a calibration of XXX is performed. Here, the models are numerically implemented and compared, providing an applied perspective and illustrating the strengths and challenges associated with each approach. Some final conclusions are left for Chapter 4, where a general overview of the work done will be given, along with reflections on possible directions for further research.

Chapter 1

Stochastic and Financial Foundations

We assume the reader is familiar with the foundations of Probability Theory. Standard references include, for example, Billingsley’s classic text [4], as well as the treatments in [15] and [24]. Nevertheless, we briefly recall the basic probabilistic framework in order to fix notation and ensure clarity.

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with Ω the sample space, $\mathcal{F}$ a σ -algebra of events, and $\mathbb{P}$ a probability measure. A **random variable** is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$, the target being $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ in most of our applications.

Convention 1.0.1. Every probability space in this dissertation is assumed *complete*: if $A \in \mathcal{F}$ has $\mathbb{P}(A) = 0$, then every $B \subseteq A$ lies in $\mathcal{F}$ and has $\mathbb{P}(B) = 0$.

Completeness is not a formality. It is what will allow us, throughout, to modify a process on a null set — as we must whenever a theorem produces “a continuous version” of an object defined only up to indistinguishability — without leaving the space we started in. This is the first of several points at which the measure-theoretic language earns its place: it is not a technical convenience but the only setting in which path regularity, filtrations and martingale properties can be discussed at all.

1.1 Probabilistic Preliminaries

The notion of *stochastic process* generalizes that of a single random variable by allowing evolution in an external parameter, usually interpreted as time. To accommodate both discrete-time and continuous-time settings in a unified framework, we let $\mathbb{T}$ denote the **index set**, representing time. Typically, $\mathbb{T}$ is a subset of $\mathbb{R}$, such as:

$$\mathbb{T} \in \{\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, [0, T], [0, \infty), \mathbb{R}\},$$

depending on the application.

Definition 1.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, and let $\mathbb{T}$ be a time index set. A **stochastic process** is a collection of random variables $\{X_t : t \in \mathbb{T}\}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in a measurable space $(E, \mathcal{E})$, such that the map

$$X : \mathbb{T} \times \Omega \rightarrow E, \quad (t, \omega) \mapsto X_t(\omega)$$

is jointly measurable with respect to $\mathcal{B}(\mathbb{T}) \otimes \mathcal{F}$. In particular:

- For each fixed $t \in \mathbb{T}$, the function $\omega \mapsto X_t(\omega)$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
- For each fixed $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called the *sample path* or *trajectory* of the process.

We may denote the process by $X = \{X_t\}_{t \in \mathbb{T}}$, and abbreviate $X_t(\omega)$ as $X(t)$ or simply X_t when context allows. We say X is *discrete* whenever $\mathbb{T}$ is $\mathbb{N}$ or $\mathbb{Z}$ and *continuous* if $\mathbb{T}$ is $[0, T]$ for some $T > 0$ or $[0, \infty)$.

This formulation includes many important examples such as Markov chains, Poisson processes, and Brownian motion. Note that the nature of the index set $\mathbb{T}$ often dictates the regularity assumptions or path properties we may require (e.g., right-continuity, càdlàg, continuity in probability, etc.). Analyzing the dynamic evolution of these processes requires us to quantify their expected future behavior given currently available information. This idea is contained in the **conditional expectation**. Its existence relies on the following profound result:

Theorem 1.1.2 (Radon–Nikodym). *Let ν and λ be two σ -finite measures on the same measurable space $(\Omega, \mathcal{F})$. Suppose that ν is absolutely continuous with respect to λ , denoted $\nu \ll \lambda$; that is,*

$$\lambda(A) = 0 \quad \Rightarrow \quad \nu(A) = 0, \quad \text{for all } A \in \mathcal{F}. \quad (1.1)$$

Then there exists a measurable function $f : \Omega \rightarrow [0, \infty)$, unique up to λ -null sets, such that for all $A \in \mathcal{F}$,

$$\nu(A) = \int_A f \, d\lambda. \quad (1.2)$$

This function f is called the Radon–Nikodym derivative of ν with respect to λ , and is denoted by $f = d\nu/d\lambda$. ■

The letter μ is deliberately avoided here. From Section 1.2 onwards it will denote a drift coefficient and nothing else; abstract measures are ν and λ throughout.

We shall need the statement in slightly greater generality than above, since the measure we are about to construct is not positive.

Corollary 1.1.3 (Signed version). *Let ν be a finite signed measure on $(\Omega, \mathcal{F})$ and λ a σ -finite positive measure with $\nu \ll \lambda$. Then there exists $f \in L^1(\Omega, \mathcal{F}, \lambda)$, real-valued and unique up to λ -null sets, satisfying (1.2).*

Proof: Apply the Jordan decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. Both $\nu^\pm \ll \lambda$, so Theorem 1.1.2 yields non-negative densities $f^\pm$, and $f := f^+ - f^-$ works; integrability follows from $\int (f^+ + f^-) \, d\lambda = |\nu|(\Omega) < \infty$. ■

As a consequence, if $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ (that is, $\int |X| \, d\mathbb{P}$ is finite) and letting $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra, the *conditional expectation* of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to $\mathbb{P}$ -a.e. equality) $\mathcal{G}$ -measurable function satisfying:

$$\int_G \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_G X \, d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \quad (1.3)$$

This definition is structurally equivalent to the Radon–Nikodym theorem. Define the set function $\nu : \mathcal{G} \rightarrow \mathbb{R}$ by

$$\nu(G) := \int_G X \, d\mathbb{P}, \quad \text{for } G \in \mathcal{G}. \quad (1.4)$$



Since X is integrable, ν is a finite *signed* measure on $(\Omega, \mathcal{G})$, and it is absolutely continuous with respect to $\mathbb{P}|_{\mathcal{G}}$, the restriction of $\mathbb{P}$ to $\mathcal{G}$. It is Corollary 1.1.3, rather than Theorem 1.1.2 itself, that applies here: X need not be non-negative, so neither is ν , and the density produced must be allowed to change sign. With that proviso (see [4] or [38]), there exists a $\mathcal{G}$ -measurable function, namely $\mathbb{E}[X | \mathcal{G}]$, such that:

$$\int_G \mathbb{E}[X | \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \quad (1.5)$$

Conditional expectation inherits the properties one would expect of an averaging operator: it preserves expectations and order, it is linear, it acts as the identity on $\mathcal{G}$ -measurable variables and as the plain expectation on independent ones, known factors may be taken out of it, and it satisfies the tower property $\mathbb{E}[X | \mathcal{H}] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$. These, together with the conditional forms of the Fatou, monotone and dominated convergence theorems, are recorded as Proposition A.1.1 in Appendix A; we shall use them throughout without comment. Two further properties, however, are used more than all the others combined and are stated here.

Proposition 1.1.4 (Conditional Jensen and the L^2 projection). *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -field. Then, almost surely:*

8. (**Conditional Jensen's inequality**) *If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and both X and $\phi(X)$ are integrable, then*

$$\phi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\phi(X) | \mathcal{G}].$$

9. (**L^2 projection**) *If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[X | \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Equivalently, it is the unique $\mathcal{G}$ -measurable Y minimising $\mathbb{E}[(X - Y)^2]$, and*

$$\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}]) Z] = 0 \quad \text{for every } Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}).$$

■

The numbering is retained from Proposition A.1.1 for ease of reference. Property 8 is what guarantees that X^2 is a submartingale whenever X is a martingale — a fact invoked twice in what follows, in Subsection 1.1.3 and again in the Doob-Meyer decomposition — and it is the reason convexity is such a persistent theme in option pricing.

Property 9 deserves more than emphasis. It says that conditioning is nothing more exotic than orthogonal projection in a Hilbert space, and once this is seen, a good deal of what follows stops looking like a series of coincidences. The Itô isometry of Theorem 1.2.5 will be an isometry between two Hilbert spaces; the density argument that extends the stochastic integral from step processes to all of L^2_{ad} is the standard extension of a bounded operator from a dense subspace; and the Martingale Representation Theorem of Subsection 1.2.5 will say that a certain family of stochastic integrals is not merely orthogonal but *complete*. The geometry is the same throughout.

There is one final structural component that needs to be described. In the study of stochastic processes, it is essential to formalize how information accumulates over time. This is done through the concept of a **filtration**, which models the *evolution of knowledge* or *information available* up to each time t .

Definition 1.1.5 (Filtration). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be a time index set. A **filtration** is a family $\{\mathcal{F}_t\}_{t \in \mathbb{T}}$ of sub- σ -algebras of $\mathcal{F}$ satisfying:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all } s \leq t. \quad (1.6)$$

That is, the filtration is *increasing* in t . We interpret $\mathcal{F}_t$ as the collection of events whose outcomes are known or observable up to t . A stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$ is said to be **adapted** to the filtration $\{\mathcal{F}_t\}$ if, for each $t \in \mathbb{T}$, the random variable X_t is measurable with respect to $\mathcal{F}_t$. We say that $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ is a **filtered probability space**.

Bare adaptedness turns out to be slightly too weak for comfort. Two further requirements are so universally imposed that they have acquired a name of their own, and we impose them here once and for all rather than invoking them piecemeal later.

Definition 1.1.6 (The usual conditions). A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ is said to satisfy the **usual conditions** if

- i) $\mathcal{F}_0$ contains every $\mathbb{P}$ -null set of $\mathcal{F}$ (*completeness*); and
- ii) the filtration is *right-continuous*, that is,

$$\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{n \geq 1} \mathcal{F}_{t+1/n}, \quad t \in [0, T]. \quad (1.7)$$

Convention 1.1.7. From this point onwards, every filtered probability space is assumed to satisfy the usual conditions, and this will not be restated.

Neither requirement is cosmetic. Completeness is what allows us to modify a process on a null set — as we shall do whenever we choose a continuous version of an object defined only up to indistinguishability — without leaving the filtration. Right-continuity is what makes first hitting times of open sets stopping times, and it is used explicitly in Proposition 1.1.11 below and again in the Doob-Meyer decomposition of Subsection 1.2.5. Neither is automatic for a natural filtration $\mathcal{F}_t^X$, but both can always be arranged: one replaces $\mathcal{F}_t^X$ by the $\mathbb{P}$ -completion of $\mathcal{F}_{t+}^X$, the so-called *augmented* filtration, and for the filtrations generated by the processes we shall study this augmentation changes nothing of substance.

Remark 1.1.8 (Modification versus indistinguishability). A related piece of hygiene, which we shall need whenever a theorem produces “a continuous version” of a process. Two processes X and Y on the same space are said to be *modifications* of each other if $\mathbb{P}(X_t = Y_t) = 1$ for every fixed t , and *indistinguishable* if $\mathbb{P}(X_t = Y_t \text{ for all } t) = 1$. The second is strictly stronger: the exceptional null set is allowed to depend on t in the first case, and a union of uncountably many null sets need not be null. The two notions do coincide when both processes have right-continuous paths, which is the situation we shall almost always be in — but the coincidence is a small theorem, not a definition, and we shall be careful to say which of the two we mean.

Recall that if χ is a real-valued random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the σ -algebra generated by χ is denoted $\sigma(\chi)$ and defined as:

$$\sigma(\chi) := \{A \in \mathcal{F} : A = \chi^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R})\}. \quad (1.8)$$

Example 1.1.9 (Natural Filtration). Given a stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$, the **natural filtration** generated by X is defined by:

$$\mathcal{F}_t^X := \sigma(X_s : s \leq t), \quad (1.9)$$

the smallest σ -algebra with respect to which all variables X_s for $s \leq t$ are measurable. This filtration encodes exactly the information revealed by observing the process up to time t .



Definition 1.1.10 (Stopping time). Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and a filtration $\{\mathcal{F}_t\}$. A mapping $\tau : \Omega \rightarrow [0, T]$ is a *stopping time* if, for each $t \in [0, T]$ (respect. $t \in [0, \infty)$), $\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$.

Proposition 1.1.11. Let $(\mathcal{F}_t)_{t \in [0, T]}$ be a right-continuous filtration. A random variable $\tau : \Omega \rightarrow [0, T]$ is a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$ if and only if

$$\{\omega \in \Omega : \tau(\omega) < t\} \in \mathcal{F}_t, \quad \text{for all } t \in [0, T].$$

Proof: Suppose first that τ is a stopping time. Since $\{\tau < t\} = \bigcup_{n \geq 1} \{\tau \leq t - 1/n\}$ and each $\{\tau \leq t - 1/n\}$ lies in $\mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, the union lies in $\mathcal{F}_t$. This implication uses nothing beyond monotonicity of the filtration.

Conversely, suppose $\{\tau < t\} \in \mathcal{F}_t$ for every t . Then for each $n \geq 1$ we have $\{\tau < t + 1/n\} \in \mathcal{F}_{t+1/n}$, and since $\{\tau \leq t\} = \bigcap_{n \geq 1} \{\tau < t + 1/n\}$,

$$\{\tau \leq t\} \in \bigcap_{n \geq 1} \mathcal{F}_{t+1/n} = \mathcal{F}_{t^+} = \mathcal{F}_t,$$

the last equality being exactly the right-continuity (1.7). ■

Note where the work was done: the converse consumes right-continuity in full, and fails without it. This is the first of several places where the usual conditions of Definition 1.1.6 do real work rather than merely tidy up.

Example 1.1.12. Let X be a measurable adapted process on $[0, T]$ with $\int_0^T |X_s|^2 ds < \infty$ almost surely. For $n \in \mathbb{N}$ set

$$A_n(\omega) = \left\{ t \in [0, T] : \int_0^t |X_s(\omega)|^2 ds > n \right\},$$

and let $\tau_n(\omega) = \inf A_n(\omega)$ if $A_n(\omega) \neq \emptyset$, and $\tau_n(\omega) = T$ otherwise. Since $t \mapsto \int_0^t |X_s|^2 ds$ is continuous and non-decreasing, the threshold n is exceeded strictly before t if and only if it is exceeded at some rational time strictly before t , so that for $t \in (0, T]$

$$\{\tau_n < t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}}} \left\{ \int_0^r |X_s|^2 ds > n \right\} \in \mathcal{F}_t, \quad (1.10)$$

whereas $\{\tau_n < 0\} = \emptyset \in \mathcal{F}_0$. By Proposition 1.1.11, τ_n is a stopping time.

This sequence is not an idle illustration. It is exactly the localising sequence that will allow us, in Subsection 1.2.1, to define the stochastic integral for integrands that are square-integrable pathwise but not in expectation, and it is the prototype of every localisation argument in this chapter.

Example 1.1.13 (First hitting time). Let X_t be a right continuous process adapted to $\{\mathcal{F}_t\}$. If U is an open set of $\mathbb{R}$, then the *first hitting time*, τ_U is defined as

$$\tau_U(\omega) = \inf\{t > 0 : X_t(\omega) \in U\} \quad (1.11)$$

is a stopping time.

1.1.1 Fundamental Distributions

Before presenting some cornerstone distributions, recall that the **characteristic function** of a (real) random variable X is defined by

$$\varphi_X(u) := \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}. \quad (1.12)$$

It is the Fourier transform of the probability distribution of X , and it uniquely determines the law of X .

Its elementary properties — normalisation, uniform continuity, Hermitian symmetry, behaviour under affine maps and under sums of independent variables, and the correspondence between differentiability at the origin and existence of moments — are collected in Proposition A.2.1 of Appendix A, and are used below without further comment.

Theorem 1.1.14 (Fourier inversion). *Let X be a random variable with distribution function F_X . If φ_X is integrable, then F_X admits a continuous density f_X given by*

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iux} \varphi_X(u) du. \quad (1.13)$$

In the general case, without assuming a density, the distribution function F_X can still be recovered via the *Lévy inversion formula*: for any $\alpha < \beta$ that are *continuity points* of F_X ,

$$F_X(\beta) - F_X(\alpha) = \lim_{R \rightarrow \infty} \frac{1}{2\pi} \int_{-R}^R \frac{e^{-i\alpha u} - e^{-i\beta u}}{iu} \varphi_X(u) du. \quad (1.14)$$

The restriction to continuity points is necessary and not a technicality: at a jump of F_X the right-hand side converges to the average of the left and right limits rather than to F_X itself. Since F_X is monotone it has at most countably many discontinuities, so continuity points are dense and the formula determines F_X everywhere by right-continuity. Note also that the truncation parameter R is unrelated to the time horizon T fixed in the notation table; the symbol has been changed to avoid the collision.

We deduce that if two random variables X and Y have the same characteristic function, i.e. $\varphi_X(u) = \varphi_Y(u)$ for all $u \in \mathbb{R}$, then X and Y have the same distribution. ■

Theorem 1.1.15 (Lévy's Continuity Theorem). *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables with characteristic functions φ_{X_n} .*

- *If $X_n \xrightarrow{d} X$ for some random variable X , then $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$ for every $u \in \mathbb{R}$.*
- *Conversely, suppose $\varphi_{X_n}(u) \rightarrow \varphi(u)$ for every $u \in \mathbb{R}$, where $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ is some function continuous at $u = 0$. Then φ is the characteristic function of a random variable X , and $X_n \xrightarrow{d} X$.* ■

The asymmetry between the two halves is deliberate and is where the theorem earns its keep. In the converse direction one is *not* allowed to assume that the limit φ is a characteristic function — that is the conclusion, not a hypothesis. Continuity at the origin is what rules out mass escaping to infinity, and it is a genuine restriction on φ ; note that if one already knew φ to be a characteristic function the requirement would be vacuous, since characteristic functions are automatically uniformly continuous. It is precisely this form of the statement that makes the Central Limit Theorem below a two-line computation: one expands φ_{X_n} , observes that the limit is $e^{-u^2/2}$, and only then concludes that the limit is Gaussian.



We now describe certain distributions of widespread importance and usefulness. The first place must be given to the normal distribution.

Example 1.1.16 (Normal Distribution). Let $\mu \in \mathbb{R}$ and $\sigma > 0$, and define the real-valued random variable $X: \Omega \rightarrow \mathbb{R}$ with distribution $X \sim \mathcal{N}(\mu, \sigma^2)$, that is, X has the probability density function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (1.15)$$

Then X is said to be normally distributed with mean μ and variance σ^2 ; thus, a normal distribution is characterized by its first two moments.

The normal distribution is fundamental in probability theory and statistics. It appears as the limiting distribution in the Central Limit Theorem and models many naturally occurring phenomena. Its characteristic function adopts the form

$$\varphi_X(u) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right). \quad (1.16)$$

Its resemblance to the *heat kernel* (see [16]) is not a coincidence and will be explored in the next section.

Let $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ be two *independent* normal random variables. The following are easily verified through Theorem 1.1.15:

- For $a, b \in \mathbb{R}$, $aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$; if instead X and Y are jointly normal with correlation ρ , the variance becomes $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y$. We write $Z \sim \mathcal{N}(\mu, 0)$ for the constant variable μ , so that the identity persists when constants are combined with normal variables.
- Its characteristic function is entire analytic, and $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$ converges in distribution to $\mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$ if and only if $\mu_n \rightarrow \mu$ and $\sigma_n^2 \rightarrow \sigma^2$.
- The normal distribution maximises **entropy** among distributions with fixed first two moments, making it the *least informative* among them.
- Independence implies zero correlation, as always; for normal variables the converse holds too, *provided the pair is jointly normal*.

The independence and joint-normality hypotheses above deserve a word, because the two statements they guard are among the most frequently misquoted in the subject, and a single counterexample disposes of both at once.

Remark 1.1.17. Let $X \sim \mathcal{N}(0, 1)$ and let ε be independent of X with $\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}$. Put $Y := \varepsilon X$. Conditioning on ε shows at once that $Y \sim \mathcal{N}(0, 1)$, and

$$\mathbb{E}[XY] = \mathbb{E}[\varepsilon] \mathbb{E}[X^2] = 0,$$

so X and Y are uncorrelated standard normal variables. They are very far from independent: $|X| = |Y|$ almost surely. Moreover $X + Y = (1 + \varepsilon)X$ vanishes with probability $\frac{1}{2}$, hence is not normal at all.

The moral is that “normal” is a property of a *law on* $\mathbb{R}^d$, not a property of a list of marginals. The pair (X, Y) above has normal marginals but its joint law is not multivariate normal in

the sense of Example 1.1.19, and every statement in the list above fails for it. Whenever we invoke Gaussianity for several variables at once — as we shall constantly do for the increments of Brownian motion, and for the driving noises of the multi-factor models of Section 1.5 — it is joint normality that is meant.

However, the most compelling reason why the normal distribution is so important rests on the following fundamental result, in a relatively weak form:

Theorem 1.1.18 (Central Limit Theorem). *Let $\{X_n\}_{n=1}^\infty$ be a sequence of independent, identically distributed (hereafter, i.i.d.) random variables with finite mean μ and variance $\sigma^2 > 0$. Then*

$$\frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad \blacksquare \quad (1.17)$$

The multivariate case is the one we shall actually need, since Brownian increments and multi-factor noises are always Gaussian several variables at a time.

Example 1.1.19 (Multivariate Normal Distribution). An $\mathbb{R}^d$ -valued random vector X follows a **multivariate normal distribution** with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$, written $X \sim \mathcal{N}_d(\mu, \Sigma)$, if $a^\top X \sim \mathcal{N}(a^\top \mu, a^\top \Sigma a)$ for every $a \in \mathbb{R}^d$.

Here Σ is necessarily symmetric positive semi-definite. If it is invertible the law is absolutely continuous on $\mathbb{R}^d$ with density $f_X(x) = (2\pi)^{-d/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$; if it is singular, X is supported on a lower-dimensional affine subspace. In either case the characteristic function is

$$\varphi_X(u) = \mathbb{E}[e^{iu^\top X}] = \exp\left(iu^\top \mu - \frac{1}{2}u^\top \Sigma u\right), \quad u \in \mathbb{R}^d. \quad (1.18)$$

For a multivariate normal vector — and, as Remark 1.1.17 showed, only then — uncorrelatedness does imply independence. Indeed, if Σ is block-diagonal then (1.18) factorises as a product of the characteristic functions of the corresponding sub-vectors, which is exactly independence.

Example 1.1.20 (Log-Normal Distribution). A random variable X is said to follow a **log-normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted

$$X \sim \text{Log}\mathcal{N}(\mu, \sigma^2), \quad (1.19)$$

if the logarithm of X is normally distributed, i.e.,

$$\log X \sim \mathcal{N}(\mu, \sigma^2). \quad (1.20)$$

The probability density function of $X \sim \text{Log}\mathcal{N}(\mu, \sigma^2)$, after a straightforward change-of-variable calculation, is

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \quad x > 0. \quad (1.21)$$

All positive moments exist and are given by $\mathbb{E}[X^k] = \exp(k\mu + \frac{1}{2}k^2\sigma^2)$; in particular $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathbb{V}[X] = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$. All real moments therefore exist, and yet the log-normal distribution has two unpleasant analytical features that will matter later.

The first is that its characteristic function admits no closed-form expression. This is a genuine inconvenience — it is precisely why the Black-Scholes model, despite its explicit pricing formula, is not amenable to the Fourier techniques that make the Heston model of Subsection 1.5.2 tractable — but it is not a pathology. Note that φ_X is defined and bounded by 1 for every *real*



u , as it is for any random variable; what fails is the *moment generating function* $\mathbb{E}[e^{uX}]$, which is infinite for every $u > 0$. The two objects are easily confused and we shall keep them apart.

The second is more surprising. Despite having moments of all orders, **the log-normal distribution is not determined by them**: there is an explicit one-parameter family of mutually distinct densities on $(0, \infty)$ sharing every moment with it, exhibited by Heyde [22] and reproduced in Appendix A.3. It is worth being precise about what does *not* fail here, since the point is easy to garble. Lévy's inversion formula (1.14) holds without exception, and the characteristic function does determine the log-normal law uniquely; what fails is the passage from the *moment sequence* to the characteristic function, the moments growing as $e^{k^2\sigma^2/2}$, fast enough to violate Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$. The lesson is one we shall meet again in Chapter 2: knowing every moment of an object is strictly weaker than knowing the object, and the signature transform will be interesting precisely because, for paths, the analogous statement can be made to go the right way.

Two elementary processes will be needed later, and we record them together.

Example 1.1.21 (Random walk and Poisson process). i) Let $\{\xi_n\}$ be i.i.d. with $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2}$. The **simple symmetric random walk** is $X_0 := 0$, $X_n := \sum_{k=1}^n \xi_k$: a discrete-time, integer-valued process with stationary and independent increments. Since $\varphi_\xi(u) = \cos(u)$, independence gives $\varphi_{X_n}(u) = \cos^n(u)$.

ii) Fix $\lambda > 0$. A process $N = \{N_t\}_{t \geq 0}$ with values in $\mathbb{N}_0$ is a **Poisson process with rate λ** if $N_0 = 0$ almost surely, N has independent increments, and

$$\mathbb{P}(N_{t+s} - N_s = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k \in \mathbb{N}_0, \quad s, t \geq 0. \quad (1.22)$$

Its paths are piecewise constant and non-decreasing, with unit jumps at random times; $\mathbb{E}[N_t] = \mathbb{V}[N_t] = \lambda t$ and $\varphi_{N_t}(u) = e^{\lambda t(e^{iu} - 1)}$.

We shall return to i) in Theorem 1.1.35, where the jump size and the waiting time between jumps are adjusted simultaneously and a new process emerges in the limit. Part ii) will serve throughout as the standard example of a process which is *continuous in time*, since $\mathbb{T} = [0, \infty)$, while having discontinuous sample paths — a distinction worth keeping straight. It is the elementary building block for markets in which prices genuinely jump, such as electricity, and it will reappear as the canonical process that is *not* a diffusion.

Remark 1.1.22. Despite the importance of discrete processes, for most of this dissertation we will assume that the process is continuous, unless otherwise stated.

1.1.2 Markov Processes

Markov processes are characterized by a specific form of *memorylessness*. They are particularly suited to describe the evolution of systems where the future depends only on the present state and not on the past history. Some of these processes arise as *diffusions* and in that case, a lot can be said about the distributions.

Definition 1.1.23 (Markov Process). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X = \{X_t\}_{t \in \mathbb{T}}$ be a stochastic process taking values in a measurable space $(E, \mathcal{E})$. We say that X is a **Markov process** if for every $s < t$ and every measurable set $A \in \mathcal{E}$,

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s) \quad \text{almost surely,} \quad (1.23)$$

where $\mathcal{F}_s = \sigma(X_r : r \leq s)$ is the σ -algebra generated by the history of the process up to time s .

The **transition probability** of a Markov process is $\mathbb{P}(X_t \in A \mid X_s = x)$; in order to integrate against the measure it defines we write $P_{s,x}(t, dy) = \mathbb{P}(X_t \in dy \mid X_s = x)$.

There is a rather sharp distinction between finite-state Markov processes and those with infinitely many states. In the finite case, the theory is built around transition matrices: the probability of moving from one state to another in a single step is recorded in a matrix P_t , so that the evolution is the product $\prod_{i=1}^n P_i$, and long-run behaviour can be analysed by classical linear-algebraic tools, especially in the stationary case, that is, when the transition probability is independent of t .

In the infinite-state setting the dynamics are instead described by operators acting on spaces of functions or measures, so that the study of these processes relies on functional analysis and differential equations as much as on probability. That apparatus — the transition semigroup $(P_t)_{t \geq 0}$, its infinitesimal generator, and the Kolmogorov equations they satisfy — is developed in Subsection 1.2.6, once diffusions are available to make it concrete; we do not anticipate it here.

1.1.3 Martingales, semimartingales and local martingales

Martingales are processes for which the expected value of a future instant is precisely the state of the process up to present time. Martingales are absolutely key in the study of stochastic processes. Some generalizations are needed as well. We keep the notation used in the previous section.

Definition 1.1.24. (Martingale) Let X_t be a stochastic process, adapted to a given filtration $\mathcal{F}_t$. We say it is a **martingale** provided:

- $\mathbb{E}[|X_t|] < \infty$ for almost all t .
- $X_s = \mathbb{E}[X_t | \mathcal{F}_s]$ for $s < t$.

We say that X_t is a **supermartingale** if the second condition is replaced by $X_s \geq \mathbb{E}[X_t | \mathcal{F}_s]$ and a **submartingale** if $X_s \leq \mathbb{E}[X_t | \mathcal{F}_s]$.

The martingale property is stated at deterministic times, but a great deal of what we shall do — localisation in Section 1.2, and the early exercise of American options in Subsection 1.5.3 — involves evaluating processes at *random* times. For this to make sense we must first say what information is available at a stopping time.

Definition 1.1.25 (The σ -algebra $\mathcal{F}_\tau$). Let τ be a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$. The σ -algebra of events observable up to τ is

$$\mathcal{F}_\tau := \left\{ A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t \in [0, T] \right\}. \quad (1.24)$$

The definition reads oddly at first but says something simple: an event belongs to $\mathcal{F}_\tau$ if, whenever τ has occurred by time t , an observer at time t can already tell whether the event happened. One checks readily that $\mathcal{F}_\tau$ is indeed a σ -algebra, that τ is $\mathcal{F}_\tau$ -measurable, that $\mathcal{F}_\tau = \mathcal{F}_t$ when $\tau \equiv t$ is constant, and that $\sigma \leq \tau$ implies $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$. If X is right-continuous and adapted, then X_τ is $\mathcal{F}_\tau$ -measurable on $\{\tau < \infty\}$.



Theorem 1.1.26 (Doob's optional stopping theorem). *Let X be a right-continuous martingale with respect to $(\mathcal{F}_t)_{t \in [0, T]}$, and let $\sigma \leq \tau$ be two stopping times. If any one of the following holds*

- i) τ is bounded, i.e. $\tau \leq T$ almost surely for the finite horizon T ;*
- ii) X is uniformly integrable; or*
- iii) there exists $Y \in L^1(\Omega)$ with $|X_t| \leq Y$ for all t ,*

— *then X_σ and X_τ are integrable and*

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] = X_\sigma \quad \text{almost surely.} \quad (1.25)$$

In particular $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$. The corresponding statements with $=$ replaced by $\leq$ or $\geq$ hold for sub- and supermartingales respectively. ■

Equation (1.25) is the assertion that a martingale remains a martingale when observed along a random clock, and it is one of those results whose hypotheses are more instructive than its conclusion. Some hypothesis is indispensable: let W be a Brownian motion and $\tau = \inf\{t : W_t = 1\}$, which is finite almost surely. Then $W_\tau = 1$ identically, so $\mathbb{E}[W_\tau] = 1 \neq 0 = \mathbb{E}[W_0]$, and (1.25) fails. Nothing is wrong with the theorem; the stopping time is simply unbounded and $W_{t \wedge \tau}$ is not uniformly integrable. The financial reading of this example is exact and worth stating: a strategy that waits however long is necessary to be up by one unit does indeed win with certainty, and it is not an arbitrage only because executing it may require unbounded losses along the way. That is precisely what the admissibility constraint of Definition 1.3.2 will forbid.

Because $\{X_{t \wedge \tau}\}$ is a martingale whenever X is — which is case i) applied at each t — optional stopping is also the mechanism behind localisation. A local martingale is by Definition 1.1.29 a process which becomes a martingale after stopping; the content of that definition is that all the martingale machinery, including (1.25) and Proposition 1.1.27, remains available on each piece.

The following result will be useful when constructing stochastic processes from Itô integrals. Notice that if X_t is a martingale, then by Jensen's inequality (Proposition 1.1.4, item 8) X_t^2 is a submartingale.

Proposition 1.1.27 (Doob's submartingale inequality). *Let $Z = \{Z_t\}_{t \in [0, T]}$ be a right-continuous submartingale. Then, for any $\varepsilon > 0$,*

$$\mathbb{P}\left(\sup_{0 \leq t \leq T} Z_t \geq \varepsilon\right) \leq \frac{\mathbb{E}[\max\{Z_T, 0\}]}{\varepsilon}.$$

In particular, for a right-continuous martingale X , $\mathbb{P}(\sup_{0 \leq t \leq T} |X_t| \geq \varepsilon) \leq \mathbb{E}[|X_T|]/\varepsilon$.

Proof: The argument is given in Appendix B.1; see also [26, Thm. 4.5.1]. ■

The bound is in terms of the *terminal* value X_T , not of a running X_t ; this is the whole content of the inequality, since it converts information at a single time into control of the entire path. For the L^2 theory of the Itô integral we shall need the following stronger, L^p form, which is proved by integrating the tail estimate above.

Proposition 1.1.28 (Doob's L^p maximal inequality). *Let X be a right-continuous martingale and let $p > 1$. Then*

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} |X_t|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_T|^p].$$

In particular, for $p = 2$, $\mathbb{E}[\sup_{t \leq T} |X_t|^2] \leq 4\mathbb{E}[|X_T|^2]$. ■

The case $p = 2$ is the one we shall use repeatedly. It says that for a square-integrable martingale, controlling the terminal second moment controls the second moment of the whole trajectory — which is exactly what is needed to upgrade convergence of the Itô integral in $L^2(\Omega)$, at each fixed time, to uniform convergence along a subsequence, and hence to continuity of sample paths (Theorem 1.2.7).

The notion of a martingale is often too restrictive, particularly when considering processes such as stochastic exponentials or certain diffusion models. A relaxation is therefore introduced.

Definition 1.1.29 (Local Martingale). A process $X = \{X_t\}_{t \geq 0}$ is a **local martingale** if there exists an increasing sequence of stopping times $(\tau_n)_{n \geq 1}$ with $\tau_n \uparrow \infty$ almost surely, such that each stopped process $X_t^{\tau_n} := X_{t \wedge \tau_n}$ is a martingale.

The idea is that X behaves like a martingale, but only up to random horizons. An analogy could be drawn from ODEs: there are solutions that explode to infinity in finite time, and those that can be extended to all times. Here, the integrability condition can be somewhat replaced if we work with sequences of stopping times. This mathematical object deserves its own study but lies beyond the scope of this dissertation.

Finally, there is another related concept, broad enough to serve as a class of *good integrators*. In particular, it extends the role played by Brownian motion as a canonical integrator, while retaining enough structure to allow a general theory of stochastic calculus.

Definition 1.1.30 (Semimartingale). A process X is called a **semimartingale** if it can be decomposed as

$$X_t = M_t + A_t,$$

where M_t is a local martingale and A_t is *continue à droite, avec limite à gauche* (càdlàg) and of locally bounded variation.

This last definition is the correct level of generality for stochastic integration, and it is worth understanding why the decomposition takes the shape it does. The two summands are exactly the two mechanisms by which an integrator can be usable: A has bounded variation, so $\int g dA$ exists pathwise as a Riemann–Stieltjes integral in the classical sense; and M is a local martingale, so $\int g dM$ exists in the Itô sense, by the L^2 construction of Section 1.2 together with localisation. A semimartingale is precisely a process that can be integrated against by splitting the work between deterministic analysis and probability, and the Bichteler–Dellacherie theorem makes this converse precise: semimartingales are, in a suitable topological sense, the *only* processes admitting a reasonable theory of stochastic integration; see [35].

We shall not develop that general theory — every integrator in Part I is continuous, and for continuous processes the decomposition is unique — but the definition is recorded because of what happens when it fails. Not every process of interest is a semimartingale: fractional Brownian motion with Hurst parameter $H \neq \frac{1}{2}$ is not one, a result of Rogers [37], and consequently neither the Itô integral nor any of the machinery built on it is available for it. Since such processes are exactly what the empirical evidence on volatility will point towards,



the reader should register Definition 1.1.30 less as a technical generalisation than as a boundary: it marks the outer limit of the theory developed in this chapter, and Part II begins where it ends.

1.1.4 Brownian Motion

Brownian motion is a fundamental stochastic process whose importance has grown throughout many areas of mathematics. It is complex enough to generate a new class of *stochastic differential equations* and to address numerous applied problems involving randomness, yet simple enough to allow explicit calculations. Its properties also enable efficient simulation, making it a cornerstone of Monte Carlo methods across disciplines.

Definition 1.1.31 (Brownian Motion). A stochastic process $W : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}$ is called a **Brownian motion** (or **Wiener process**) if the following conditions hold:

$$\text{i)} \quad \mathbb{P}(W_0 = 0) = 1. \quad (1.26)$$

$$\text{ii)} \quad \mathbb{P}(\{\omega : t \mapsto W_t(\omega) \text{ is continuous}\}) = 1. \quad (1.27)$$

$$\text{iii)} \quad \text{For } 0 \leq s < t, \quad W_t - W_s \sim \mathcal{N}(0, t - s). \quad (1.28)$$

iv) The process has independent increments: for $0 \leq t_1 < t_2 < \dots < t_n$, the random variables

$$\{W_{t_1}, W_{t_2} - W_{t_1}, W_{t_3} - W_{t_2}, \dots, W_{t_n} - W_{t_{n-1}}\} \quad (1.29)$$

are independent.

Some properties of Brownian motion follow immediately from the definition and are presented below. The last one illustrates the property known as *stochastic self-similarity*.

In order to work with the broadest possible class of stochastic processes, we introduce the following general result.

Proposition 1.1.32. *Let $S(t)$, $t \geq 0$, be a process adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$ such that for any $0 \leq s < t$, the increment $S(t) - S(s)$ is independent of $\mathcal{F}_s$. Then $S(t)$ has independent increments.*

Proof. Let $0 \leq t_1 < t_2 < \dots < t_n$ and write $\Delta_k = S(t_k) - S(t_{k-1})$. We show by induction on n that the vector $(\Delta_1, \dots, \Delta_n)$ has independent components. For $n = 1$ there is nothing to prove. Assume the claim for $n - 1$. Each of $\Delta_1, \dots, \Delta_{n-1}$ is $\mathcal{F}_{t_{n-1}}$ -measurable, since S is adapted and $t_k \leq t_{n-1}$ for $k \leq n - 1$; hence so is any Borel function of them. By hypothesis Δ_n is independent of $\mathcal{F}_{t_{n-1}}$, and therefore independent of $(\Delta_1, \dots, \Delta_{n-1})$. For bounded Borel $g_1, \dots, g_n$ this gives

$$\mathbb{E} \left[\prod_{k=1}^n g_k(\Delta_k) \right] = \mathbb{E} [g_n(\Delta_n)] \mathbb{E} \left[\prod_{k=1}^{n-1} g_k(\Delta_k) \right] = \prod_{k=1}^n \mathbb{E} [g_k(\Delta_k)],$$

the last equality by the induction hypothesis. Since this holds for all such g_k , the increments are independent. ■

Therefore, a process $W(t)$ satisfying the conditions of Definition 1.1.31, together with a filtration satisfying Proposition 1.1.32, is a Brownian motion.¹ Throughout, we will work with processes adapted to their natural filtration, i.e. W_t is $\mathcal{F}_t$ -measurable and increments $W(t) - W(s)$ are independent of $\mathcal{F}_s$ whenever $s \leq t$.

The next definition and proposition highlight some of the structural richness of such processes. In particular, the last property quantifies the *roughness* of sample paths, a key feature of Brownian motion.

Definition 1.1.33 (Quadratic Variation). Let $f : [0, T] \rightarrow \mathbb{R}$, and for a partition $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = t\}$ of $[0, t]$ write

$$\|\mathcal{P}\| = \max_{1 \leq k \leq n} (t_k - t_{k-1}). \quad (1.30)$$

If the limit

$$[f]_t := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{k=1}^n \left(f(t_k) - f(t_{k-1}) \right)^2 \quad (1.31)$$

exists, it is called the **quadratic variation** of f on $[0, t]$.

Two cautions are in order. First, the limit in (1.31) need not exist for an arbitrary f , which is why the definition is phrased conditionally. Second — and this is the point that will matter — for a *stochastic* process the sums in (1.31) are random variables, and the limit must be specified as a limit of random variables. We shall always understand it in $L^2(\Omega)$, which for our purposes is both the easiest to obtain and strong enough to imply convergence in probability, and hence almost sure convergence along a subsequence.

It is worth pausing on how strange (1.31) is from the point of view of ordinary calculus. If f is continuously differentiable, then $|f(t_k) - f(t_{k-1})| \leq C(t_k - t_{k-1})$ and the sum is bounded by $C^2 \|\mathcal{P}\| t \rightarrow 0$: smooth functions have zero quadratic variation. The same computation applies to any function of bounded variation that is continuous. A non-zero quadratic variation is therefore a certificate of irregularity, and the following proposition asserts that Brownian motion carries exactly such a certificate.

Proposition 1.1.34. *Let $(W_t)_{t \geq 0}$ be a Brownian motion. Then:*

a) *For $t > 0$, $W_t \sim \mathcal{N}(0, t)$. Moreover, for all $s, t \geq 0$,*

$$\mathbb{E}[W_s W_t] = \min\{s, t\}. \quad (1.32)$$

b) *For $t_0 \geq 0$, the process*

$$Y(t) = W(t + t_0) - W(t_0) \quad (1.33)$$

is also a Brownian motion.

c) *For any $\lambda > 0$, the process*

$$Z(t) = \frac{1}{\sqrt{\lambda}} W(\lambda t) \quad (1.34)$$

is also a Brownian motion.

d) *Brownian motion is both a Markov process and a martingale.*

¹Analogous considerations hold for a Poisson process; see [15].



e) The quadratic variation of W over $[0, t]$ exists and equals

$$[W]_t = t \quad \text{the limit being taken in } L^2(\Omega). \quad (1.35)$$

f) The process $W_t^2 - t$ is a martingale.

Properties b) and c) are called translation invariance and scaling invariance, respectively.

Proof. a) By i) we have $W_t = W_t - W_0$ almost surely, so $W_t \sim \mathcal{N}(0, t)$ by iii). For $s < t$, write $W_t = (W_t - W_s) + W_s$ and use that $W_t - W_s$ is independent of W_s with zero mean:

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s = \min\{s, t\}.$$

b) $Y_0 = 0$ and the paths of Y are continuous, since those of W are. For $s < t$ we have $Y_t - Y_s = W_{t+t_0} - W_{s+t_0} \sim \mathcal{N}(0, t-s)$, and increments of Y over disjoint intervals are increments of W over disjoint intervals, hence independent.

c) Again $Z_0 = 0$ and the paths are continuous. For $s < t$, $Z_t - Z_s = \lambda^{-1/2}(W_{\lambda t} - W_{\lambda s})$ is normal with mean 0 and variance $\lambda^{-1}(\lambda t - \lambda s) = t - s$; independence of increments is inherited from W because $s \mapsto \lambda s$ preserves disjointness of intervals.

d) Integrability is clear from a). For $s < t$, using independence of $W_t - W_s$ from $\mathcal{F}_s$ and $\mathcal{F}_s$ -measurability of W_s ,

$$\mathbb{E}[W_t | \mathcal{F}_s] = \mathbb{E}[W_t - W_s | \mathcal{F}_s] + \mathbb{E}[W_s | \mathcal{F}_s] = \mathbb{E}[W_t - W_s] + W_s = W_s,$$

so W is a martingale; the same computation with $\mathbb{P}(W_t \in A | \mathcal{F}_s)$ in place of $\mathbb{E}[W_t | \mathcal{F}_s]$ shows that the conditional law of W_t given $\mathcal{F}_s$ depends on ω only through W_s , which is the Markov property.

e) Let $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$ and set $X_k = (W_{t_k} - W_{t_{k-1}})^2 - (t_k - t_{k-1})$. The X_k are independent with $\mathbb{E}[X_k] = 0$, and since $\mathbb{E}[Z^4] = 3\sigma^4$ for $Z \sim \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}[X_k^2] = 3(t_k - t_{k-1})^2 - 2(t_k - t_{k-1})^2 + (t_k - t_{k-1})^2 = 2(t_k - t_{k-1})^2.$$

Hence, by independence and orthogonality,

$$\mathbb{E}\left[\left(\sum_k (W_{t_k} - W_{t_{k-1}})^2 - t\right)^2\right] = \sum_k \mathbb{E}[X_k^2] = 2 \sum_k (t_k - t_{k-1})^2 \leq 2\|\mathcal{P}\| t \rightarrow 0$$

as $\|\mathcal{P}\| \rightarrow 0$, which is (1.35). See also [26, Thm. 4.1.2].

f) Integrability follows from a). For $s < t$, expanding $W_t^2 = (W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2$ and conditioning,

$$\mathbb{E}[W_t^2 | \mathcal{F}_s] = (t - s) + 2W_s \cdot 0 + W_s^2,$$

so $\mathbb{E}[W_t^2 - t | \mathcal{F}_s] = W_s^2 - s$. ■

One might wonder whether such a process actually exists, given the strong set of requirements in Definition 1.1.31. The question is not idle: we have demanded independent Gaussian increments *and* continuity of every sample path, and there is no reason a priori why the two should be compatible. A full construction would take us too far afield, but it would be unsatisfying to leave the matter entirely unaddressed, so we indicate the two results which settle it — both of which we shall have occasion to use again for their own sake.

The first supplies the process. Recall the simple symmetric random walk of Example 1.1.21, where we promised to return once the jump size and the waiting time between jumps had been adjusted. This is that return.

Theorem 1.1.35 (Donsker's invariance principle). *Let $\{\xi_k\}_{k \geq 1}$ be i.i.d. with $\mathbb{E}[\xi_1] = 0$ and $\mathbb{V}[\xi_1] = 1$, and let $X_n = \sum_{k=1}^n \xi_k$ be the associated random walk. Define the rescaled, linearly interpolated process*

$$W_t^{(n)} := \frac{1}{\sqrt{n}} \left(X_{[nt]} + (nt - [nt]) \xi_{[nt]+1} \right), \quad t \in [0, T]. \quad (1.36)$$

Then $W^{(n)} \Rightarrow W$ as $n \rightarrow \infty$, where the convergence is in distribution on the space $C[0, T]$ of continuous functions with the uniform topology, and W is a Brownian motion. ■

The scaling in (1.36) is the whole content of the theorem: space is compressed by $\sqrt{n}$ while time is compressed by n . That is exactly the relation space $\sim \sqrt{\text{time}}$ which Definition 1.1.31 imposes through $W_t - W_s \sim \mathcal{N}(0, t - s)$, and which reappears as the scaling invariance of Proposition 1.1.34(c). Donsker's theorem also explains why Brownian motion is *universal* in the same sense that the Gaussian distribution is: the limit does not depend on the law of ξ_1 beyond its first two moments, so any accumulation of small, independent, finite-variance shocks produces the same object. This is the honest reason Brownian motion is ubiquitous in modelling, and it is worth keeping in mind that the same universality argument gives no warrant at all for the shocks being independent — a point the empirical evidence of Part II will press hard.

The second result supplies the continuity, and does so from a moment condition alone.

Theorem 1.1.36 (Kolmogorov's continuity criterion). *Let $X = \{X_t\}_{t \in [0, T]}$ be a stochastic process for which there exist constants $\alpha, \beta, C > 0$ with*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C |t - s|^{1+\beta}, \quad s, t \in [0, T]. \quad (1.37)$$

Then X admits a continuous modification $\tilde{X}$. Moreover, the paths of $\tilde{X}$ are almost surely locally γ -Hölder continuous for every $\gamma \in (0, \beta/\alpha)$. ■

Applied to Brownian motion the criterion is immediate. Since $W_t - W_s \sim \mathcal{N}(0, t - s)$, the Gaussian moments give $\mathbb{E}[|W_t - W_s|^{2m}] = c_m |t - s|^m$ for every $m \in \mathbb{N}$, so (1.37) holds with $\alpha = 2m$ and $\beta = m - 1$. A continuous modification therefore exists, which is what was needed; and the Hölder conclusion yields exponents $\gamma < (m - 1)/2m$, which as $m \rightarrow \infty$ approach $1/2$ from below but never reach it. That the supremum is exactly $1/2$, and that it is not attained, is the subject of the next subsection.

Note the word *modification*: what Theorem 1.1.36 produces is a process agreeing with X at each fixed time almost surely, not one agreeing with X along whole paths (Remark 1.1.8). This is not a defect of the theorem but a necessity, since (1.37) constrains only two-dimensional distributions and cannot possibly determine path behaviour on its own. As is customary, we shall always work with the continuous modification and cease to mention the distinction.

1.1.5 Path regularity

We now examine how irregular the trajectories of Brownian motion actually are. This subsection is short, and every statement in it is classical, but it is the most consequential material in the chapter: the construction of the Itô integral in Section 1.2 is a *response* to what follows, and the theory developed in Part II is a response to what happens when these estimates are pushed slightly further.

We begin with the observation that quadratic variation and total variation are in direct competition.



Definition 1.1.37 (Total variation). The **total variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_1(f)_t := \sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|, \quad (1.38)$$

the supremum being over all partitions $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$. If $V_1(f)_t < \infty$ we say f has **bounded variation** on $[0, t]$.

Bounded variation is exactly the condition under which the Riemann–Stieltjes integral $\int_0^t g df$ is defined for every continuous g ; equivalently, it is the condition under which f induces a signed measure df . It is therefore the natural first thing to ask of a prospective integrator.

Proposition 1.1.38. *Let W be a Brownian motion. Then, almost surely, $V_1(W)_t = \infty$ for every $t > 0$.*

Proof: Fix $t > 0$ and a sequence of partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| \rightarrow 0$. On any partition,

$$\sum_k (W_{t_k} - W_{t_{k-1}})^2 \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) \cdot \sum_k |W_{t_k} - W_{t_{k-1}}| \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) V_1(W)_t. \quad (1.39)$$

Suppose, for contradiction, that $V_1(W)_t < \infty$ on an event of positive probability. Sample paths of W are continuous, hence uniformly continuous on the compact $[0, t]$, so the maximal increment on the right of (1.39) tends to 0 as $\|\mathcal{P}_n\| \rightarrow 0$. The whole right-hand side therefore tends to 0, forcing $[W]_t = 0$ on that event. This contradicts (1.35), which gives $[W]_t = t > 0$ in $L^2(\Omega)$ and hence almost surely along a subsequence. ■

This is the single most important fact in the chapter, and it is worth stating its consequence without any hedging. **The integral $\int_0^t f dW$ cannot be defined pathwise as a Riemann–Stieltjes integral**, for any non-trivial class of integrands, because W does not induce a measure on $[0, t]$ along any path. Every difficulty in Section 1.2 — the restriction to adapted integrands, the insistence on evaluating at the left endpoint, the approximation in $L^2(\Omega)$ rather than pathwise, the appearance of a second-order term in the chain rule — traces back to (1.39). Itô’s construction is not one possible approach among several; it is a way of exploiting probabilistic structure to recover an object which deterministic analysis cannot reach.

The same competition can be expressed more finely by asking, not whether the sum of increments converges, but for which exponent p the sum of p -th powers does.

Definition 1.1.39 (p -variation). Let $p \geq 1$. The **p -variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_p(f)_t := \left(\sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|^p \right)^{1/p}, \quad (1.40)$$

and f is said to have *finite p -variation* on $[0, t]$ if $V_p(f)_t < \infty$. We write $f \in \mathcal{C}^{p\text{-var}}([0, t])$.

For $p = 1$ this is Definition 1.1.37, and for $p = 2$ it is a supremum version of the quadratic variation of Definition 1.1.33. The scale is monotone: if f is continuous and $V_p(f)_t < \infty$, then $V_q(f)_t < \infty$ for every $q > p$, since $|f(t_k) - f(t_{k-1})|^q \leq (\max_k |f(t_k) - f(t_{k-1})|)^{q-p} |f(t_k) - f(t_{k-1})|^p$. Each function thus has a critical exponent below which its p -variation is infinite and above which it is finite, and this exponent is a precise measure of roughness.

Theorem 1.1.40 (Regularity of Brownian paths). *Let W be a Brownian motion. Then, almost surely:*

- i) the sample path $t \mapsto W_t$ is locally γ -Hölder continuous for every $\gamma < \frac{1}{2}$, and is locally γ -Hölder for no $\gamma \geq \frac{1}{2}$;
- ii) $V_p(W)_t < \infty$ for every $p > 2$ and every t , whereas $V_p(W)_t = \infty$ for every $p \leq 2$;
- iii) the sample path is nowhere differentiable: for almost every ω , there is no t at which $t \mapsto W_t(\omega)$ possesses a finite derivative. ■

Part i) was half-proved above via Kolmogorov's criterion; the negative half, like iii), is due to Paley, Wiener and Zygmund, and proofs of all three may be found in [1] or [25]. Part iii) is worth a moment's reflection. A continuous function which is differentiable nowhere is already a celebrated pathology in real analysis, exhibited with some effort by Weierstrass; the theorem says that if one picks a Brownian path at random one obtains such a function with probability one. The pathological case is the generic case.

It is part ii) which will matter most. Let us record it in the form we shall use.

Remark 1.1.41 (The critical exponent). Brownian motion has finite p -variation exactly for $p > 2$. The number 2 is a threshold in a strong sense, and the reason is the following. For a path of finite p -variation with $p < 2$, the integral $\int g df$ can be constructed *pathwise*, by purely deterministic means, provided g has finite q -variation with $1/p + 1/q > 1$; this is Young's integral, and no probability is involved. Brownian motion misses this regime — but only just, and by an amount that no refinement of the estimate can recover, since $V_2(W)_t = \infty$.

The Itô integral of Section 1.2 is precisely the tool that bridges this gap, and it does so by spending probabilistic structure: the integrand must be adapted, the approximation is in $L^2(\Omega)$ rather than pathwise, and the resulting object is defined only up to null sets. That is a real price. It buys us the entire theory of Part I, but it also means the theory is tied to a specific process and a specific filtration, and it is exactly this dependence which becomes an obstacle when the driving signal is rougher than Brownian — when even $p = 2$ is out of reach, and neither Young nor Itô applies. The theory that operates in that regime is the subject of Part II.

[FIGURE HERE] Simulated Brownian path with, alongside it, the partition sums for total variation (diverging) and quadratic variation (converging to t) as the mesh is refined. This is the visual statement of Proposition 1.1.38.

Finally, we record the relation between the pathwise quadratic variation of Definition 1.1.33 and the predictable version produced by the Doob–Meyer decomposition of Subsection 1.2.5, since both notations occur in the literature and we shall use them interchangeably.

Proposition 1.1.42. *Let M be a continuous square-integrable martingale. Then the pathwise quadratic variation $[M]_t$ of Definition 1.1.33 exists as a limit in $L^2(\Omega)$ and coincides, up to indistinguishability, with the compensator $\langle M \rangle_t$ of Theorem 1.2.26. Furthermore, the polarised quantity*

$$\langle M, N \rangle_t := \frac{1}{2} \left(\langle M + N \rangle_t - \langle M \rangle_t - \langle N \rangle_t \right) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_k (M_{t_k} - M_{t_{k-1}}) (N_{t_k} - N_{t_{k-1}}) \quad (1.41)$$

*defines the **quadratic covariation** of two continuous square-integrable martingales, is bilinear and symmetric, and satisfies the Kunita–Watanabe inequality*

$$\int_0^t |d\langle M, N \rangle_s| \leq \sqrt{\langle M \rangle_t} \sqrt{\langle N \rangle_t} \quad \text{almost surely.} \quad \blacksquare \quad (1.42)$$

The distinction between $[M]$ and $\langle M \rangle$ is genuine for processes with jumps, where the two differ by the accumulated squared jumps; for the continuous processes of this dissertation it is empty, and we shall write $\langle M \rangle$ throughout. Quadratic covariation is what gives meaning to the off-diagonal entries $\rho_{ij} dt$ of the multiplication table in Subsection 1.2.4, and it is the object that the correlation parameter of a stochastic volatility model actually controls.

1.2 Stochastic Calculus

Stochastic Calculus is a wide area in Mathematics that allows, for example, the incorporation of randomness and uncertainty into differential equations. One of our main aims is to give meaning to expressions of the kind

$$dX_t = a(t, X_t) dt + b(t, X_t) dW_t, \quad X_0 \in \mathbb{R}. \quad (1.43)$$

Here dW_t will be specified in the next subsection. Notice that this generalizes the deterministic case: if $b \equiv 0$, then X_t reduces to the solution of the ODE $dY_t = a(t, Y_t) dt$.

We shall first construct a rigorous integral interpretation of such expressions, providing concrete meaning to the stochastic term. Afterwards, we will introduce the stochastic chain rule (Itô's Lemma), which differs from the classical one in a somewhat surprising way, and then proceed to study the existence and uniqueness of solutions, and some of their properties.

It must be highlighted that the following dissertation can be expanded up to the most general setting, however we will stick to the most classical presentation of the subject, leaving the most general case a worthy space at the end, but without letting the trees hide the forest.

1.2.1 The Itô Integral

The stochastic differential equation (1.43) can be reformulated in integral form as

$$X_t - X_0 = \int_0^t a(s, X_s) ds + \int_0^t b(s, X_s) dW_s. \quad (1.44)$$

For each $\omega \in \Omega$, the first integral on the right-hand side is nothing new. Regarding the second one, we may ask whether there is a sensible way of defining

$$M_t = \int_0^t f(s, \omega) dW_s, \quad 0 \leq t \leq T,$$

so that $\{M_t\}_{t \in [0, T]}$ is a martingale. Another important consideration is that the integrand must be *adapted* to the filtration associated with W_t . These requirements naturally lead to the following definitions.

Definition 1.2.1 (Adapted square-integrable processes). Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ be a filtered probability space.

- i) The space of stochastic processes on $[0, T]$, adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ and such that

$$\int_0^T \mathbb{E}[|f(t)|^2] dt < \infty,$$

will be denoted by $L_{ad}^2([0, T] \times \Omega)$. This is a Hilbert space with the inner product

$$\langle f, g \rangle = \int_0^T \mathbb{E}[f(t)g(t)] dt.$$

ii) We say $f \in L^2_{ad}([0, T] \times \Omega)$ is a **step stochastic process** if

$$f(t, \omega) = \sum_{i=1}^n \varphi_\omega(t_{i-1}) \mathbf{1}_{[t_{i-1}, t_i)}(t),$$

where $\varphi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and has finite variance ($\mathbb{E}[\varphi(t_{i-1})^2] < \infty$).

The construction of the stochastic integral begins by defining it on step processes and extending by density. For such f , take $t_0 = 0$, $t_n = T$, and define²

$$I(f) = \int_0^T f(t) dW_t := \sum_{i=1}^n \varphi(t_{i-1}) (W(t_i) - W(t_{i-1})). \quad (1.45)$$

This mapping $f \mapsto I(f)$ is linear. The following lemmas are key.

Lemma 1.2.2. *If f is a step stochastic process, then*

$$\mathbb{E}[I(f)] = 0, \quad \mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

Proof: Write $\Delta_i W = W(t_i) - W(t_{i-1})$, so that $I(f) = \sum_{i=1}^n \varphi(t_{i-1}) \Delta_i W$. Since $\varphi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and $\Delta_i W$ is independent of $\mathcal{F}_{t_{i-1}}$ with mean zero, conditioning gives

$$\mathbb{E}[\varphi(t_{i-1}) \Delta_i W] = \mathbb{E}[\varphi(t_{i-1}) \mathbb{E}[\Delta_i W | \mathcal{F}_{t_{i-1}}]] = 0,$$

and summing yields $\mathbb{E}[I(f)] = 0$. For the second moment, expand the square. If $i < j$ then $\varphi(t_{i-1}) \varphi(t_{j-1}) \Delta_i W$ is $\mathcal{F}_{t_{j-1}}$ -measurable while $\Delta_j W$ is independent of it with mean zero, so the cross terms vanish by the same argument. The diagonal terms give, using independence of $\Delta_i W$ from $\mathcal{F}_{t_{i-1}}$ once more,

$$\mathbb{E}[\varphi(t_{i-1})^2 (\Delta_i W)^2] = \mathbb{E}[\varphi(t_{i-1})^2] (t_i - t_{i-1}) = \int_{t_{i-1}}^{t_i} \mathbb{E}[f(t)^2] dt,$$

since $f \equiv \varphi(t_{i-1})$ on $[t_{i-1}, t_i)$. Summing over i gives $\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt$. ■

Note that both computations rest on exactly one structural fact: the integrand is evaluated at the *left* endpoint, which is what makes $\varphi(t_{i-1})$ independent of the increment it multiplies. Evaluating anywhere else destroys both identities.

Lemma 1.2.3 (Density of step processes). *For every $f \in L^2_{ad}([0, T] \times \Omega)$, there exists a sequence $\{f_n\}_{n=1}^\infty$ of step stochastic processes such that*

$$\lim_{n \rightarrow \infty} \int_0^T \mathbb{E}[(f(t) - f_n(t))^2] dt = 0.$$

That is, step processes are dense in $L^2_{ad}([0, T] \times \Omega)$.

Proof: The argument is given in Appendix B.2; see also [26, Lemma 4.3.3]. ■

We are now ready to extend the definition.

²The choice of the left endpoint t_{i-1} ensures that the stochastic integral preserves the martingale property. Variants with different evaluation points lead to different calculi, e.g. Stratonovich calculus, where the classical chain rule holds (see [26]).



Definition 1.2.4 (Stochastic Integral). Let $f \in L^2_{ad}([0, T] \times \Omega)$. We define its *stochastic integral* over $[0, T]$ as the limit in $L^2(\Omega)$ of $I(f_n)$ for any approximating sequence of step processes $\{f_n\}$ with $f_n \rightarrow f$ in $L^2_{ad}([0, T] \times \Omega)$.

The following fundamental result follows.

Theorem 1.2.5 (Itô's isometry). Let $f \in L^2_{ad}([0, T] \times \Omega)$.

i) The stochastic integral $I(f)$ is a centered random variable and

$$\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

ii) For another $g \in L^2_{ad}([0, T] \times \Omega)$,

$$\mathbb{E} \left[\left(\int_0^T f(t) dW_t \right) \left(\int_0^T g(t) dW_t \right) \right] = \int_0^T \mathbb{E}[f(t)g(t)] dt.$$

Proof: i) Let $\{f_n\}$ be step processes with $f_n \rightarrow f$ in $L^2_{ad}([0, T] \times \Omega)$, as furnished by Lemma 1.2.3. By Lemma 1.2.2 applied to $f_n - f_m$,

$$\mathbb{E}[(I(f_n) - I(f_m))^2] = \int_0^T \mathbb{E}[(f_n(t) - f_m(t))^2] dt \xrightarrow{n, m \rightarrow \infty} 0,$$

so $\{I(f_n)\}$ is Cauchy in $L^2(\Omega)$ and its limit $I(f)$ exists and is independent of the approximating sequence. Both $f \mapsto \mathbb{E}[I(f)]$ and $f \mapsto \mathbb{E}[I(f)^2]$ are continuous for the L^2 norms involved, so the identities $\mathbb{E}[I(f_n)] = 0$ and $\mathbb{E}[I(f_n)^2] = \int_0^T \mathbb{E}[f_n^2] dt$ pass to the limit.

ii) Apply i) to $f + g$ and expand: $\mathbb{E}[I(f + g)^2] = \mathbb{E}[I(f)^2] + 2\mathbb{E}[I(f)I(g)] + \mathbb{E}[I(g)^2]$ by linearity of I , while the right-hand side expands identically. Cancelling the square terms gives the stated polarisation. ■

In the language of Proposition 1.1.4, item 9, this is precisely the statement that I is an isometry from $L^2_{ad}([0, T] \times \Omega)$ into $L^2(\Omega)$, extended from a dense subspace by continuity — the standard extension of a bounded operator, and the reason the construction works at all.

It is worth noting that if f is deterministic (i.e. does not depend on ω), then

$$I(f) \sim \mathcal{N}\left(0, \int_0^T f(t)^2 dt\right).$$

For $0 \leq t \leq T$, we have $\int_0^t \mathbb{E}[f(s)^2] ds \leq \int_0^T \mathbb{E}[f(s)^2] ds$, so $f \in L^2_{ad}([0, t] \times \Omega)$ and the stochastic integral is well defined on $[0, t]$. Consider then the process $X_t = \int_0^t f(s) dW(s)$. We know this process has finite first two moments, and so we can take the conditional expectation of X_t with respect to the given filtration.

Theorem 1.2.6 (Martingale property). Take $f \in L^2_{ad}([0, T] \times \Omega)$. Then the process given by $X_t = I(f\mathbb{1}_{[0, t]})$ is a martingale with respect to the filtration $\mathcal{F}_t$.

Proof: Integrability follows from Itô's isometry, since $\mathbb{E}[X_t^2] = \int_0^t \mathbb{E}[f(s)^2] ds < \infty$ and $L^2(\Omega) \subset L^1(\Omega)$ on a probability space. Adaptedness is inherited from the approximating step processes.

For the martingale identity, take first f a step process and $s < t$ lying in the partition defining it (refining the partition if necessary, which changes nothing). Then $X_t - X_s = \sum_i \varphi(t_{i-1})(W(t_i) -$

$W(t_{i-1}))$, the sum running over the subintervals of $[s, t]$. For each such i we have $t_{i-1} \geq s$, so conditioning first on $\mathcal{F}_{t_{i-1}}$ and then on $\mathcal{F}_s$ gives

$$\mathbb{E}[\varphi(t_{i-1})(W(t_i) - W(t_{i-1})) \mid \mathcal{F}_s] = \mathbb{E}[\varphi(t_{i-1}) \mathbb{E}[W(t_i) - W(t_{i-1}) \mid \mathcal{F}_{t_{i-1}}] \mid \mathcal{F}_s] = 0,$$

by the tower property (Proposition 1.1.4, item 5) and independence of the increment from $\mathcal{F}_{t_{i-1}}$. Hence $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$.

For general $f \in L^2_{ad}$, take step processes $f_n \rightarrow f$ and write X^n for the corresponding integrals. Itô's isometry gives $X^n_t \rightarrow X_t$ in $L^2(\Omega)$, hence in $L^1(\Omega)$, and conditional expectation is a contraction on L^1 (Proposition 1.1.4, item 7), so $\mathbb{E}[X^n_t \mid \mathcal{F}_s] \rightarrow \mathbb{E}[X_t \mid \mathcal{F}_s]$ in $L^1(\Omega)$. The left-hand side equals $X^n_s \rightarrow X_s$, and the martingale identity passes to the limit. ■

The Markov property also holds, but we will return to that when we see these processes as diffusions in Subsection 1.2.6.

We shall now address the continuity of the process X_t . We must keep in mind that this integral is not defined for each ω in a Riemann sense. The following result is therefore not as elementary as in real analysis.

Theorem 1.2.7 (Continuity property). *Suppose $f \in L^2_{ad}([0, T] \times \Omega)$. Then, almost all paths of $X_t = \int_0^t f(s) dW(s)$ are continuous on $[0, T]$.*

Proof: The argument is given in Appendix B.3; see also [26, Thm. 4.6.2]. ■

This is the payoff of Proposition 1.1.28 announced earlier: the maximal inequality is exactly what upgrades convergence at each fixed time to uniform convergence along a subsequence, and hence L^2 convergence to a pathwise statement.

Finally, a result that actually links Riemann sums with stochastic integrals will be helpful in the next section.

Theorem 1.2.8. *Suppose $f \in L^2_{ad}([0, T] \times \Omega)$ and assume that $\mathbb{E}[f(t)f(s)]$ is a continuous function of t and s . Then, for partitions of $[0, T]$, given by $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = T\}$,*

$$\int_0^T f(t) dW(t) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^n f(t_{i-1})(W(t_i) - W(t_{i-1})) \quad (1.46)$$

in $L^2(\Omega)$, where the limit is taken as in Definition 1.1.33.

Proof: The argument is given in Appendix B.4; see also [26, Thm. 4.7.1]. ■

The continuity hypothesis on $\mathbb{E}[f(t)f(s)]$ is what makes this work and cannot simply be dropped: without it the left-endpoint approximation need not converge, and the Riemann-sum interpretation of the Itô integral fails.

More genearely, we can consider a larger family of integrands. Let $\mathcal{L}(\Omega, L^2[0, T])$ be the family of processes f such that

- $f(t)$ is $\mathcal{F}_t$ -adapted.
- $\int_0^T |f(t)|^2 dt < \infty$ almost surely.



Notice how we dropped the condition of having finite expectation, so that we have indeed a larger class of processes. Consider for example the process $f(t) = e^{W(t)^2}$. It follows that

$$\mathbb{E}[f(t)^2] = \begin{cases} \frac{1}{\sqrt{1-4t}} & \text{if } 0 \leq t < 1/4 \\ \infty & \text{if } t \geq 1/4. \end{cases} \quad (1.47)$$

Hence $\int_0^1 e^{W(t)^2} dW_t$ does not make sense with the previous construction. It can be shown, however, that these processes are still pretty worth the attention. Beyond the scope of our purposes, it can be shown that:

Theorem 1.2.9. (*Local Martingale Property*) *Let $f \in \mathcal{L}(\Omega, L^2[0, T])$ and $t \in [0, T]$. Then*

$$X_t = \int_0^t f(s) dW_s$$

is a local martingale. Furthermore, it has a continuous realization. ■

1.2.2 Itô's Chain Rule

One of the most shocking facts about this new integration is a sophisticated version of the Chain Rule, that generalizes the familiar expression $d(f(g))(x) = f'(g(x))dg(x)$. The key underlying fact for this new integration rule is the non-zero quadratic variation of Brownian motion. This fantastic new calculus allows us to find some new relations and solve new equations that naturally generalize the already rich world of differential equations.

Definition 1.2.10. (Itô process)

Given an initial random variable X_0 , an Itô process is a stochastic process on $[0, T]$ of the form

$$X_t = X_0 + \int_0^t g(s) ds + \int_0^t f(s) dW_s \iff dX_t = g(s) ds + f(s) dW_s, \quad X(0) = X_0$$

where we emphasize the equivalent formulations, since the latter is just a way of writing the former. We impose X_0 being $\mathcal{F}_0$ -measurable, $g \in \mathcal{L}(\Omega, L^1[0, T])$ and $f \in \mathcal{L}(\Omega, L^2[0, T])$. Note how the first integral is just a Lebesgue integral for each ω , that is, for each path of W . The second is a proper Itô integral.

We directly state the more general form of Itô's new calculus rule, in the convenient *differential formulation*. A complete proof is beyond the scope of this Thesis. However, such a renowned Theorem has been addressed by several authors, for example in [26].

Theorem 1.2.11 (Itô's Rule). *Let $dX_t = \mu(t) dt + \sigma(t) dW_t$ be an Itô process, and let $V(t, x)$ be a continuous function on $[0, T] \times \mathbb{R}$ with continuous partial derivatives $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial x}$ and $\frac{\partial^2 V}{\partial x^2}$. Then $V(t, X_t)$ is also an Itô process, and*

$$dV(t, X_t) = \left(\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma^2(t) \frac{\partial^2 V}{\partial x^2}(t, X_t) \right) dt + \sigma(t) \frac{\partial V}{\partial x}(t, X_t) dW_t. \quad (1.48)$$

A convenient way to derive Equation (1.48) is through a symbolic derivation using the Taylor expansion to first order in dt and second order in dX_t , together with the *Itô table*:

$$\begin{array}{c|cc} \times & dW_t & dt \\ \hline dW_t & dt & 0 \\ dt & 0 & 0 \end{array}$$

Applying the Taylor expansion to $V(t, X_t)$ gives

$$dV(t, X_t) = \frac{\partial V}{\partial t}(t, X_t) dt + \frac{\partial V}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2}(t, X_t) (dX_t)^2.$$

Using the Itô table, we have $(dX_t)^2 = \sigma(t)^2 dt$, so using $dX_t = \mu(t) dt + \sigma(t) dW_t$ yields

$$dV(t, X_t) = \frac{\partial V}{\partial x}(t, X_t) \sigma(t) dW_t + \left[\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma(t)^2 \frac{\partial^2 V}{\partial x^2}(t, X_t) \right] dt,$$

which is precisely Equation (1.48).

We stress that the computation just performed is a *mnemonic*, not a proof: the symbols dW_t and $(dW_t)^2$ have been manipulated as though they were infinitesimals, which they are not. What makes the manipulation legitimate is the content of (1.35) — the second-order term survives precisely because W has non-vanishing quadratic variation — and a genuine proof proceeds by Taylor-expanding along a partition and controlling the remainder in $L^2(\Omega)$. We refer to [26] for the details. The value of the Itô table is that it makes the bookkeeping automatic once one accepts that $\langle W \rangle_t = t$, and we shall use it in that spirit throughout.

Just like ordinary Chain Rule, this has opened plenty of possibilities in order to calculate some stochastic integrals in terms of others.

Example 1.2.12. We can use this result to easily obtain a simpler form for the process $W_t dW_t$. We can simply consider $g(x) = x^2$ and define $H_t = g(W_t)$ to obtain

$$dH_t = 2W_t dW_t + \frac{2}{2} (dW_t)^2 = dt + 2W_t dW_t \quad (1.49)$$

so that $\int_0^t W_s dW_s = \frac{1}{2} (W_t^2 - t)$, which is a much simpler expression, and clearly a martingale.

Example 1.2.13. In Finance, the Black-Scholes model relies on the *log-normal* equation, given by $dS_t = rS_t dt + \sigma S_t dW_t$, where $r, \sigma \in \mathbb{R}$ and $\sigma > 0$. It can be solved by taking $H_t = \ln(S_t)$, so that

$$dH_t = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 = r dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt \quad (1.50)$$

obtaining $H_t = H_0 + (r - \sigma^2/2)t + \sigma W_t \implies S_t = S_0 e^{(r - \sigma^2/2)t + \sigma W_t}$.

Example 1.2.14. (Lamperti's Transform)

Consider a general process $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ and assume σ is differentiable once with respect to time and twice with respect to x . Define the function

$$\Gamma(t, x) = \int_{X_0}^x \frac{1}{\sigma(t, s)} ds \quad (1.51)$$



and consider the process $H_t = \Gamma(t, X_t)$. After a few calculations, we arrive to

$$\begin{aligned} dH_t &= \left(\frac{\partial \Gamma}{\partial t} + \mu(t, X_t) \frac{\partial \Gamma}{\partial x} + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 \Gamma}{\partial x^2} \right) dt + \underbrace{\sigma(t, X_t) \frac{\partial \Gamma}{\partial x}}_{=1} dW_t \\ &= \left(\frac{\partial \Gamma}{\partial t} + \frac{\mu(t, X_t)}{\sigma(t, X_t)} - \frac{1}{2} \frac{\partial \sigma}{\partial x}(t, X_t) \right) dt + dW_t \end{aligned} \quad (1.52)$$

so that all the complexity of the original process is contained in the drift. This transformation is convenient because it creates a process of constant volatility, which enables much faster and accurate numerical methods, such as derivative pricing on recombining trees.

[TODO] cross-ref to Ch. 3 numerics

1.2.3 Stochastic Differential Equations

The main aim of this section is to provide necessary and sufficient conditions for the existence of Itô processes. We will follow a direct approach and state the existence and uniqueness theorem. In addition, notice how we will be working with the differential form of the equations, however realising the integral sense they have been provided with. We need a precise definition for *stochastic differential equation* or *SDE* and thus some preliminary results.

Given an $\mathcal{F}_0$ -measurable random variable Θ of finite variance, consider the following formal equation on $[0, T]$:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t \quad X(0) = \Theta. \quad (1.53)$$

Definition 1.2.15 (Solution of an SDE). A stochastic process X_t , $0 \leq t \leq T$, is called a solution of (1.53) if it satisfies:

- $\sigma(t, X_t) \in \mathcal{L}(\Omega, L^2[0, T])$ so its Itô integral makes sense for each $t \in [0, T]$.
- Almost all sample paths of $\mu(t, X_t)$ belong to $L^1[0, T]$, that is, its expected value is finite.
- Equation 1.53 holds almost surely.

We need to establish some growth conditions on μ and σ for the to make sense on, at least, an open interval, just like in ordinary differential equations. Let us define the following conditions:

Definition 1.2.16. Let $g(t, x)$ a measurable function on $[0, T] \times \mathbb{R}$.

- It is said to satisfy a Lipschitz condition in x if there exists $K > 0$ such that $|g(t, x) - g(t, y)| \leq K|x - y|$ for all $0 \leq t \leq T$, $x, y \in \mathbb{R}$.
- It is said to satisfy a linear growth condition in x if there exists $K > 0$ such that $|g(t, x)| \leq K(1 + |x|)$ for all $0 \leq t \leq T$, $x \in \mathbb{R}$.

Now, notice that for $x \geq 0$, it holds that $1 + x^2 \leq (1 + x)^2 \leq 2(1 + x^2)$ so that the linear growth condition can be restated in terms of a new constant M such that $|g(t, x)| \leq M\sqrt{1 + x^2}$.

The proof of existence and uniqueness rests on a Picard iteration, exactly as for ordinary differential equations, and the comparison step is furnished by Grönwall's inequality (Lemma A.4.1 in Appendix A); since the argument is deterministic and entirely standard we do not reproduce it here.

Theorem 1.2.17. *Let μ and σ be measurable functions on $[0, T] \times \mathbb{R}$ satisfying both the Lipschitz and the linear growth conditions in x of Definition 1.2.16, and let the initial condition Θ be $\mathcal{F}_0$ -measurable with $\mathbb{E}[\Theta^2] < \infty$. Then the stochastic differential equation 1.53 admits a solution, which is unique in the following sense: any two solutions are indistinguishable. Moreover the solution has continuous sample paths.*

Proof: We shall only prove the uniqueness part, the existence being a Picard iteration as indicated above. Let X and Y be two continuous solutions and put $Z_t = X_t - Y_t$, so that

$$Z_t = \int_0^t [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s + \int_0^t [\mu(s, X_s) - \mu(s, Y_s)] ds.$$

We cannot yet take expectations in this identity. By Definition 1.2.15 the integrand of the stochastic term lies only in the pathwise class $\mathcal{L}(\Omega, L^2[0, T])$, on which the Itô integral is constructed by localisation and is a local martingale rather than a martingale; Itô's isometry, a statement about $L^2_{ad}([0, T] \times \Omega)$, is therefore not available. Nor do we know that $\mathbb{E}[Z_t^2]$ is finite, and an inequality between infinite quantities would tell us nothing. Both obstructions are removed by the same device. Define, for $n \in \mathbb{N}$,

$$\tau_n := \inf \{t \in [0, T] : |X_t| \geq n \text{ or } |Y_t| \geq n\} \wedge T, \quad (1.54)$$

which is a stopping time by Proposition 1.1.11, since X and Y are continuous and adapted. As their paths are continuous, they are bounded on $[0, T]$ for almost every ω , whence $\tau_n \uparrow T$ almost surely.

Fix n and stop the identity above at τ_n . On $\{s \leq \tau_n\}$ the Lipschitz condition gives $|\sigma(s, X_s) - \sigma(s, Y_s)| \leq K|Z_s| \leq 2Kn$, so the stopped integrand $\mathbf{1}_{[0, \tau_n]}(s) [\sigma(s, X_s) - \sigma(s, Y_s)]$ is bounded, adapted, and hence genuinely an element of $L^2_{ad}([0, T] \times \Omega)$. Itô's isometry now applies to it, and together with the Lipschitz condition yields

$$\mathbb{E} \left[\left(\int_0^{t \wedge \tau_n} [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s \right)^2 \right] = \int_0^t \mathbb{E} [(\sigma(s, X_s) - \sigma(s, Y_s))^2 \mathbf{1}_{\{s \leq \tau_n\}}] ds \leq K^2 \int_0^t \mathbb{E} [Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}}] ds.$$

For the Lebesgue term, the Cauchy-Schwarz inequality followed by the Lipschitz condition gives, pathwise,

$$\left(\int_0^{t \wedge \tau_n} [\mu(s, X_s) - \mu(s, Y_s)] ds \right)^2 \leq T \int_0^t (\mu(s, X_s) - \mu(s, Y_s))^2 \mathbf{1}_{\{s \leq \tau_n\}} ds \leq TK^2 \int_0^t Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}} ds.$$

Write $u_n(t) := \mathbb{E}[Z_{t \wedge \tau_n}^2]$. Since $|Z_{t \wedge \tau_n}| \leq 2n$ we have $u_n(t) \leq 4n^2 < \infty$ for every t , which is the finiteness the argument was missing. Moreover $Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}} \leq Z_{s \wedge \tau_n}^2$ pointwise, so combining the two estimates through $(a + b)^2 \leq 2(a^2 + b^2)$ and setting $\lambda := 2K^2(1 + T)$ gives

$$u_n(t) \leq \lambda \int_0^t u_n(s) ds, \quad 0 \leq t \leq T.$$

This is a Grönwall inequality with $g \equiv 0$ applied to a finite, non-negative function, so Lemma A.4.1 of Appendix A forces $u_n \equiv 0$. Hence $Z_{t \wedge \tau_n} = 0$ almost surely for each t and each n ;



letting $n \rightarrow \infty$ and using $\tau_n \uparrow T$ together with the continuity of Z gives $Z_t = 0$ almost surely for each fixed $t \in [0, T]$.

It remains to upgrade this to indistinguishability, and the step is not vacuous: the null set so far depends on t (Remark 1.1.8). Let $\{r_n\}$ enumerate the rationals of $[0, T]$ and let Ω_0 be the intersection of the full-measure events $\{Z_{r_n} = 0\}$ with the full-measure event on which Z has continuous paths. Then $\mathbb{P}(\Omega_0) = 1$, and for $\omega \in \Omega_0$ the function $t \mapsto Z_t(\omega)$ is continuous and vanishes on a dense set, hence vanishes identically. Thus X and Y are indistinguishable. ■

The localisation in (1.54) is worth pausing over, since it is the argument Example 1.1.12 was introduced to prepare. The pattern is invariably the same: one cannot integrate or take expectations in the class where the object naturally lives, so one stops the process at the first time it leaves a bounded region, works in the well-behaved class there, and lets the region exhaust the space. Without it the proof above would appeal to an isometry that does not hold for these integrands and to a Grönwall inequality that could be vacuous.

Three comments on the hypotheses, none of which is pedantry.

First, both conditions are needed, and they do different jobs: the Lipschitz condition delivers uniqueness, whereas the linear growth condition prevents the solution from exploding in finite time — exactly as $\dot{y} = y^2$ does in the deterministic theory. If μ and σ are globally Lipschitz in x uniformly in t , and $|\mu(t, 0)|$ and $|\sigma(t, 0)|$ are bounded on $[0, T]$, then linear growth follows automatically, so the second hypothesis is often not stated separately; we state it because the multidimensional version, Theorem 1.2.25, does.

Second, the square-integrability of the initial condition is not decoration. The whole construction of the Itô integral took place in L^2 , and the Picard iterates are estimated there; without $\mathbb{E}[\Theta^2] < \infty$ the very first iterate need not lie in the space where the argument is conducted.

Third, and most consequentially for what follows, the word “unique” conceals a distinction that this formulation is too coarse to see. We have asserted *pathwise* uniqueness: given the Brownian motion, the solution is determined. One may instead ask only that the *law* of the solution be determined, which is a weaker requirement and corresponds to the notion of a *weak* solution — one where the driving Brownian motion is part of the output rather than the input. The two notions genuinely differ, and the gap between them is not a curiosity: the Lipschitz hypothesis above is violated by one of the most widely used models in this dissertation, the square-root variance process of Subsection 1.5.2, whose diffusion coefficient $\xi\sqrt{v}$ is only Hölder continuous of order $\frac{1}{2}$ at the origin. Theorem 1.2.17 as stated therefore does *not* cover the Heston model. We now repair this.

Beyond Lipschitz: strong and weak solutions

Let us first make the distinction precise, since it is systematically blurred in the applied literature and the blurring causes real confusion.

Definition 1.2.18 (Strong and weak solutions). Consider the equation $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ with $X_0 = \Theta$.

- i) A **strong solution** is a process X , adapted to the augmented filtration generated by a *given* Brownian motion W and the initial condition, satisfying the equation almost surely. *Pathwise uniqueness* holds if any two strong solutions on the same space, driven by the same W , are indistinguishable.

- ii) A **weak solution** is a pair (X, W) together with a filtered probability space, such that W is a Brownian motion for that filtration and the equation holds. *Uniqueness in law* holds if any two weak solutions induce the same law on $C[0, T]$.

The distinction is one of information. For a strong solution the noise is handed to us and the trajectory is a measurable functional of it — one may, at least in principle, simulate the solution from a given stream of random numbers. For a weak solution we are permitted to construct the noise ourselves, and all we control is the distribution of the result. For pricing this is often enough, since prices are expectations and expectations depend only on the law; for hedging, and for any pathwise statement, it is not.

The relation between the four notions is settled by the Yamada–Watanabe theorem: pathwise uniqueness together with existence of a weak solution implies existence of a strong solution and uniqueness in law. What we require is the companion criterion, which relaxes the Lipschitz condition in exactly the direction needed.

Theorem 1.2.19 (Yamada–Watanabe). *Consider the one-dimensional equation $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$. Suppose there exist a strictly increasing function $h : [0, \infty) \rightarrow [0, \infty)$ with $h(0) = 0$ and*

$$\int_{0+} \frac{du}{h(u)^2} = \infty, \quad (1.55)$$

and a strictly increasing concave κ with $\kappa(0) = 0$ and $\int_{0+} du/\kappa(u) = \infty$, such that for all t and all x, y

$$|\sigma(t, x) - \sigma(t, y)| \leq h(|x - y|), \quad |\mu(t, x) - \mu(t, y)| \leq \kappa(|x - y|).$$

Then pathwise uniqueness holds for the equation [42]. ■

The point of (1.55) is that it admits $h(u) = u^\gamma$ for every $\gamma \geq \frac{1}{2}$, since $\int_{0+} u^{-2\gamma} du$ diverges precisely when $2\gamma \geq 1$. Hölder continuity of order $\frac{1}{2}$ is thus permitted for the diffusion coefficient — and only just, the exponent $\frac{1}{2}$ being critical. The drift is treated separately and more leniently, needing only an Osgood-type modulus.

Example 1.2.20 (The square-root process). Consider the process which will serve as the variance in the Heston model of Subsection 1.5.2,

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t, \quad v_0 > 0, \quad (1.56)$$

with $\kappa, \theta, \xi > 0$. The drift is affine, hence Lipschitz. The diffusion coefficient $\sigma(v) = \xi\sqrt{v}$ is not Lipschitz at the origin, but it satisfies $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$, so Theorem 1.2.19 applies with $h(u) = \xi\sqrt{u}$, and pathwise uniqueness holds. A non-negative solution exists, and (1.56) is therefore well posed — but by Yamada–Watanabe, not by Theorem 1.2.17.

Well-posedness is not the end of the story for (1.56), because we also need the solution to stay strictly positive: it is to be a variance, and $\sqrt{v_t}$ must make sense. Note that the diffusion coefficient vanishes at $v = 0$ while the drift there equals $\kappa\theta > 0$, so the origin is a point where the noise switches off and the drift pushes upward. Whether that upward push is strong enough is a quantitative question with a sharp answer.

Proposition 1.2.21 (Feller condition). *Let v solve (1.56) with $v_0 > 0$. If*

$$2\kappa\theta \geq \xi^2, \quad (1.57)$$

then $\mathbb{P}(v_t > 0 \text{ for all } t \in [0, T]) = 1$. If $2\kappa\theta < \xi^2$, the process reaches 0 with positive probability, though it is instantaneously reflecting and does not become negative [17]. ■



Condition (1.57) has a transparent reading: mean reversion must be strong enough, and the volatility of volatility small enough, that the drift dominates the noise near the origin. It is worth knowing that in practice it is very often *violated* by calibrated parameters — fitting a steep short-maturity skew tends to demand a large ξ — and that this is a well-known source of difficulty in numerical work, since a naive Euler discretisation of (1.56) will produce negative values and then fail on $\sqrt{v_t}$. Practitioners resort to truncation or reflection schemes, or to exact simulation of the non-central χ^2 transition law. We flag the point here because it is a genuine instance of the theory in this section constraining what can be done in Chapter 3, rather than a technicality to be waved through.

1.2.4 The multidimensional case

For the time being, we have only addressed processes in dimension 1. Almost every result already covered can be restated for multidimensional processes or has a direct analogue.

By a multidimensional Brownian motion on $\mathbb{R}^n$ we mean a stochastic process of the form $\mathbf{W}(t) = (W_1(t), W_2(t), \dots, W_n(t))$ formed by n independent Brownian motions on $\mathbb{R}$. Similarly, we can set them to start at $x_0 \in \mathbb{R}^n$ just by a simple translation. It is a martingale and a Markov process that, starting at $\vec{0}$, has a transition density function given by $p_t(x) = (2\pi t)^{-n/2} e^{-\|x\|^2/2t}$.

There are certain properties of these processes that dramatically change with dimension. For example, for $n = 1$, it returns to 0 infinitely often, and for $n = 2$ enters arbitrarily small neighborhoods of 0 infinitely often with probability 1, however the probability that it reaches $\vec{0}$ is 0. For $n \geq 3$, the process is much more sparse, satisfying $\lim_{t \rightarrow \infty} \|\mathbf{W}(t)\| = \infty$ almost surely.

In terms of fractal dimensions, the *range* $\mathbf{W}([0, t])$ has Hausdorff dimension 2 for $n \geq 2$, and dimension 1 for $n = 1$ (where it is simply an interval). The *graph* $\{(s, \mathbf{W}(s)) : s \in [0, t]\} \subset \mathbb{R}^{1+n}$ behaves differently: it has dimension 3/2 for $n = 1$, a classical result of Taylor [40], and dimension 2 for $n \geq 2$. These fascinating properties of Brownian motion can be found in [1].

The exponent 3/2 in the one-dimensional graph is not an isolated curiosity. A graph of dimension 3/2 is exactly what one obtains from a function which is α -Hölder for every $\alpha < 1/2$ and for no $\alpha \geq 1/2$, and that critical exponent 1/2 — together with what happens on either side of it — will return in Part II as the organising theme of the whole dissertation.

We can therefore define a multidimensional stochastic integral (expressed in differential form) as

$$dX_t = \mu dt + \sum_{i=1}^n \sigma_i dW_t^i \quad (1.58)$$

where each integral must be considered separately. Let $W_1(t), W_2(t), \dots, W_m(t)$ be independent Brownian motions. Consider n Itô processes $X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(n)}$ given by

$$X_t^{(i)} = X_0^{(i)} + \int_0^t \mu_i(s) ds + \sum_{j=1}^m \int_0^t \sigma_{ij}(s) dW_j(s), \quad 1 \leq i \leq n, \quad (1.59)$$

where $\mu_i \in \mathcal{L}(\Omega, L^1[0, T])$ and $\sigma_{ij} \in \mathcal{L}(\Omega, L^2[0, T])$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. Collecting $W = (W_1, \dots, W_m)^\top$ and $X_t = (X_t^{(1)}, \dots, X_t^{(n)})^\top$ into column vectors, $\mu = (\mu_1, \dots, \mu_n)^\top$ likewise, and $\sigma = [\sigma_{ij}]$ into an $n \times m$ matrix, equation (1.59) becomes

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW(s), \quad 0 \leq t \leq T,$$

the integrals being taken componentwise. Growth conditions on μ_i and σ_{ij} must again be imposed for the solution to make sense, and Itô's formula holds for $\theta(X_t)$ as before, now accounting for the crossed second derivatives $\partial^2\theta/\partial x_l\partial x_h$.

Before writing the corresponding multiplication table we must introduce an object which the table presupposes and which the definition above does not provide. So far, a multidimensional Brownian motion has *independent* components. In modelling one almost always wants the components to be correlated — a falling equity price accompanied by rising volatility, two rates moving together, an exchange rate and the foreign interest rate — and the following makes that precise.

Definition 1.2.22 (Correlated Brownian motion). Let $P = (\rho_{ij}) \in \mathbb{R}^{n \times n}$ be symmetric and positive semi-definite with $\rho_{ii} = 1$ for all i . An n -dimensional process $\widetilde{\mathbf{W}} = (\widetilde{W}^1, \dots, \widetilde{W}^n)$ is a **Brownian motion with correlation matrix** P if each $\widetilde{W}^i$ is a one-dimensional Brownian motion and

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \rho_{ij} t, \quad 1 \leq i, j \leq n, \quad t \in [0, T]. \quad (1.60)$$

Such a process always exists, and is obtained from an independent one by a linear map. Since P is symmetric positive semi-definite it admits a factorisation $P = LL^\top$ — the Cholesky factorisation when P is positive definite, and a square root more generally. Setting $\widetilde{\mathbf{W}} := L \mathbf{W}$ for an independent $\mathbf{W}$, bilinearity of the quadratic covariation (1.41) gives

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \sum_{k,l} L_{ik} L_{jl} \langle W^k, W^l \rangle_t = \sum_k L_{ik} L_{jk} t = (LL^\top)_{ij} t = \rho_{ij} t,$$

and each component is a continuous local martingale with $\langle \widetilde{W}^i \rangle_t = t$, hence a Brownian motion by Theorem 1.2.27.

The positive semi-definiteness of P is a genuine constraint and not a formality. It is what makes the factorisation possible, and it is a real restriction as soon as $n \geq 3$: one cannot independently choose ρ_{12} , ρ_{13} and ρ_{23} , since a strongly positive correlation between the first two pairs forbids a strongly negative one between the third. This is a familiar nuisance in practice, where correlation matrices estimated entrywise from market data frequently fail to be positive semi-definite and must be repaired before they can be used.

With this in hand, the Itô multiplication table extends as follows.

$\times$	dt	dW_t^1	dW_t^2	$\dots$	dW_t^n
dt	0	0	0	$\dots$	0
dW_t^1	0	dt	$\rho_{12} dt$	$\dots$	$\rho_{1n} dt$
dW_t^2	0	$\rho_{21} dt$	dt	$\dots$	$\rho_{2n} dt$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
dW_t^n	0	$\rho_{n1} dt$	$\rho_{n2} dt$	$\dots$	dt

Table 1.1: Itô multiplication table for correlated n -dimensional Brownian motion.

The entries are not a new convention but a restatement of (1.60): the symbol $dW_t^i dW_t^j$ abbreviates $d\langle W^i, W^j \rangle_t = \rho_{ij} dt$. The independent case is recovered by taking $P = \mathbb{I}_{n \times n}$.

Example 1.2.23. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ a $\mathcal{C}^2$ function and let $\nabla f = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n})$ and $\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}$ to obtain

$$f(\mathbf{W}(t)) = f(\mathbf{W}(0)) + \int_0^t \nabla f(\mathbf{W}(s)) \cdot d\mathbf{W}_s + \frac{1}{2} \int_0^t \Delta f(\mathbf{W}(s)) ds.$$



This equation exhibits the deep connection between Brownian motion and diffusions, showing how *one half of the Laplace operator* is the generator of Brownian motion; see Subsection 1.2.6, where the same $\frac{1}{2}$ reappears from the semigroup side. The factor is not decorative: it is the quadratic variation $\langle W^i \rangle_t = t$ entering through the second-order term of Itô's rule, and it is the reason the heat equation associated with Brownian motion is $\partial_t u = \frac{1}{2} \Delta u$ rather than $\partial_t u = \Delta u$.

Example 1.2.24. An important example is given by $\theta(x, y) = xy$. Clearly, $\partial\theta/\partial x = y$ and also $\partial\theta/\partial y = x$. Therefore, if we have processes X_t and Z_t , we obtain

$$d(X_t Z_t) = Z_t dX_t + X_t dZ_t + \frac{1}{2} dX_t dZ_t + \frac{1}{2} dX_t dZ_t = Z_t dX_t + X_t dZ_t + dX_t dZ_t \quad (1.61)$$

what is usually known as Itô product formula, understanding $dX_t dZ_t$ as in table 1.1.

The following generalization holds.

Theorem 1.2.25. *Let $\sigma(t, x)$ be an $n \times m$ matrix-valued function and $\mu(t, x)$ an $\mathbb{R}^n$ -valued function. Assume there exists $C > 0$ such that for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,*

$$\|\sigma(t, x) - \sigma(t, y)\| \leq C|x - y| \quad , \quad |\mu(t, x) - \mu(t, y)| \leq C|x - y|$$

and

$$\|\sigma(t, x)\|^2 \leq C(1 + |x|^2) \quad , \quad |\mu(t, x)|^2 \leq C(1 + |x|^2)$$

where $\|\sigma\|^2 = \sum_i \sum_j \sigma_{ij}^2$. For any $\mathcal{F}_a$ -measurable random vector Θ of finite variance, the SDE $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, $X_0 = \Theta$ has a unique continuous solution. ■

1.2.5 Probability measure changes

Another important topic we must cover has to do with probability changes. Brownian motion depends crucially on the underlying probability $\mathbb{P}$. Given a process X_t , it seems legit to ask whether there is a probability $\mathbb{Q}$ so that X_t is a Brownian motion with respect to $\mathbb{Q}$. We assume the σ -algebra and the filtration remain unchanged.

The Doob–Meyer theorem below is stated in the language of *predictable* processes, and one word is owed on why the notion appears at all, since for the processes studied here it will make no difference whatsoever. Writing $\mathbb{L}$ for the adapted processes with almost all sample paths left-continuous, the **predictable** σ -algebra is the smallest σ -algebra on $[0, T] \times \Omega$ making every element of $\mathbb{L}$ measurable, and a process is predictable if it is measurable with respect to it.

When the integrator has continuous sample paths — as Brownian motion does, and as every model in Part I does — predictable and merely adapted integrands give rise to the same stochastic integrals, which is why the construction of Subsection 1.2.1 could proceed without the notion. The distinction becomes essential the moment the integrator is allowed to jump, and for a reason about information rather than technique: if the integrand may “see” a jump as it happens, one can bet on the jump at the instant it occurs and the resulting integral fails to be a martingale. Predictability forbids exactly this, and its financial reading is that one may not trade on information one does not yet have. This is not idle for genuinely discontinuous underlyings — electricity spot prices, several commodities, and credit-sensitive instruments where default is by nature a jump — but we shall not need it again.

Theorem 1.2.26 (Doob–Meyer Decomposition Theorem). *Let $M(t)$, $t \in [0, T]$ be a càdlàg, square integrable martingale. Then there exists a unique decomposition of the form*

$$M_t^2 = R(t) + \langle M_t \rangle \quad (1.62)$$

where $R(t)$ is càdlàg martingale and $\langle M_t \rangle$ is a predictable, right continuous and increasing process such that $\langle M_0 \rangle = 0$ and $\mathbb{E}[\langle M_t \rangle] < \infty$ for all $t \in [0, T]$. We call $\langle M_t \rangle$ the **compensator** of M_t^2 and we say it is the predictable quadratic variation of M_t . ■

The theorem holds in a way more general form for submartingales, but if M_t is a square integrable martingale, then, by Jensen's inequality, M_t^2 is a submartingale and it suffices for our purposes. For a proof of the Doob–Meyer decomposition theorem, see [26] or [39].

Theorem 1.2.27 (Lévy's Characterisation Theorem). *Let X be a continuous local martingale on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ with $X_0 = 0$ and $\langle X \rangle_t = t$ almost surely for every $t \in [0, T]$. Then X is an $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion under $\mathbb{P}$.*

More generally, let $X = (X^1, \dots, X^n)$ be a vector of continuous local martingales with $X_0 = 0$ and

$$\langle X^i, X^j \rangle_t = \delta_{ij} t \quad \text{almost surely, for all } i, j \text{ and all } t,$$

where δ_{ij} is the Kronecker delta. Then X is an n -dimensional $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion with independent components.

Proof: The argument is given in Appendix B.5; see also [26, Thm. 8.4.2]. ■

The statement deserves to be read slowly, because it is considerably stronger than it looks. Being a Brownian motion is, by Definition 1.1.31, a statement about the *joint law* of the whole family $\{X_t\}$: independent, stationary, Gaussian increments. Lévy's theorem says that all of this is already implied by two facts about second moments — that X is a local martingale and that its quadratic variation is exactly t . Gaussianity is not assumed anywhere; it is a conclusion.

The multidimensional form, with the condition $\langle X^i, X^j \rangle_t = \delta_{ij}t$, is the one we shall actually need. It is what will allow us, in the proof of Girsanov's theorem, to recognise a process as a Brownian motion under a *new* measure without recomputing any distributions — one need only check that quadratic variation is preserved, which it is, since quadratic variation is a pathwise notion and equivalent measures agree on which paths occur.

Definition 1.2.28. (Exponential Process) Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. We define its exponential process by

$$\mathcal{E}_\alpha(t) = e^{\int_0^t \alpha(s) dW(s) - \frac{1}{2} \int_0^t \alpha^2(s) ds} \quad (1.63)$$

which is precisely the solution of $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW(t)$, $\mathcal{E}_\alpha(0) = 1$ so that $\mathcal{E}_\alpha(t)$ is a local martingale.

This exponential process is important because it leads to the construction of a new probability $\mathbb{Q}$ for which $\mathcal{E}_\alpha(t)$ is actually a martingale.

Theorem 1.2.29. *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then $\mathcal{E}_\alpha(t)$ is a martingale if, and only if, $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ for all $t \in [0, T]$.*

Proof: Necessity is immediate: if $\mathcal{E}_\alpha$ is a martingale then $\mathbb{E}[\mathcal{E}_\alpha(t)] = \mathbb{E}[\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_0]] = \mathbb{E}[\mathcal{E}_\alpha(0)] = 1$ for every t .

For sufficiency, recall from the definition that $\mathcal{E}_\alpha(t) = 1 + \int_0^t \alpha(s)\mathcal{E}_\alpha(s) dW_s$, so $\mathcal{E}_\alpha$ is a non-negative local martingale, and let (τ_n) be a localising sequence. A non-negative local martingale is always a supermartingale — apply the conditional Fatou lemma (Proposition A.1.2) to $\mathcal{E}_\alpha(t \wedge \tau_n) \rightarrow \mathcal{E}_\alpha(t)$ — so for $s < t$,

$$\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_s] \leq \mathcal{E}_\alpha(s) \quad \text{almost surely.}$$

Both sides are non-negative and, by hypothesis, have the same expectation, namely 1. A non-negative random variable dominated by another of equal expectation is almost surely equal to it, so the inequality is an equality and $\mathcal{E}_\alpha$ is a martingale. See [26, Thm. 8.7.3]. ■

The previous result is tautological and hard to check in practice, so we need a more practical condition. The following result is known as Novikov's condition.



Theorem 1.2.30 (Novikov's condition). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. If $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \alpha^2(s) ds} \right] < \infty$, then $\mathcal{E}_\alpha(t)$ is a martingale. A proof may be found in [26, Sec. 8.7].* ■

This is a much more practical way to check whether $\mathcal{E}_\alpha(t)$ is a martingale, and therefore whether we can construct a new probability $\mathbb{Q}$ such that $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$. For the following Theorem, we need to make use of two auxiliary lemmas.

Lemma 1.2.31 (Conditional expectation under a change of measure). *Let γ be a nonnegative integrable random variable, so that $d\mathbb{Q} = \gamma d\mathbb{P}$ defines a new probability measure (see 1.2). Then, for any σ -algebra $\mathcal{G} \subset \mathcal{F}$ and Θ such that $\mathbb{E}_\mathbb{Q}[\Theta] < \infty$, it follows that*

$$\mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}] = \frac{\mathbb{E}_\mathbb{P}[\gamma\Theta | \mathcal{G}]}{\mathbb{E}_\mathbb{P}[\gamma | \mathcal{G}]}, \quad \mathbb{Q} - \text{almost surely.} \quad (1.64)$$

Proof: Integrability holds because $\mathbb{E}_\mathbb{P}[|\Theta|\gamma] = \mathbb{E}_\mathbb{Q}[|\Theta|] < \infty$, so $\mathbb{E}_\mathbb{P}[\gamma\Theta | \mathcal{G}]$ is defined. Fix $G \in \mathcal{G}$. By the defining property of conditional expectation under $\mathbb{P}$ and the definition of $\mathbb{Q}$,

$$\int_G \mathbb{E}_\mathbb{P}[\gamma\Theta | \mathcal{G}] d\mathbb{P} = \int_G \gamma\Theta d\mathbb{P} = \int_G \Theta d\mathbb{Q} = \int_G \mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}] d\mathbb{Q}.$$

On the other hand, writing $d\mathbb{Q} = \gamma d\mathbb{P}$ and using that $\mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}]$ is $\mathcal{G}$ -measurable, so that it may be taken out of a conditional expectation given $\mathcal{G}$ (Proposition 1.1.4, item 4),

$$\int_G \mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}] d\mathbb{Q} = \int_G \mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}] \gamma d\mathbb{P} = \int_G \mathbb{E}_\mathbb{Q}[\Theta | \mathcal{G}] \mathbb{E}_\mathbb{P}[\gamma | \mathcal{G}] d\mathbb{P}.$$

The two displays agree for every $G \in \mathcal{G}$, and both integrands are $\mathcal{G}$ -measurable, so they coincide almost surely. Dividing by $\mathbb{E}_\mathbb{P}[\gamma | \mathcal{G}]$, which is strictly positive $\mathbb{Q}$ -almost surely, gives the result [26, Lemma 8.9.2]. ■

Then, if $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ satisfies the condition on Theorem 1.2.29, it follows that $\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale. Let $\mathbb{Q}$ be defined by $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$, that is, $\mathbb{Q}(A) = \int_A \mathcal{E}_\alpha(T) d\mathbb{P}$ for every $A \in \mathcal{F}$.

Lemma 1.2.32 (Martingale transfer). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then a $\{\mathcal{F}_t\}$ -adapted process X_t , $t \in [0, T]$ is a $\mathbb{Q}$ -martingale if, and only if, $X(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale.*

Proof: Apply Lemma 1.2.31 with $\gamma = \mathcal{E}_\alpha(T)$ and $\mathcal{G} = \mathcal{F}_s$. Since $\mathcal{E}_\alpha$ is a $\mathbb{P}$ -martingale, $\mathbb{E}_\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathcal{E}_\alpha(s)$, so for $s \leq t$

$$\mathbb{E}_\mathbb{Q}[X(t) | \mathcal{F}_s] = \frac{\mathbb{E}_\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s]}{\mathcal{E}_\alpha(s)}. \quad (1.65)$$

The numerator simplifies by conditioning on $\mathcal{F}_t$ first and using that $X(t)$ is $\mathcal{F}_t$ -measurable:

$$\mathbb{E}_\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathbb{E}_\mathbb{P}\left[X(t) \mathbb{E}_\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_t] \middle| \mathcal{F}_s\right] = \mathbb{E}_\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s].$$

Hence $\mathbb{E}_\mathbb{Q}[X(t) | \mathcal{F}_s] = X(s)$ if and only if $\mathbb{E}_\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s] = X(s)\mathcal{E}_\alpha(s)$, which is the assertion [26, Lemma 8.9.3]. ■

Finally, the following key result holds from Lemmas 1.2.31 and 1.2.32.

Theorem 1.2.33. (*Girsanov's Theorem*) *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ such that $\mathcal{E}_\alpha$ is a martingale. Then the process*

$$Z(t) = W(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T] \quad (1.66)$$

is a Brownian motion with respect to the measure $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$.

Proof: The argument is given in Appendix B.6; see also [26, Thm. 8.9.4]. ■

Girsanov's theorem is the formal statement of something we shall use constantly: *changing the measure changes the drift and leaves the volatility alone*. The Brownian motion Z under $\mathbb{Q}$ is the old W with a drift subtracted, and nothing else has moved. The reason is already contained in Theorem 1.2.27: to verify that Z is a $\mathbb{Q}$ -Brownian motion one need only check that it is a continuous local martingale with $\langle Z \rangle_t = t$, and quadratic variation is a pathwise limit, so it is unaffected by passing to an equivalent measure. Two equivalent measures disagree about how likely a path is, never about which paths exist. This observation is the mathematical content of the statement — which we shall meet again in Section 1.4 — that an option's price does not depend on the expected return of the underlying, while it depends critically on its volatility. For the multi-factor models of Section 1.5 we require the vector-valued form.

Theorem 1.2.34 (Girsanov's Theorem, multidimensional). *Let $\mathbf{W} = (W^1, \dots, W^n)$ be an n -dimensional Brownian motion and let $\alpha = (\alpha^1, \dots, \alpha^n)$ be an adapted process with $\int_0^T \|\alpha(s)\|^2 ds < \infty$ almost surely. Define*

$$\mathcal{E}_\alpha(t) = \exp \left(\int_0^t \alpha(s)^\top d\mathbf{W}(s) - \frac{1}{2} \int_0^t \|\alpha(s)\|^2 ds \right), \quad (1.67)$$

and suppose Novikov's condition holds in the form $\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \|\alpha(s)\|^2 ds \right) \right] < \infty$. Then $\mathcal{E}_\alpha$ is a martingale, $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$ defines a probability measure equivalent to $\mathbb{P}$, and

$$\mathbf{Z}(t) = \mathbf{W}(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T], \quad (1.68)$$

is an n -dimensional $\mathbb{Q}$ -Brownian motion with independent components. ■

That the components of $\mathbf{Z}$ remain independent under $\mathbb{Q}$ deserves emphasis, since it is not obvious and is exactly what the multidimensional Lévy characterisation delivers: the cross-variations $\langle Z^i, Z^j \rangle_t = \delta_{ij}t$ are pathwise quantities and survive the change of measure untouched. Note also that this is a statement about *independent* driving factors; when a model is specified with correlated Brownian motions, as the Heston model of Subsection 1.5.2 is, one applies the theorem after decorrelating, in the manner of Definition 1.2.22 above.

The Martingale Representation Theorem

Girsanov's theorem tells us which measures are available. The following tells us which *payoffs* are attainable, and it is the deepest result of this subsection. Together the two constitute the entire mathematical content of the fundamental theorems of asset pricing in Section 1.3.

Theorem 1.2.35 (Itô's Martingale Representation Theorem). *Let $\mathbf{W}$ be an n -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $(\mathcal{F}_t^W)_{t \in [0, T]}$ be its augmented natural filtration. Let M be a square-integrable $\mathbb{P}$ -martingale with respect to that filtration. Then there exists a unique adapted process $\varphi = (\varphi^1, \dots, \varphi^n) \in L_{\text{ad}}^2([0, T] \times \Omega)$ such that*

$$M_t = M_0 + \int_0^t \varphi(s)^\top d\mathbf{W}(s), \quad t \in [0, T], \quad (1.69)$$

almost surely. If M is only a local martingale, the same holds with φ in the larger class $\mathcal{L}(\Omega, L^2[0, T])$. ■

The theorem should be read as a converse to Theorem 1.2.6. There we showed that every Itô integral is a martingale; here we learn that, provided the filtration is the one generated by $\mathbf{W}$ and nothing more, *every* martingale is an Itô integral. In the language of Proposition 1.1.4,



item 9, the stochastic integrals do not merely sit inside the space of martingales: they exhaust it.

Two aspects of the hypothesis deserve to be underlined, because both will be load-bearing.

The first is that the filtration must be generated by the Brownian motion itself. If $(\mathcal{F}_t)_{t \in [0, T]}$ carries additional information — another independent Brownian motion, a Poisson process, an independent random variable revealed at time T — the conclusion fails outright, and it fails for an obvious reason: a martingale driven by that extra randomness cannot be written as an integral against $\mathbf{W}$, because the integral would be measurable with respect to the wrong σ -algebra. The theorem is a statement about there being *no more randomness in the model than the driving Brownian motion supplies*.

The second is that φ is not constructed. The proof is an orthogonality argument: one shows that the stochastic integrals form a closed subspace of $L^2(\Omega, \mathcal{F}_T)$ whose orthogonal complement contains only constants, using the fact that exponential martingales $\mathcal{E}_\alpha(T)$ span a dense set. Existence is therefore obtained without any formula for φ , which is worth knowing in advance: the theorem will guarantee that a replicating strategy exists while saying nothing whatever about how to compute it. Recovering φ in concrete cases is precisely the delta-hedging computation of Section 1.4, and in models where no closed form is available it is a numerical problem.

We can already state the consequence for which we shall need it, though the vocabulary is only introduced in the next section.

Remark 1.2.36 (Why this matters for pricing). Let H_T be a square-integrable contingent claim and let $\mathbb{Q}$ be an equivalent martingale measure. The discounted price process $M_t := \mathbb{E}^\mathbb{Q}[H_T/S_T^0 \mid \mathcal{F}_t]$ is by construction a $\mathbb{Q}$ -martingale — this is the tower property, nothing more. If the filtration is generated by the Brownian motion driving the traded assets, Theorem 1.2.35 applied under $\mathbb{Q}$ produces φ with $M_t = M_0 + \int_0^t \varphi^\top d\mathbf{Z}$, and φ is — after rewriting the integral in terms of the discounted asset prices — a self-financing strategy replicating H_T exactly.

Every claim is thus attainable, so the market is *complete*, and the price of the claim is not merely a plausible expectation but the unique cost of a portfolio that reproduces it in every state of the world. This is the substance behind the Second Fundamental Theorem (Theorem 1.3.6), and it is the reason the Black–Scholes model is complete (Section 1.4) while the stochastic volatility models of Subsection 1.5.2 are not: in the latter, the filtration is generated by *two* Brownian motions while only *one* asset is traded, the hypothesis of Theorem 1.2.35 fails, and the volatility risk cannot be hedged away.

1.2.6 Itô Diffusions

Recall that, for a Markov process, the *transition probability* $\mathbb{P}_{s,x}(t, A) = \mathbb{P}(X_t \in A \mid X_s = x)$, for A measurable, $0 \leq s \leq t \leq T$ and $x \in \mathbb{R}^n$. The Chapman-Kolmogorov equation states that

$$\mathbb{P}_{s,x}(t, A) = \int_{\mathbb{R}^n} \mathbb{P}_{u,z}(t, A) \mathbb{P}_{s,x}(u, dz).$$

Definition 1.2.37. (Diffusion Process) A $\mathbb{R}^n$ -valued process X_t , $t \in [0, T]$ is called a **diffusion process** if its transition probabilities $\{\mathbb{P}_{s,x}(t, \cdot)\}$ satisfy the following three conditions for every $t \in [0, T]$, every $x \in \mathbb{R}^n$ and every $\delta > 0$:

- i) $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| > \delta\}} \mathbb{P}_{t,x}(t + \varepsilon, dy) = 0$
- ii) $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y - x) \mathbb{P}_{t,x}(t + \varepsilon, dy) = \mu(t, x) \in \mathbb{R}^n$ exists,

$$\text{iii)} \quad \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y-x)(y-x)^\top \mathbb{P}_{t,x}(t+\varepsilon, dy) = \Sigma(t, x) \in \mathbb{R}^{n \times n} \text{ exists,}$$

where the vectors in $\mathbb{R}^n$ are implicitly taken as columns, so that $(y-x)(y-x)^\top$ yields a symmetric positive semi-definite matrix. μ and Σ are called the *drift* and the *diffusion* coefficient, respectively.

The restriction of the domain of integration to a ball around x is essential and not a technicality: the integral of a probability measure over the whole of $\mathbb{R}^n$ is 1, so dropping the constraint in i) would make the limit $+\infty$ rather than 0. Read correctly, the three conditions say that in a short time ε the process moves a distance of order $\sqrt{\varepsilon}$ with overwhelming probability, that the mean of that displacement is of order ε , and that its covariance is of order ε as well. When conditions ii) and iii) hold, the truncation at level δ turns out not to affect the limits, so we may and shall write them informally as

$$\mu(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[X_{t+\varepsilon} - x \mid X_t = x], \quad \Sigma(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[(X_{t+\varepsilon} - x)(X_{t+\varepsilon} - x)^\top \mid X_t = x].$$

Note how condition i) forbids the process from having jumps: it is precisely the statement that a displacement of size δ in time ε has probability $o(\varepsilon)$. Therefore the Poisson process, for which such a displacement has probability $\lambda\varepsilon + o(\varepsilon)$, is not a diffusion.

Example 1.2.38. Take an $\mathbb{R}^n$ -valued Brownian motion W_t . Note that $\|W_t\|^2$ is a sum of n independent squares of normal variables, so that $\mathbb{E}\|W_\varepsilon\|^4 = n(n+2)\varepsilon^2$. Hence, for any $c > 0$, Markov's inequality applied to the fourth power gives

$$\mathbb{P}(\|W_{t+\varepsilon} - x\| \geq c \mid W_t = x) = \mathbb{P}(\|W_{t+\varepsilon} - W_t\| \geq c) \leq c^{-4} \mathbb{E}\|W_\varepsilon\|^4 = c^{-4} n(n+2)\varepsilon^2,$$

which is $o(\varepsilon)$. Condition i) of Definition 1.2.37 is therefore satisfied, and a direct computation of the remaining two limits gives $\mu(t, x) = 0$ and $\Sigma(t, x) = \mathbb{I}_{n \times n}$.

The following result finally connects the previous concepts and gives an unambiguous meaning to μ .

Theorem 1.2.39. *Let $\sigma(t, x)$ and $\mu(t, x)$ be functions as in Theorem 1.2.25. Assume they are also continuous on $[0, T] \times \mathbb{R}^n$. The solution to the SDE there defined is a diffusion process where $\mu(t, x)$ is identified and $\Sigma(t, x) = \sigma(t, x) \cdot \sigma(t, x)^t$.*

The proof of this Theorem is beyond the scope of our purposes and can be found in [26].

We have reached a critical point where diffusions, SDE and semigroups intereseect. These seemingly unrelated mathematical objects give rise to a set of (parabolic) partial differential equations (PDEs) that are crucial for computational purposes and pricing financial derivatives. In this context, powerful mathematical tools can be applied.

The first of the approaches, where we consider a semigroup of operators, is included for completeness purposes, showing how its infinitesimal generators may be used to construct diffusion processes. It is also included to illustrate how Functional Analysis arises in this context.

1. Semigroup approach

We restrict to the **stationary case**, in which μ and Σ depend on x alone, and use standard Banach-space language; see [9].

A family $\{T_t\}_{t \geq 0}$ of linear operators on a Banach space $\mathbb{B}$ is a **contraction semigroup** if $T_0 = \text{I}$ and $T_{t+s} = T_t T_s$ (semigroup property), $T_t v \rightarrow v$ as $t \downarrow 0$ for every v (strong continuity), and $\|T_t v\| \leq \|v\|$ (contraction). Its **infinitesimal generator** is

$$\mathcal{A}(v) = \lim_{t \downarrow 0} \frac{1}{t} (T_t(v) - v), \quad (1.70)$$



defined on those v for which the limit exists strongly; it is typically unbounded, as already for the translation semigroup $T_t f(x) = f(x + t)$ on $C_0[0, \infty)$, whose generator is $\mathcal{A}f = f'$. The choice of space matters here: on the larger space $C_b[0, \infty)$ the same family fails to be strongly continuous, since $\|T_t f - f\|_\infty \rightarrow 0$ is precisely uniform continuity of f , which bounded continuous functions need not possess — $f(x) = \sin(x^2)$ is the standard witness. Restricting to C_0 , or to the bounded uniformly continuous functions, repairs this.

Given a stationary diffusion X_t driven by μ and Σ , one sets $T_t(f)(x) = \int_{\mathbb{R}^n} f(y) \mathbb{P}_t(x, dy)$. Under mild regularity on the transition probabilities — essentially that mass neither escapes to infinity in finite time nor moves a macroscopic distance instantaneously, and that $T_t f$ is continuous — the family $\{T_t\}$ is a contraction semigroup on $C_0(\mathbb{R}^n)$, the continuous functions vanishing at infinity, and its generator is the second-order elliptic operator

$$\mathcal{A}(f)(x) = \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} + \sum_{i=1}^n \mu_i(x) \frac{\partial f}{\partial x_i}. \quad (1.71)$$

For Brownian motion, where $\Sigma = \mathbb{I}_{n \times n}$ and $\mu = 0$, this is $\mathcal{A}f = \frac{1}{2} \Delta f$.

We shall not develop the semigroup picture further, having obtained from it what we needed: the identification of the generator, and the confirmation that the $\frac{1}{2}$ appearing in Itô's rule is the same $\frac{1}{2}$ appearing in $\frac{1}{2} \Delta$. It is worth recording, however, what the semigroup property actually encodes, because it is the first of the three approaches to fail in Part II. The identity $P_{t+s} = P_t \circ P_s$ is a restatement of the Markov property: the evolution over $t + s$ can be decomposed into evolution over s followed by evolution over t , because at the intermediate instant the present state is a sufficient statistic for the future. When the driving noise has memory — as fractional Brownian motion does — this decomposition is unavailable, the family (P_t) is no longer a semigroup, and the generator ceases to exist as a local operator. Neither the PDE approach nor the semigroup approach survives intact; only the pathwise, integral formulation does, which is a large part of why the theory of Part II is built the way it is.

2. Partial Differential Equations Approach

Let $\mathbb{P}_{s,x}(t, \cdot)$ be the transition probabilities of a diffusion X_t driven by μ and Σ , and let $F_{s,x}$ be the corresponding distribution function. The argument is a Taylor expansion inside the Chapman–Kolmogorov equation

$$F_{s,x}(t, y) = \int_{\mathbb{R}^n} F_{u,z}(t, y) dF_{s,x}(u, z),$$

and runs as follows. Fix t and y , put $u = s + \varepsilon$, and expand the integrand to second order in z about x . The zeroth-order term gives $F_{s+\varepsilon,x}$; the first-order term contributes $\nabla_x F_{s+\varepsilon,x} \cdot \mu(s, x) \varepsilon$, since by Definition 1.2.37(ii) the mean displacement over $[s, s + \varepsilon]$ is $\mu(s, x) \varepsilon + o(\varepsilon)$; and the second-order term contributes $\frac{1}{2} \text{Tr}(H_x F_{s+\varepsilon,x} \Sigma(s, x)) \varepsilon$ by Definition 1.2.37(iii). Rearranging, dividing by ε and letting $\varepsilon \rightarrow 0$ yields

$$-\frac{\partial F_{s,x}}{\partial s} = \sum_{i=1}^n \mu_i \frac{\partial F_{s,x}}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij} \frac{\partial^2 F_{s,x}}{\partial x_i \partial x_j}, \quad (1.72)$$

together with the terminal condition

$$\lim_{s \uparrow t} F_{s,x}(t, y) = \mathbb{1}_{\{x \leq y\}}. \quad (1.73)$$

Equation (1.72) is called the **Kolmogorov backward equation**, because the variable s is the starting time, effectively moving backwards from the terminal time t . However, we often aim for a formula that enables *forward* calculations. Directly linked to this equation, the Feynman-Kac

formula states that the solution to the Kolmogorov backward equation can be represented as an expected value of a functional of the diffusion process X_t . This will be stated in the next section.

Notice that the Kolmogorov backward equation derived in Equation (1.72) is effectively a special case of the Feynman-Kac formula where the interest rate $r(t, x) = 0$ and the terminal condition is the indicator function $h(x) = \mathbb{1}_{\{x \leq y\}}$. In the context of quantitative finance, the Feynman-Kac theorem is a cornerstone result: it establishes the fundamental bridge between the risk-neutral pricing of a derivative (computed as a discounted expected payoff under a martingale measure) and the deterministic PDE that the derivative's price must satisfy to preclude arbitrage.

We shall stay in the one-dimensional case for simplicity now. Assume the transition probabilities of a Markov process X_t possess density functions, that is,

$$P_{s,x}(t, dy) = p_{s,x}(t, y) dy, \quad (1.74)$$

so that the Chapman-Kolmogorov equation takes the form

$$p_{s,x}(t, y) = \int_{\mathbb{R}} p_{u,z}(t, y) p_{s,x}(u, z) dz. \quad (1.75)$$

By using this identity and some auxiliary test functions (integration by parts), it can be proven that

$$\frac{\partial p_{s_0, x_0}(t, y)}{\partial t} = -\frac{\partial(\mu(t, y)p_{s_0, x_0}(t, y))}{\partial y} + \frac{1}{2} \frac{\partial^2(\Sigma(t, y)p_{s_0, x_0}(t, y))}{\partial y^2} \quad (1.76)$$

which, together with the initial condition

$$\lim_{t \downarrow s_0} p_{s_0, x_0}(t, y) = \delta_{x_0}(y) \quad (1.77)$$

is known as either the **Kolmogorov forward equation** or the **Fokker-Planck equation**. This equation is much more useful in a financial context because it allows calculations *forward* in time, i.e., evolving from known current market variables.

Theorem 1.2.40. *Let Σ and μ satisfy the linear growth and Lipschitz conditions in x . Assume, additionally, that there exists $c > 0$ such that $w^T \Sigma(t, y) w \geq c \|w\|^2$ for all $w \in \mathbb{R}^n$ (uniform ellipticity). Then:*

- a) *The Kolmogorov backward equation has a unique solution. Moreover, there exists a continuous Markov process X_t whose transition probabilities satisfy $\mathbb{P}_{s,x}(t_0, dy_0) = dF_{s,x}(t_0, y_0)$.*
- b) *Assume additionally that the derivatives of μ and Σ up to second order satisfy the Lipschitz and linear growth conditions. Then the Kolmogorov forward equation has a unique solution.*

In terms of the infinitesimal generator, we can summarize the previous discussion in the following table:

Forward Equation	Backward Equation
$\partial \rho / \partial t = \mathcal{A}^* \rho$	$-\partial u / \partial s = \mathcal{A} u$

where $\mathcal{A}^*$ represents the adjoint of the generator $\mathcal{A}$; see [25] for the functional-analytic details. The duality is worth reading off the table: the backward equation propagates a *payoff* backwards from maturity and is the natural object for pricing a single contract, whereas



the forward equation propagates a *density* forwards from today and is the natural object for calibration, since it evolves from known current market data. Dupire's formula in Subsection 1.5.1 is precisely an application of the second.

3. Stochastic Differential Equations Approach

In this case, take μ , Σ and σ such that $\sigma\sigma^T = \Sigma$. Notice that σU is also a valid solution for any matrix U such that $UU^T = \mathbf{I}_{n \times n}$ (orthogonal matrix). Therefore, the choice of σ is unique up to multiplication by an orthogonal matrix. Since $UW(t)$ is also a Brownian motion, the transition probabilities of X_t do not depend on the particular choice of σ within its class. We call this class of matrices the *square root* of Σ .

Now, suppose Σ and μ satisfy the linear growth and Lipschitz conditions. It is clear that any square root of Σ satisfies the growth condition. Moreover, if Σ is uniformly elliptic (bounded away from zero), σ also satisfies the Lipschitz conditions. In the 1D case, this can be seen via the inequality:

$$|\sigma(t, x) - \sigma(t, y)| = \frac{|\Sigma(t, x) - \Sigma(t, y)|}{|\sigma(t, x)| + |\sigma(t, y)|} \leq \frac{|\Sigma(t, x) - \Sigma(t, y)|}{2\sqrt{c}}. \quad (1.78)$$

Having this, we can directly state the main theorem corresponding to the SDE approach.

Theorem 1.2.41. *The unique solution to*

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_{t_0} = \Theta \quad (1.79)$$

is a diffusion process, whose transition distribution function $F_{s,x}(t, y)$ satisfies the Kolmogorov backward equation. If the additional conditions on part b) of the previous Theorem are satisfied, then X_t has density functions $p_{s,x}(t, y)$, satisfying the corresponding Kolmogorov forward equation.

Example 1.2.42. We illustrate the previous discussion with the example that will occupy the rest of this chapter. Take $n = 1$ and consider geometric Brownian motion, $dX_t = rX_t dt + \sigma X_t dW_t$ on $(0, \infty)$, so that $\mu(x) = rx$ and $\Sigma(x) = \sigma^2 x^2$. By the theorem above its generator is

$$\mathcal{A}f(x) = \frac{1}{2}\sigma^2 x^2 f''(x) + rx f'(x), \quad (1.80)$$

and the backward equation $-\partial u / \partial s = \mathcal{A}u$ is, after adding the discounting term $-ru$ supplied by Feynman–Kac, exactly the Black–Scholes equation (1.104) of Section 1.4. Its adjoint,

$$\mathcal{A}^* \rho(x) = \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2 x^2 \rho(x)) - \frac{\partial}{\partial x} (rx \rho(x)), \quad (1.81)$$

governs the forward equation (1.76), and it is the object Dupire inverts in Subsection 1.5.1 to read a local volatility function off market data. The two equations of the table are thus not an abstraction: they are the pricing equation and the calibration equation of the entire second half of this chapter.

1.2.7 References to Section 1.2

The theoretical framework presented in Sections 1.2 and the preliminaries preceding it draws primarily from two authoritative sources. The general Probability Theory concepts follow the classic exposition of P. Billingsley ([4]), while the construction of the Stochastic Integral is based on the comprehensive treatment by H. Kuo ([26]). Additionally, references for the underlying Functional Analysis theory, implicit in the PDE treatment of diffusions, can be found in the seminal work of H. Brezis ([9]).

Certain advanced topics, such as the general formulation of Stochastic Integration with respect to semimartingales or deeper pathwise properties of Brownian motion, have been omitted as they lie beyond the scope of this work. However, the selected contents provide a rigorous and sufficient foundation for the financial applications discussed in subsequent sections and chapters.

1.3 Financial Derivatives

In this section, we introduce the main ideas behind Mathematical Finance and the role played by Mathematics in this field. We first informally introduce the concept of *derivative* and give some examples of the main quoted derivatives. The rigorous mathematical treatment of hedging and market completeness is addressed afterwards.

1.3.1 The Concept of a Derivative

In the context of modern mathematical finance, a financial derivative is formally defined as a *contingent claim*. Its value is derived from—or “contingent” upon—the state of some other underlying variables.

Let us fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$, where $\mathbb{P}$ represents the physical (or real-world) probability measure, and the filtration $(\mathcal{F}_t)_{t \geq 0}$ models the flow of information over time up to a finite time horizon T . Mathematically, a financial derivative with maturity T is an $\mathcal{F}_T$ -measurable random variable, denoted by H_T . There are certain derivatives, such as perpetual bonds, which have an infinite time horizon ($T = \infty$), but these fall outside the scope of our finite-horizon models, as they lack standard liquidity and do not constitute a core part of the derivatives market.

This measurability condition implies that the value of the derivative at maturity is fully determined by the information available at time T . The function determining this value is often referred to as the *payoff function*, denoted by Φ_H . The key idea behind the subsequent theory is being able to assign *fair prices* to those contingent claims, deduced from quoted (and liquid) market data.

The prices of such claims are not independent of one another. A small set of standardised, highly liquid contracts carries enough information to determine the price of a great many bespoke ones, and the principle enforcing this consistency is the absence of **arbitrage**: informally, the impossibility of a strictly positive risk-free profit from zero net initial investment.

A single example fixes the idea. Let a non-dividend-paying stock trade at S_0 and let a forward contract on it have delivery price $K > S_0 e^{rT}$, with r the continuous risk-free rate. Borrowing S_0 , buying the stock and shorting the forward produces, at maturity, a delivery of the stock against K and a repayment of $S_0 e^{rT}$, for a certain profit of $K - S_0 e^{rT} > 0$. Consistency therefore forces $K = S_0 e^{rT}$.

Real markets impose frictions — transaction costs, bid-ask spreads, asymmetric borrowing and lending rates, taxes — which widen this identity into a band rather than a point. But for the fungible assets and electronic markets we shall be concerned with, the band is tight, and it is precisely the narrowness of that band which makes the mathematics worth doing: not only to price consistently, but to quantify the residual risk and to construct the dynamic strategies that **hedge** it.

The Underlying Asset

The process driving the derivative’s value is the *underlying asset*, an $(\mathcal{F}_t)$ -adapted process $S = (S_t)_{0 \leq t \leq T}$; a derivative may depend on several. In practice S_t is the price of an equity or



index, a bond or interest rate (often modelled through a short-rate process r_t), a commodity such as oil, gold or electricity, or an exchange rate. Derivatives are then categorised by the functional form of their terminal payoff H_T .

1. Linear Derivatives (Forwards and Futures) These instruments possess a payoff that is a linear function of the underlying asset price at maturity. A prime example is the **Forward Contract**, which is an agreement to buy or sell the asset at time T for a pre-specified delivery price K . The payoff at time T for the holder of a long position is:

$$H_T = S_T - K \quad (1.82)$$

Note that this payoff ranges over $\mathbb{R}$, implying a potential liability for the holder if the asset price falls below K .

2. Non-Linear Derivatives (Options) Options provide the *right*, but not the *obligation*, to transact. This asymmetry introduces convexity (non-linearity) into the payoff structure, truncating downside risk.

- **European Call Option:** The right to buy S_T at strike price K . The payoff is defined as:

$$H_T = (S_T - K)^+ = \max(S_T - K, 0) \quad (1.83)$$

- **European Put Option:** The right to sell S_T at strike price K . The payoff is defined as:

$$H_T = (K - S_T)^+ = \max(K - S_T, 0) \quad (1.84)$$

- **American Options:** American call or put options are similar to their European counterparts, with the key difference being that the option can be *exercised* at any time up to maturity T .

From a theoretical standpoint, derivatives serve two primary functions in the market:

1. **Hedging:** They allow agents to transfer specific risks (e.g., volatility or tail risk) to those willing to bear them.
2. **Market Completeness:** In an idealized complete market, derivatives allow for the replication of any payoff profile, effectively spanning the state space of future uncertainties.

These concepts will be rigorously defined in the next subsection. However, the main question remains: what price should these contracts hold? Before entering the mathematical machinery, we can establish bounds based on physical intuition, much like conservation laws:

- If $T = 0$, meaning we are pricing a claim that expires immediately (and only depends on the underlying at maturity, not on its trajectory), the price must equal the payoff. For a European Call Option with strike K and underlying S_0 , its price must be $(S_0 - K)^+$.
- If an asset gives its buyer strictly more optionality compared to another derivative, its price must be greater than or equal to the latter. For example, the price of an American Call Option must be bounded below by the price of a European Call Option with same strike and maturity.

1.3.2 Derivative Pricing

We now turn to the rigorous problem of pricing contingent claims. While elementary treatments often assume asset prices follow geometric Brownian motions and interest rates are constant, these assumptions are insufficient for a general mathematical theory.

We work in a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ satisfying the usual conditions (right-continuity and completeness). The time horizon T is finite.

Let the market consist of $n + 1$ assets denoted by the vector process $S = (S^0, S^1, \dots, S^n)$. We assume that each component S^i is an **Itô process**. While the most abstract fundamental theorems of asset pricing rely on the broader class of general semimartingales, Itô processes provide the precise mathematical framework required for the diffusion models prevalent in this thesis, seamlessly connecting with the stochastic calculus developed in Section 1.2.

We distinguish the 0-th asset, S^0 , as the **numeraire**.

Definition 1.3.1 (Numeraire). A numeraire is a price process S^0 that is strictly positive ($S_t^0 > 0$ almost surely for all t) and non-dividend paying. It serves as the unit of account for the market.

While S^0 is often chosen to be a bank account $B_t = \exp(\int_0^t r_s ds)$, the theory holds for *any* valid Itô process numeraire (e.g., a foreign currency or a stock). The bank account is merely a convenient particular case because it explicitly models the time value of money.

We define the **discounted price process** X as the prices of the primary assets expressed in units of the numeraire:

$$X_t = \left(1, \frac{S_t^1}{S_t^0}, \dots, \frac{S_t^n}{S_t^0}\right) \quad (1.85)$$

Because the denominator S_t^0 is strictly positive almost surely, the transformation $f(x, y) = x/y$ is smooth. By a direct application of the multidimensional Itô's formula, the discounted price processes X_t^i are themselves Itô processes.

A trading strategy is a predictable process $H = (H^0, H^1, \dots, H^n)$, where H_t^i represents the number of units of asset i held at time t . The value of the portfolio at time t is the scalar process:

$$V_t(H) = \sum_{i=0}^n H_t^i S_t^i \quad (1.86)$$

For the strategy to be economically meaningful, it must be **self-financing**. In the language of stochastic integration, this implies that changes in wealth are generated solely by the fluctuations of the asset prices, with no external injection or extraction of capital:

$$dV_t(H) = \sum_{i=0}^n H_t^i dS_t^i \quad (1.87)$$

When working with discounted prices X , the self-financing condition simplifies. Applying Itô's product rule, it can be shown that a strategy is self-financing if and only if the discounted wealth process V_t/S_t^0 satisfies:

$$\frac{V_t(H)}{S_t^0} = \frac{V_0(H)}{S_0^0} + \int_0^t \sum_{i=1}^n H_u^i dX_u^i = \frac{V_0}{S_0^0} + (H \cdot X)_t \quad (1.88)$$

where $(H \cdot X)$ denotes the vector stochastic integral.

In continuous time, the standard definition of arbitrage is insufficient because "doubling strategies" (integrands with extremely large negative values) can technically generate profit from nothing in local time intervals. To preclude this, we restrict ourselves to certain strategies.



Definition 1.3.2 (Admissible Strategy). A self-financing strategy H is called a -admissible if its discounted value process is uniformly bounded from below by a constant $-a$ almost surely (i.e., $(H \cdot X)_t \geq -a$ for all t). A strategy is admissible if it is a -admissible for some $a \geq 0$.

The classical definition of "No Arbitrage" fails to be closed in the topology of L^∞ . Delbaen and Schachermayer (1994) proved that the appropriate condition for the existence of an equivalent martingale measure is *No Free Lunch with Vanishing Risk*.

Let K be the set of random variables representing the terminal wealth of all admissible strategies starting with zero capital:

$$K = \{(H \cdot X)_T \mid H \text{ is admissible and } H_0 = 0\} \quad (1.89)$$

Let C be the cone of claims dominated by elements in K , i.e., $C = \{g \in L^\infty(\mathcal{F}_T) \mid \exists f \in K, g \leq f\}$. The market satisfies (at least tries to) **NFLVR** if:

$$\overline{C} \cap L_+^\infty = \{0\} \quad (1.90)$$

where $\overline{C}$ is the closure of C in the L^∞ norm. Intuitively, this condition prevents not just exact arbitrage, but also sequences of strategies that approximate an arbitrage opportunity with risk converging to zero.

We are now equipped to state the core result linking market topology to measure theory.

Definition 1.3.3 (Equivalent Local Martingale Measure). A probability measure $\mathbb{Q}$ is an Equivalent Local Martingale Measure (ELMM) for the numeraire S^0 if:

1. $\mathbb{Q} \sim \mathbb{P}$ (they share the same null sets).
2. The discounted price process $X_t = S_t/S_t^0$ is a **local martingale** under $\mathbb{Q}$.

Theorem 1.3.4 (First Fundamental Theorem of Asset Pricing). *Assume the process X is locally bounded. The market satisfies the NFLVR condition if and only if there exists at least one Equivalent Local Martingale Measure $\mathbb{Q}$.*

The other reasonable question to ask ourselves is whether every contingent claim can be replicated by a self-financing strategy. Before stating the second theorem, we must rigorously define market completeness.

Definition 1.3.5 (Market Completeness). A market is **complete** if every contingent claim $\xi \in L^\infty(\mathcal{F}_T)$ is *attainable*. That is, for every bounded random variable ξ , there exists an admissible self-financing strategy H and an initial capital $x \in \mathbb{R}$ such that:

$$\frac{\xi}{S_T^0} = x + (H \cdot X)_T \quad \mathbb{P}\text{-almost surely.} \quad (1.91)$$

Note that the statement is made under the physical measure $\mathbb{P}$. This is deliberate: at this stage $\mathbb{Q}$ has not been shown to be unique — indeed its uniqueness is exactly what the next theorem characterises — so quantifying over “ $\mathbb{Q}$ -almost surely” would be ambiguous. Since every equivalent local martingale measure shares its null sets with $\mathbb{P}$, the choice is harmless, but it is the honest one.

Theorem 1.3.6 (Second Fundamental Theorem of Asset Pricing). *Assume that the set of equivalent local martingale measures $\mathcal{M}^e$ is non-empty (i.e., the market is arbitrage-free). The market is complete if and only if $\mathcal{M}^e$ is a singleton set (i.e., $\mathbb{Q}$ is unique).*

If the market is arbitrage-free and complete (i.e., there exists a unique $\mathbb{Q}$), the fair price at time t of a generic derivative paying H_T at maturity is given by the expectation of the discounted payoff:

$$\frac{V_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}} \left[\frac{H_T}{S_T^0} \middle| \mathcal{F}_t \right] \implies V_t = S_t^0 \mathbb{E}^{\mathbb{Q}} \left[\frac{H_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad (1.92)$$

This formulation reveals that "risk-neutral probabilities" are not absolute; they are linked to the choice of numeraire.

Let N be a different numeraire (strictly positive semimartingale). Let $\mathbb{Q}^S$ be the measure associated with numeraire S^0 , and $\mathbb{Q}^N$ be the measure associated with numeraire N . Prices must be consistent regardless of the unit of account. Therefore, for any asset Z :

$$\frac{Z_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}^S} \left[\frac{Z_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad \text{and} \quad \frac{Z_t}{N_t} = \mathbb{E}^{\mathbb{Q}^N} \left[\frac{Z_T}{N_T} \middle| \mathcal{F}_t \right] \quad (1.93)$$

The change of measure from $\mathbb{Q}^S$ to $\mathbb{Q}^N$ is defined by the Radon-Nikodym derivative, which turns out to be exactly the normalized ratio of the numeraires:

$$\frac{d\mathbb{Q}^N}{d\mathbb{Q}^S} \bigg|_{\mathcal{F}_T} = \frac{N_T/S_T^0}{N_0/S_0^0} \quad (1.94)$$

This elegantly shows that changing the numeraire is mathematically equivalent to performing a change of probability measure. This technique is particularly powerful in interest rate theory (e.g., switching to the T -forward measure to efficiently price bond options and swaptions).

Finally, we can now state the Feynman-Kac formula, which provides a powerful link between the PDE approach and the probabilistic representation of derivative prices.

Theorem 1.3.7 (Feynman-Kac Formula). *Let X_τ be an n -dimensional Itô diffusion process on the interval $[t, T]$ driven by the stochastic differential equation*

$$dX_\tau = \mu(\tau, X_\tau)d\tau + \sigma(\tau, X_\tau)dW_\tau, \quad \tau \in [t, T], \quad (1.95)$$

with initial condition $X_t = x$, where W_τ is a standard n -dimensional Brownian motion, and the diffusion matrix is given by $\Sigma(\tau, x) = \sigma(\tau, x)\sigma(\tau, x)^\top$. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be a Borel-measurable terminal payoff function, and $r : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function representing the continuously compounded interest rate.

Assume $u(t, x)$ is a sufficiently regular function (typically $u \in C^{1,2}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$) that satisfies the following linear parabolic partial differential equation:

$$\frac{\partial u}{\partial t}(t, x) + \sum_{i=1}^n \mu_i(t, x) \frac{\partial u}{\partial x_i}(t, x) + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(t, x) \frac{\partial^2 u}{\partial x_i \partial x_j}(t, x) - r(t, x)u(t, x) = 0, \quad (1.96)$$

for all $(t, x) \in [0, T] \times \mathbb{R}^n$, subject to the terminal condition:

$$u(T, x) = h(x). \quad (1.97)$$

If $u(t, x)$ satisfies a suitable polynomial growth condition preventing the existence of explosive solutions, then $u(t, x)$ admits the stochastic representation:

$$u(t, x) = \mathbb{E} \left[e^{-\int_t^T r(\tau, X_\tau) d\tau} h(X_T) \middle| X_t = x \right]. \quad (1.98)$$

1.4 The Black-Scholes Framework

Having established the general theory of arbitrage-free pricing and martingale measures, we now specialize our market model to the most famous and widely used framework in financial history: the Black-Scholes-Merton model.



1.4.1 The Model

We assume a frictionless market: no transaction costs, no bid-ask spread, unrestricted short selling, and a constant risk-free rate r at which one may borrow and lend freely. Under the *physical* measure $\mathbb{P}$ — the measure describing what actually happens — the underlying is modelled by a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (1.99)$$

where $\mu \in \mathbb{R}$ is the expected rate of return, $\sigma > 0$ the volatility, and W a $\mathbb{P}$ -Brownian motion. By Example 1.2.13 the solution is $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$, which is positive for all t — as a limited-liability equity should be.

It is essential to begin under $\mathbb{P}$ and not under a risk-neutral measure. Doing so makes the central result of the theory visible rather than assumed, and that result is the following: **the price of the derivative does not depend on μ** . Two investors who disagree completely about the expected return of the stock, but agree on its volatility, must agree on the price of every option written on it. This is not intuitive, it is not a modelling convention, and it is worth seeing emerge from the computation rather than being handed the risk-neutral dynamics with μ already replaced by r .

The hedging argument

Consider a derivative whose value is a smooth function $V(t, S)$ of time and the current spot. Form the portfolio

$$\Pi_t = -V(t, S_t) + \Delta_t S_t,$$

consisting of a short position in the derivative and Δ_t units of the underlying. Assuming the position is self-financing in the sense of (1.87), its value evolves as

$$d\Pi_t = -dV_t + \Delta_t dS_t. \quad (1.100)$$

Itô's rule (Theorem 1.2.11) applied to $V(t, S_t)$, using $(dS_t)^2 = \sigma^2 S_t^2 dt$ from Table 1.1, gives

$$dV_t = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t. \quad (1.101)$$

Substituting (1.99) and (1.101) into (1.100) and collecting terms,

$$d\Pi_t = \underbrace{\left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \mu S_t \left[\Delta_t - \frac{\partial V}{\partial S} \right] \right)}_{\text{drift}} dt + \underbrace{\sigma S_t \left(\Delta_t - \frac{\partial V}{\partial S} \right)}_{\text{diffusion}} dW_t. \quad (1.102)$$

The portfolio is exposed to randomness only through the second term. Choosing

$$\Delta_t = \frac{\partial V}{\partial S}(t, S_t) \quad (1.103)$$

annihilates it. This is the *delta* of the derivative, and (1.103) is the hedge.

Now observe what (1.102) has become. The same choice (1.103) that kills the diffusion term also kills the bracket multiplying μ in the drift. **The expected return of the stock has cancelled**, and it has done so for a structural reason rather than by accident: μ enters dS_t and dV_t through the identical channel, so a position that is neutral to the random part of S is automatically neutral to its drift as well. What remains,

$$d\Pi_t = \left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt,$$

is riskless. A riskless self-financing portfolio must earn the risk-free rate — otherwise, by borrowing or lending against it, one manufactures an arbitrage in the sense of Subsection 1.3.2 — so $d\Pi_t = r\Pi_t dt$ with $\Pi_t = -V + S_t \partial V / \partial S$. Equating the two expressions and rearranging yields the Black–Scholes equation.

Theorem 1.4.1 (Black–Scholes–Merton). *Under the dynamics (1.99), the value $V(t, S)$ of a European derivative with payoff $V(T, S) = \Phi(S)$ satisfies the linear parabolic partial differential equation*

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \quad (t, S) \in [0, T) \times (0, \infty), \quad (1.104)$$

together with the terminal condition $V(T, S) = \Phi(S)$. ■

Equation (1.104) contains r but not μ , which is the claim made above, now proved. It is also exactly the Feynman–Kac equation of Theorem 1.3.7 for a diffusion with drift rS and diffusion coefficient σS , whence the probabilistic representation

$$V(t, S) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S], \quad dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (1.105)$$

The passage from (1.99) to the dynamics in (1.105) is precisely Girsanov’s theorem with $\alpha = (\mu - r)/\sigma$, the *market price of risk*: the measure change subtracts a drift and leaves σ untouched, exactly as discussed after Theorem 1.2.33. That the hedging argument and the martingale argument give the same answer is not a coincidence but the content of Feynman–Kac, and the two viewpoints will alternate throughout what follows — the PDE when we want to compute, the expectation when we want to reason.

Finally, the market is complete: by Theorem 1.2.35 and Remark 1.2.36, the filtration is generated by the single Brownian motion driving the single traded asset, so every square-integrable claim is attainable and $\mathbb{Q}$ is unique. The delta (1.103) is the integrand φ whose abstract existence the representation theorem guarantees; here we have it explicitly.

The closed-form solution

For a European call, $\Phi(S) = (S - K)^+$, evaluating (1.105) against the log-normal density gives

$$C_{BS}(t, S) = S N(d_+) - K e^{-r(T-t)} N(d_-), \quad d_{\pm} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad (1.106)$$

with $\tau = T - t$ and N the standard normal distribution function. The corresponding put price follows from put–call parity, Proposition 1.4.4 below.

One might instead assume **normal** (Bachelier) dynamics $dS_t = \mu dt + \sigma dW_t$, as in the original 1900 thesis. This permits negative prices, which is why it was criticised for equities, but it is standard for interest rates and for spread options where negative values are economically meaningful. For equities the log-normal convention prevails, on account of the limited liability of shareholders.

Dividends, and why the forward is the right variable

The derivation above quietly assumed that holding the stock produces no cash flows. Real equities pay dividends, and the way they do so is awkward enough to force a change of variable that we shall keep for the rest of the dissertation.

Suppose first that dividends are paid *continuously* at a constant proportional rate q . The holder of the stock receives $qS_t dt$ over $[t, t + dt]$, so the self-financing condition (1.100) acquires the extra term $\Delta_t q S_t dt$; repeating the argument verbatim replaces the coefficient r of $\partial V / \partial S$ in



(1.104) by $r - q$, and r by $r - q$ in $d_{\pm}$. Nothing structural changes. This is the case usually presented, and it is the case that does not occur.

Actual equity dividends are *discrete cash amounts*, announced in advance and paid on known dates. If an amount D_i is paid at time t_i , then no-arbitrage forces the spot to drop by that amount across the payment:

$$S_{t_i} = S_{t_i^-} - D_i. \quad (1.107)$$

This is genuinely destructive of the framework above, and it is worth being precise about why. Equation (1.107) says S has a *predictable jump*: the process is not continuous, so Itô's rule as stated does not apply across t_i . Worse, the jump is *additive* while geometric Brownian motion is multiplicative, so the two are structurally incompatible — one cannot write $S_t = S_0 \mathcal{E}(\cdot)$ for a positive $\mathcal{E}$ and reproduce (1.107). Worse still, a fixed cash dividend can in principle exceed the spot price, so a model treating D_i as deterministic does not even guarantee positivity. And since σ multiplies S_t in (1.99), a discontinuous drop in S produces a discontinuous drop in the absolute volatility, which is not what is observed.

The resolution is to stop modelling the spot and model the *forward* instead.

Definition 1.4.2 (Forward price). The **forward price** $F(t, T)$ is the delivery price agreed at t for a transaction in the underlying at T which has zero value at inception. Equivalently, writing $P(t, T)$ for the price at t of a zero-coupon bond maturing at T ,

$$F(t, T) = \frac{\mathbb{E}^{\mathbb{Q}^T}[S_T | \mathcal{F}_t] \cdot P(t, T)}{P(t, T)} = \mathbb{E}^{\mathbb{Q}^T}[S_T | \mathcal{F}_t], \quad (1.108)$$

where $\mathbb{Q}^T$ is the T -forward measure associated with the numeraire $P(\cdot, T)$.

Identity (1.108) says that the forward price is a $\mathbb{Q}^T$ -martingale — it is a conditional expectation — and this is the whole point. It holds by construction, with no assumption whatever on interest rates, on dividends, or on their nature. In particular we deliberately do *not* write $F(t, T) = S_t e^{(r-q)(T-t)}$, which is valid only for constant r , constant q , and purely proportional dividends. Instead we record the relation abstractly:

$$S_t = \Theta_t F(t, T), \quad (1.109)$$

where Θ_t is a strictly positive, adapted, model-independent conversion factor assembled from the discount curve, the repo or borrow cost, and the announced cash and proportional dividend schedule. Its precise construction is a market-data question, not a modelling one, and it is addressed in Chapter 3. What matters here is that (1.109) remains valid for time-dependent $r(t)$ and $q(t)$, for discrete cash dividends, for proportional dividends, and for any mixture of them.

Modelling the forward as a driftless log-normal martingale,

$$dF_t = \sigma F_t dW_t^T, \quad (1.110)$$

therefore has three advantages simultaneously. The dividend discontinuities are absorbed into Θ_t and never touch the dynamics, which stay diffusive. The drift vanishes, because a forward is already a martingale under its own measure — so there is no drift to specify and none to get wrong. And the resulting formula is calibrated directly against quantities the market quotes.

The Forward Measure and Black-76

Changing numeraire from the bank account to the zero-coupon bond $P(\cdot, T)$, in the manner of (1.94), converts (1.105) into an expectation of the undiscounted payoff under $\mathbb{Q}^T$, multiplied by a deterministic discount factor. With the dynamics (1.110) this gives Black's 1976 formula.

Proposition 1.4.3 (Black-76 [6]). *Under (1.110), the time- t value of a European call with strike K and maturity T is*

$$C(t) = P(t, T) \left[F_t N(d_+) - K N(d_-) \right], \quad d_{\pm} = \frac{\ln(F_t/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad \tau = T - t. \quad (1.111)$$

■

Formula (1.111) is formally identical to (1.106) with S replaced by F_t and the interest rate removed from $d_{\pm}$ — but note that the discount factor $P(t, T)$ remains, multiplying the whole bracket. It has not disappeared; it has merely been collected outside. We shall use (1.111) as the primitive throughout, treating (1.106) as the special case obtained when $P(t, T) = e^{-r\tau}$ and $F_t = S_t e^{r\tau}$.

Put–call parity

Before turning to implied volatility we record the one relation in this subject that holds independently of any model, and which we shall invoke repeatedly.

Proposition 1.4.4 (Put–call parity). *Let $C(t)$ and $P(t)$ be the time- t prices of European call and put options on the same underlying, with the same strike K and maturity T . Then, in any arbitrage-free market,*

$$C(t) - P(t) = P(t, T)(F(t, T) - K). \quad (1.112)$$

Proof: The payoffs satisfy the pointwise identity $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, valid for every value of S_T . Taking $\mathbb{E}^{\mathbb{Q}^T}[\cdot | \mathcal{F}_t]$ and multiplying by $P(t, T)$, the right-hand side becomes $P(t, T)(F(t, T) - K)$ by (1.108). ■

The proof used no dynamics — not (1.110), not even continuity of paths — only the linearity of expectation and an algebraic identity between payoffs. Consequently (1.112) must hold in every model in this dissertation, and it is the standard sanity check on any implementation: a pricer that violates put–call parity is wrong, and no appeal to model choice can rescue it.

Two consequences are worth noting now. First, since $C - P$ is model-free, calls and puts of the same strike carry *identical* information about volatility; a call and a put struck at K must therefore have the same implied volatility, and the smile is a single function of strike rather than two. Second, (1.112) is how the forward is extracted from quoted option prices in practice: the call-minus-put spread is linear in K with slope $-P(t, T)$ and root $K = F(t, T)$, so a regression across strikes recovers both the discount factor and the forward without any assumption about dividends. This is exactly the construction of Θ_t in (1.109), obtained from market data rather than postulated.

The Inversion Problem and Implied Volatility

In professional markets, participants do not "price" options using Black-Scholes to find the dollar value. The market price C^{mkt} is determined by supply and demand.

Instead, Black-Scholes is used as a translation mechanism. Since the pricing function $C_{BS}(\sigma)$ is strictly monotonic in σ , we can invert it to find the unique parameter σ_{imp} that matches the market price:

$$C_{BS}(S, K, T, r, q, \sigma_{\text{imp}}) = C^{mkt} \quad (1.113)$$

This implied volatility is the true "price" quoted in the market. It represents the market's consensus on the average future volatility of the underlying asset over the life of the option.



1.4.2 Smile and Skewness

If the assumptions of Black-Scholes were satisfied physically (i.e., if returns were truly log-normal with constant variance), the implied volatility σ_{imp} would be constant across all strike prices K and maturities T .

Empirical data reveals this is not the case. When we plot $\sigma_{\text{imp}}(K)$ against K , we observe distinct shapes:

- **The Skew (Smirk):** Typical in equity markets. σ_{imp} decreases as K increases. This reflects "Crash Phobia"—investors bid up the price of low-strike Puts (insurance against market drops), implying higher volatility for downside strikes.
- **The Smile:** Typical in Forex markets. σ_{imp} is lowest at the money ($K \approx S$) and increases for both deep ITM and OTM options. This reflects "Fat Tails" (leptokurtosis) in the return distribution—extreme moves are more probable than the Gaussian model predicts.

The two shapes are not merely descriptive. Both say that the market's risk-neutral distribution is not log-normal, and the following classical result makes the connection exact.

Proposition 1.4.5 (Breedon–Litzenberger [8]). *Let $C(K, T)$ denote the price of a European call of strike K and maturity T , and suppose the risk-neutral law of S_T admits a density ψ_T . Then*

$$\psi_T(K) = \frac{1}{P(0, T)} \frac{\partial^2 C}{\partial K^2}(K, T). \quad (1.114)$$

Proof: Write $C(K, T) = P(0, T) \int_K^\infty (x - K) \psi_T(x) dx$. Differentiating once under the integral sign, the boundary term vanishes because the integrand is zero at $x = K$, leaving $\partial_K C = -P(0, T) \int_K^\infty \psi_T(x) dx$. Differentiating a second time gives (1.114). ■

The result is remarkable for what it does not assume: no model, no dynamics, only that prices are discounted expectations of the payoff. A continuum of option quotes at a fixed maturity therefore determines the entire marginal distribution of S_T — the market does not merely express an opinion about volatility, it expresses a complete probability law, and the smile is the visible trace of that law failing to be log-normal.

Two consequences of (1.114) will be used constantly. First, since $\psi_T \geq 0$, arbitrage-free call prices must be *convex* in strike; a violation is a butterfly arbitrage, executable by buying two wing options and selling two at the middle strike. Second, the formula involves a *second* derivative of market data, which is numerically delicate in a way that Subsection 1.5.1 will have to confront directly.

[FIGURE HERE] Empirical implied volatility smile for a liquid index at two maturities, showing the flattening of the skew with increasing T . To be produced from the market data of Chapter 3.

1.5 Other Important Models

To account for the volatility smile and skew, we must extend the Black-Scholes framework. There are two primary approaches: making volatility a deterministic function of the state (Local Volatility) or making it a stochastic process (Stochastic Volatility).

1.5.1 Local Volatility Models

The Local Volatility (LV) framework, introduced by Dupire (1994), posits that volatility is a deterministic function of the current asset price and time: $\sigma = \sigma(S_t, t)$. The dynamics under the risk-neutral measure are:

$$dS_t = rS_t dt + \sigma(S_t, t)S_t dW_t \quad (1.115)$$

This model preserves the completeness of the market (there is only one source of randomness W_t) while allowing the model to fit *any* observed arbitrage-free implied volatility surface exactly.

The Dupire Formula and Gatheral's Transformation

Dupire's insight was to link the unknown function $\sigma_{\text{loc}}(K, T)$ directly to observed market option prices. Applying the Fokker–Planck equation (1.76) to the transition density of the diffusion and combining it with Breeden–Litzenberger yields the following.

Theorem 1.5.1 (Dupire). *Suppose call prices $C(K, T)$ are known for a continuum of strikes and maturities and are consistent with a diffusion of the form $dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(S_t, t)S_t dW_t$. Then the local volatility function is uniquely determined by*

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}}. \quad (1.116)$$

■

The content of (1.116) is striking and deserves to be stated plainly: *there is exactly one* local volatility diffusion consistent with a given arbitrage-free surface. The calibration problem has a unique solution, not a family of them, and the model can therefore reproduce every European quote simultaneously and exactly. Note also that the denominator is proportional to the risk-neutral density by Proposition 1.4.5, so (1.116) is well posed precisely when that density is strictly positive.

Static arbitrage and the implied-variance parameterisation

Formula (1.116) is, however, close to unusable as written. It requires one derivative in maturity and two in strike of raw market data, and market data are quoted on a sparse grid, carry bid-ask spreads, and are contaminated by microstructure noise. Numerical differentiation amplifies all three, and the practical symptom is a negative denominator — that is, a negative local variance, which is meaningless.

The remedy is not better differentiation but a change of variables, together with an interpolation that is guaranteed arbitrage-free by construction. Define the **log-moneyness** and the **total implied variance**

$$k := \log \frac{K}{F(t, T)}, \quad w(k, T) := \sigma_{\text{imp}}^2(k, T) T. \quad (1.117)$$

Total variance is the natural coordinate because it, rather than implied volatility itself, is what accumulates linearly in time under Black–Scholes. We now state the conditions under which a surface w is free of *static* arbitrage — that is, admits no arbitrage exploitable by a buy-and-hold portfolio of European options, with no trading in between.

Proposition 1.5.2 (Absence of static arbitrage). *A total implied variance surface $w(k, T)$ is free of static arbitrage if the following hold.*



i) (**Calendar spread**) For each fixed k , the map $T \mapsto w(k, T)$ is non-decreasing:

$$\frac{\partial w}{\partial T}(k, T) \geq 0. \quad (1.118)$$

ii) (**Butterfly**) For each fixed T , the function

$$g(k) := \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_{kk}^2 w}{2} \quad (1.119)$$

satisfies $g(k) \geq 0$ for all k .

iii) (**Wings**) $w(k, T)$ grows at most linearly in $|k|$ as $|k| \rightarrow \infty$; more precisely, the asymptotic slopes are bounded by 2, which is Lee's moment formula [27].

Each condition has a direct financial meaning. Condition i) says a longer-dated option cannot be cheaper than a shorter-dated one of the same moneyness — otherwise one buys the long and sells the short. Condition ii) is convexity of the call price in strike, expressed in the new coordinates, and its violation is a butterfly spread with negative cost and non-negative payoff. Condition iii) prevents the extrapolated wings from implying infinite moments of S_T .

In practice one first fits a parametric form guaranteeing i)–iii) — Gatheral's SVI parameterisation [19] being the standard choice, with SABR common in rates markets — and only then differentiates. Since the fitted surface is smooth and arbitrage-free by construction, the differentiation is stable and the local variance is automatically non-negative. This yields Gatheral's form of the Dupire formula, which is the version actually implemented.

Theorem 1.5.3 (Dupire's formula in implied-variance coordinates). *With k and w as in (1.117),*

$$\sigma_{\text{loc}}^2(k, T) = \frac{\frac{\partial w}{\partial T}}{1 - \frac{k}{w} \frac{\partial w}{\partial k} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2}}. \quad (1.120)$$

Formula (1.120) is worth a closer look, because the two no-arbitrage conditions of Proposition 1.5.2 are visible in it. The numerator is exactly the calendar spread quantity (1.118), and the denominator is precisely the butterfly quantity $g(k)$ of (1.119). Hence

$$\frac{\partial w}{\partial T} \geq 0 \text{ and } g(k) \geq 0 \implies \sigma_{\text{loc}}^2 \geq 0,$$

so that **an implied volatility surface free of static arbitrage yields a non-negative local variance**. The implication runs one way only, and it is worth being precise about why: a quotient is non-negative whenever numerator and denominator share a sign, so $\sigma_{\text{loc}}^2 \geq 0$ is also compatible with both being negative, a configuration which certainly does admit arbitrage. What is true is the chain

$$\text{no static arbitrage} \implies \frac{\partial w}{\partial T} \geq 0 \wedge g(k) \geq 0 \implies \sigma_{\text{loc}}^2 \geq 0,$$

in which each of the two conditions is *separately* necessary for absence of arbitrage — $\partial_T w \geq 0$ for calendar spreads, $g(k) \geq 0$ for butterflies — and only the last arrow fails to reverse.

That the denominator of (1.120) is $g(k)$ exactly, and not merely something of the same sign, is the substantive observation here, and it is what makes the two-step procedure above more than a matter of taste: arbitrage-freeness is not an aesthetic requirement imposed on the interpolation, it is the condition under which (1.120) is guaranteed to return a usable answer. It is also the reason this parameterisation, rather than the raw form (1.116), is the one calibrated in practice.

The limitation

Local volatility fits the surface perfectly, and by Theorem 1.5.1 it is the unique diffusion that does. It also preserves market completeness, since a single Brownian motion drives a single traded asset and Theorem 1.2.35 applies unchanged. These are substantial virtues.

The defect is equally fundamental, and it concerns *dynamics* rather than fit. In a local volatility model the instantaneous volatility is a deterministic function of the spot, so the future smile is entirely determined by where the spot goes. The consequence, observed universally in practice, is that the model predicts the smile flattening as the spot moves, whereas real smiles translate with the spot and retain their shape. A model may therefore reproduce every European price today and still be wrong about every European price tomorrow. This matters not at all for pricing European options — their value depends only on the terminal marginal, which is fitted exactly — and a great deal for anything path-dependent: barriers, cliquets, forward-starting options and autocallables all depend on the joint law across dates, which local volatility gets wrong. That observation is what motivates the next subsection.

1.5.2 Stochastic Volatility Models

Throughout this subsection, and in Subsection 1.5.3 which follows it, interest rates and dividend yields are taken to be *deterministic*. The assumption is worth stating once rather than repeating: it is what makes the T -forward measure coincide with the risk-neutral one, $\mathbb{Q}^T = \mathbb{Q}$, so that expectations below may be written under $\mathbb{Q}$ while still exploiting the $\mathbb{Q}^T$ -martingale property of the forward; and it is what allows the discounting term in the American pricing operator of (1.130) to be a deterministic function of time. With stochastic rates both statements require the change of numeraire of Subsection 1.4.1 to be carried through explicitly.

To capture realistic dynamics of the volatility surface itself, we introduce a second source of randomness. In Stochastic Volatility (SV) models, the variance $v_t = \sigma_t^2$ follows its own Stochastic Differential Equation. Under the risk-neutral measure $\mathbb{Q}$, the system is described by:

$$\begin{cases} dS_t = (r - q)S_t dt + \sqrt{v_t}S_t dW_t^S \\ dv_t = \alpha(v_t, t)dt + \beta(v_t, t)dW_t^v \end{cases} \quad (1.121)$$

A critical parameter is the correlation ρ between the two Brownian motions: $d\langle W^S, W^v \rangle_t = \rho dt$.

- If $\rho < 0$, a drop in price ($S \downarrow$) is associated with an increase in volatility ($v \uparrow$), naturally generating the **skew** observed in equity markets.
- The "vol of vol" parameter (in the diffusion of v_t) controls the convexity or **curvature** of the smile.

The Heston Model

The most prominent SV model is that of Heston (1993), in which the variance follows the Cox–Ingersoll–Ross square-root process already analysed in Example 1.2.20:

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \end{cases} \quad d\langle W^S, W^v \rangle_t = \rho dt, \quad (1.122)$$

where κ is the speed of mean reversion, θ the long-run variance, ξ the volatility of volatility, and ρ the correlation, understood in the sense of Definition 1.2.22. We recall from Subsection 1.2.3 that (1.122) is well posed by Yamada–Watanabe rather than by the Lipschitz theory, and that



strict positivity of v requires the Feller condition $2\kappa\theta \geq \xi^2$ of Proposition 1.2.21 — a condition frequently violated by calibrated parameters, with the numerical consequences noted there.

Unlike local volatility, stochastic volatility renders the market **incomplete**, and we are now in a position to say exactly why rather than merely asserting it. The filtration is generated by two Brownian motions while only one asset is traded, so the hypothesis of Theorem 1.2.35 fails: a claim whose value depends on W^v cannot be written as a stochastic integral against the traded asset alone. By Theorem 1.3.6 the equivalent martingale measure is therefore not unique, and pricing requires either specifying a market price of volatility risk — already embedded in the $\mathbb{Q}$ -dynamics (1.122) — or, as is universal in practice, calibrating $(\kappa, \theta, \xi, \rho, v_0)$ directly to liquid quotes. Volatility risk cannot be hedged away with the underlying; it can only be hedged with other options.

The reason Heston's model, rather than some other stochastic volatility model, became the industry standard is computational, and worth stating explicitly because Part II will build directly on it. The characteristic function of the log-price is available in closed form.

Theorem 1.5.4 (Heston's characteristic function). *Under (1.122), write $\tau = T - t$ and $F_s = S_s e^{(r-q)(T-s)}$ for the forward to T . Then*

$$\varphi(u, \tau) := \mathbb{E}^{\mathbb{Q}} \left[\exp \left(iu \log \frac{F_T}{F_t} \right) \middle| \mathcal{F}_t \right] = \exp \left(C(u, \tau) \theta + D(u, \tau) v_t \right), \quad (1.123)$$

the characteristic function of the centred log-forward $\log(F_T/F_t)$, where C and D solve the system of Riccati ordinary differential equations

$$\frac{\partial D}{\partial \tau} = \frac{\xi^2}{2} D^2 + (\rho \xi i u - \kappa) D - \frac{u^2 + iu}{2}, \quad \frac{\partial C}{\partial \tau} = \kappa D, \quad (1.124)$$

with $C(u, 0) = D(u, 0) = 0$. The system (1.124) admits an explicit solution in terms of elementary functions. ■

The centring in (1.123) is deliberate and should not be skipped over. Taking the characteristic function of $\log(F_T/F_t)$ rather than of the absolute log-spot X_T strips out the factor $e^{iu \log F_t}$, and does so for two reasons. The first is arithmetical: the inversion formula below carries F_t in its prefactor and again inside the log-moneyness $k = \log(K/F_t)$, so a characteristic function still carrying F_t would introduce it a third time and the normalisation would be applied once and undone once. The second is structural: the centred φ depends only on $(v_t, \kappa, \theta, \xi, \rho, \tau)$ and not on S_t, r or q , which is exactly the separation between model parameters and market data that makes calibration well posed. This is the same reason the forward was singled out as the right variable in Subsection 1.4.1.

The centring also supplies a cheap correctness check, worth recording because the error it catches is silent. Since F is a $\mathbb{Q}^T$ -martingale, $\mathbb{E}^{\mathbb{Q}}[F_T/F_t | \mathcal{F}_t] = 1$, so

$$\varphi(-i, \tau) = 1 \quad \text{for every } \tau \geq 0, \quad (1.125)$$

an identity satisfied by the centred characteristic function and violated by the uncentred one. Any implementation of (1.123) should be tested against (1.125) before it is tested against prices. Given (1.123), European option prices follow by Fourier inversion: the call price is recovered from the characteristic function through an integral of Lewis–Carr–Madan type [10],

$$C(K, T) = P(t, T) \left[F_t - \frac{\sqrt{F_t K}}{\pi} \int_0^\infty \operatorname{Re} \left(e^{iuk} \varphi \left(u - \frac{i}{2}, \tau \right) \right) \frac{du}{u^2 + \frac{1}{4}} \right], \quad (1.126)$$

a one-dimensional integral evaluated by quadrature in microseconds, with $k = \log(K/F_t)$ the log-moneyness of Subsection 1.5.1. Calibration — which requires pricing hundreds of options

at each step of an optimiser — is thereby feasible, whereas a model requiring Monte Carlo for each European quote would not be.

One implementation caveat deserves a line, since it is a genuine trap rather than a nuisance. The explicit solution of (1.124) involves a complex logarithm, and the naive branch choice makes φ discontinuous in τ once the integrand has wound sufficiently far, corrupting prices at long maturities while leaving short ones correct. The remedy is the standard one in the literature, namely to write the solution in the form whose principal branch is continuous; the phenomenon is documented in [3].

That the tractability of the model reduces to a system of Riccati equations is a point worth holding onto. The affine structure of (1.122) — drift and variance both affine functions of the state — is exactly what produces (1.124), and it is a structural property rather than a happy accident. When we come to rough volatility in Part II, the model will retain this affine character but lose the Markov property, and (1.124) will be replaced by a *fractional* Riccati equation. The pricing machinery (1.126) survives essentially unchanged; only the equation for D becomes non-local in time. Recording the classical case here is what will make that generalisation a short step rather than a fresh start.

What stochastic volatility does and does not fix

Stochastic volatility repairs the principal defect of local volatility: because the volatility has its own noise, the smile moves with the spot instead of flattening, and forward-starting products are priced far more sensibly. The correlation ρ generates skew and the vol-of-vol ξ generates curvature, both dynamically rather than by construction.

It does not, however, fit the surface exactly — five parameters cannot match hundreds of quotes — and it fails in one specific and instructive place. Empirically, the implied volatility skew of index options behaves like $T^{-H'}$ for small T with H' well below $\frac{1}{2}$, becoming very steep at short maturities. Any model driven by standard Brownian motion, Heston included, produces a short-maturity skew that flattens as $T \rightarrow 0$, because over short horizons a diffusive variance has had no time to move. No choice of $(\kappa, \theta, \xi, \rho, v_0)$ repairs this: the failure is not one of calibration but of the driving noise. This is the empirical observation from which Part II begins.

1.5.3 Path-Dependence and American Options

We conclude this section by briefly discussing the challenges posed by path-dependent derivatives, focusing on the most liquid and prominent example: American Options.

Because American options hold the right to be exercised at any time up to maturity, merely knowing a single implied volatility is insufficient to price them; a single volatility parameter is compatible with an infinite set of potential asset dynamics. This distinction is critically important because options on *single names* (i.e., individual corporate stocks) are almost exclusively American-style. Investors in single equities are reluctant to relinquish the flexibility of early exercise, particularly to capture discrete dividends or navigate corporate actions. In contrast, broad market indices are generally considered more stable and are traded via extremely liquid European options.

We aim to show how a Local Volatility model can be adapted to price American options. However, we must first note a fundamental limitation: Local Volatility models do not perform perfectly for all types of path-dependent derivatives. In particular, because LV models assume volatility is a deterministic function perfectly correlated with the spot price, they often misprice derivatives with discontinuous payoffs (which often lead to ill-posed pricing problems) or derivatives whose strikes are set at a future date, such as forward-starting options and cliquets. Nevertheless, to price American options accurately we must model the entire volatility surface prior to maturity, not merely the terminal marginal. The model we use, expressed under the



T -forward measure of Subsection 1.4.1, is

$$dF_t = \sigma_{\text{loc}}(F_t, t) F_t dW_t^T. \quad (1.127)$$

Working in the forward has the same three advantages set out in Subsection 1.4.1: the drift vanishes because a forward is a martingale under its own measure, the dynamics stay purely diffusive, and — crucially here — the discontinuities introduced by discrete cash dividends are absorbed into the spot-to-forward factor Θ_t of (1.109) and never enter (1.127) at all.

Optimal stopping and the Snell envelope

Before writing any partial differential equation, let us state what an American option *is* in probabilistic terms, since the PDE formulation is derived from it and is opaque without it. The holder may exercise at any stopping time of her choosing, and will choose the one maximising the value of the resulting cash flow. Writing $\mathcal{T}_{[t,T]}$ for the set of stopping times valued in $[t, T]$ and Φ for the exercise payoff, the value at time t is

$$V_t = \text{ess sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^\tau r(s) ds} \Phi(\tau, S_\tau) \middle| \mathcal{F}_t \right]. \quad (1.128)$$

The essential supremum is required because the family is uncountable; this is an optimal stopping problem, and its solution is classical.

Theorem 1.5.5 (Snell envelope). *Let Y be a right-continuous adapted process of class (D) representing the discounted payoff. Then the value process U defined by (1.128) is the **Snell envelope** of Y : that is, U is the smallest càdlàg supermartingale dominating Y . It satisfies:*

- i) $U_t \geq Y_t$ for all t (domination);
- ii) U is a $\mathbb{Q}$ -supermartingale;
- iii) U is a $\mathbb{Q}$ -martingale on the continuation region $\mathcal{C} = \{(t, \omega) : U_t > Y_t\}$;

and the stopping time $\tau^* = \inf\{s \geq t : U_s = Y_s\}$ is optimal in (1.128). A full treatment is given in [34]. ■

The three properties have an immediate financial reading, and it is worth pausing on it because the LCP below is nothing more than their restatement. Property i) says the option is never worth less than exercising it — otherwise one would buy the option, exercise immediately, and pocket the difference. Property ii) says that holding the option is, on average and after discounting, a losing proposition: waiting has a cost, which is what makes early exercise ever attractive. Property iii) says that this cost is zero exactly where continuation is optimal, so that in the region where one does not exercise, the option is a fair game. The optimal rule is then transparent: continue while $U > Y$, and stop the instant they meet.

The Linear Complementarity Problem (LCP)

Translating Theorem 1.5.5 into the language of partial differential equations gives a free boundary problem, most rigorously expressed as a linear complementarity problem. Write $\mathcal{L}$ for the pricing operator associated with (1.127),

$$\mathcal{L}V := \frac{\partial V}{\partial t} + \frac{1}{2} \sigma_{\text{loc}}^2(F, t) F^2 \frac{\partial^2 V}{\partial F^2} - r(t) V.$$

A point of care is required with the obstacle, and it is the reason this subsection is not a routine transcription. Exercise delivers the *spot*, not the forward: the holder of an American

put receives $K - S_\tau$, and by (1.109) we have $S_t = \Theta_t F_t$. Expressed in the forward variable, the obstacle is therefore

$$\Phi(t, F) = (K - \Theta_t F)^+ \quad \text{for a put,} \quad \Phi(t, F) = (\Theta_t F - K)^+ \quad \text{for a call,} \quad (1.129)$$

which is **explicitly time-dependent through Θ_t** . Writing $(K - F)^+$ instead, as is sometimes done, silently assumes $\Theta_t \equiv 1$ and is simply wrong: it misplaces the exercise boundary, and it does so by the widest margin in a name paying large discrete dividends — precisely the case where early exercise matters most. Keeping Θ_t explicit costs nothing and preserves validity for time-dependent $r(t)$ and $q(t)$, for cash and proportional dividends, and for any mixture of them.

Theorem 1.5.6 (American pricing as an LCP). *The value $V(t, F)$ of the American option satisfies, almost everywhere on $[0, T) \times (0, \infty)$, the three simultaneous conditions*

$$\begin{cases} \mathcal{L}V \leq 0 & (\text{supermartingale property, Theorem 1.5.5(ii)}) \\ V(t, F) \geq \Phi(t, F) & (\text{domination, Theorem 1.5.5(i)}) \\ (\mathcal{L}V) \cdot (V(t, F) - \Phi(t, F)) = 0 & (\text{complementarity, Theorem 1.5.5(iii)}) \end{cases} \quad (1.130)$$

together with the terminal condition $V(T, F) = \Phi(T, F)$ and Φ as in (1.129). ■

The correspondence with Theorem 1.5.5 is exact, line for line, and this is the point of having stated the probabilistic version first. The complementarity condition encodes the dichotomy: at each point either $V > \Phi$, in which case $\mathcal{L}V = 0$ and the ordinary pricing equation holds, or $V = \Phi$ and the option is exercised. The boundary separating the two regions is not known in advance — it is part of the solution — which is what makes the problem free-boundary rather than merely parabolic, and what rules out any closed-form solution.

Calibration to Single Names: De-Americanisation

A major practical hurdle arises during calibration: Dupire's formula (Theorem 1.5.1) requires the partial derivatives of *European* option prices, but single-name market quotes are American. We cannot apply Dupire directly to American prices.

Therefore, calibrating the local volatility surface to American options requires a short, iterative mapping procedure known as **De-Americanisation**:

1. **Quote Extraction:** Obtain the market prices of the liquid American options and calculate their Black-Scholes implied volatilities using an American pricer (such as a Binomial Tree or a PDE solver).
2. **Pseudo-European Mapping:** Assume these implied volatilities approximate the volatilities of European options with the exact same strikes and maturities. Use these volatilities to generate a grid of "pseudo-European" option prices.
3. **Local Volatility Construction:** Apply a smoothing parameterization (e.g., SVI) to the pseudo-European surface to ensure it is free of static arbitrage, and then apply Gatheral's form (1.120) of the Dupire formula to construct the local volatility surface $\sigma_{\text{loc}}(F, t)$.
4. **Repricing and Verification:** Substitute this $\sigma_{\text{loc}}(F, t)$ back into the LCP (Equation (1.130)) and solve it backwards using a finite difference method (such as Crank-Nicolson combined with a Projected Successive Over-Relaxation (PSOR) algorithm) to verify that the model accurately recovers the original American market quotes.



This iterative procedure effectively bridges the theoretical gap between the European foundations of Local Volatility and the practical reality of American single-name markets, ensuring that our models reflect true market dynamics without introducing arbitrage.

However, the de-Americanisation process can be inverted, and this must be of no surprise at this stage, by the so called PINNs (*Physics-Informed Neural Networks*). We will dive further into this topic in the final chapter of the present dissertation.

1.5.4 References to Sections 1.3 to 1.5

The financial framework and pricing methodologies developed in the latter half of this chapter are built upon cornerstone texts in quantitative finance and stochastic analysis. The rigorous transition from abstract probability to continuous-time asset pricing, including the formulation of equivalent martingale measures and the *No Free Lunch with Vanishing Risk* (NFLVR) condition, relies heavily on the definitive work by Delbaen and Schachermayer ([13]). This abstract measure-theoretic approach is bridged with continuous-time applications in the classic textbooks by Shreve ([39]), Øksendal ([32]), and Björk ([5]).

The explicit formulation of diffusion models is rooted in the foundational breakthroughs of Black and Scholes ([7]) and Merton ([30]). To address the empirical shortcomings of this framework, Local Volatility was formalized by Dupire ([14]), with its practical translation into the implied volatility domain extensively covered by Gatheral ([18]). For an exhaustive, practitioner-oriented treatment of volatility smile dynamics and Stochastic Volatility models (such as the Heston model [21]), Bergomi ([3]) serves as an indispensable reference.

Part II

Modern Approaches

Chapter 2

Rough Paths

2.1 Introduction

IDEA:

Series temporales y volatilidad: No nos vale con el movimiento browniano $\rightarrow$ fractional brownian motion y principales modelos de tipo rough volatility

siguiente sección: teoría de rough paths, con sus resultados fundamentales (integración, ecuaciones diferenciales, existencia, unicidad y continuidad) y la transformación de signature.

Capítulo siguiente y final: un primer ejemplo de calibración de una superficie de volatilidad implícita y obtención de volatilidad local consistente, con una red neuronal que haga de interpolador (ESTO YA LO TENGO HECHO Y TESTEADO, AL 100%)

Calibración de una serie histórica de algún tipo o algo semejante para

2.2 Rough Paths

2.2.1 Tensor Spaces

2.2.2 Multiplicative Property

2.2.3 Types of Rough Paths

2.3 Fundamental Results

2.3.1 Integration

2.3.2 Differential Equations

2.3.3 Existence

2.3.4 Uniqueness and Continuity

2.4 Signature Transformation

Chapter 3

Applications

3.1 Calibration of a Volatility Surface

3.1.1 The Model

3.2 Joint Calibration (?)

3.2.1 The Challenge (?)

3.2.2 Calibration of a (???)

3.3 Implementation

3.4 Results

Chapter 4

Conclusions

Theorem 4.0.1. *This is the main result.*

As we saw in Theorem 4.0.1, ...

$$E = mc^2 \tag{4.1}$$

We substitute into equation (4.1). also 4.1

Bibliography

- [1] Kenneth [3.] Falconer. *Fractal Geometry*. University of St Andrews, 2003.
- [2] Christian Bayer *et al.* *Rough Volatility*. Financial Mathematics. Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM), 2023, p. 291. ISBN: 9781611977776.
- [3] Lorenzo Bergomi. *Stochastic Volatility Modeling*. Chapman & Hall/CRC Financial Mathematics Series. Boca Raton, FL: CRC Press / Taylor & Francis, 2016, p. 522. ISBN: 9781482244069.
- [4] Patrick Billingsley. *Probability and Measure*. 3rd. New York: Wiley, 1995.
- [5] Tomas Björk. *Arbitrage Theory in Continuous Time*. 3rd. Oxford, UK: Oxford University Press, 2009.
- [6] Fischer Black. “The Pricing of Commodity Contracts”. In: *Journal of Financial Economics* 3.1-2 (1976), pp. 167–179.
- [7] Fischer Black and Myron Scholes. “The Pricing of Options and Corporate Liabilities”. In: *Journal of Political Economy* 81.3 (1973), pp. 637–654.
- [8] Douglas T. Breeden and Robert H. Litzenberger. “Prices of State-Contingent Claims Implicit in Option Prices”. In: *The Journal of Business* 51.4 (1978), pp. 621–651.
- [9] Haim Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. New York: Springer, 2011.
- [10] Peter Carr and Dilip B. Madan. “Option Valuation Using the Fast Fourier Transform”. In: *Journal of Computational Finance* 2.4 (1999), pp. 61–73.
- [11] Ilya Chevyrev and Andrey Kormilitzin. “A Primer on the Signature Method in Machine Learning”. In: *arXiv preprint arXiv:1603.03788* (2016). URL: <https://arxiv.org/abs/1603.03788>.
- [12] Christa Cuchiero *et al.* “Joint Calibration to SPX and VIX Options with Signature-Based Models”. In: *Mathematical Finance* 35.1 (2025), pp. 161–213. DOI: 10.1111/mafi.12442. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/mafi.12442>.
- [13] Freddy Delbaen and Walter Schachermayer. “A General Version of the Fundamental Theorem of Asset Pricing”. In: *Mathematische Annalen* 300.1 (1994), pp. 463–520.
- [14] Bruno Dupire. “Pricing with a Smile”. In: *Risk* 7.1 (1994), pp. 18–20.
- [15] Rick Durrett. *Probability: Theory and Examples*. 5th ed. Cambridge, UK: Cambridge University Press, 2019. ISBN: 978-1108425391.
- [16] Lawrence C. Evans. *Partial Differential Equations*. 2nd. Vol. 19. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2010.
- [17] William Feller. “Two Singular Diffusion Problems”. In: *Annals of Mathematics* 54.1 (1951), pp. 173–182.
- [18] Jim Gatheral. *The Volatility Surface: A Practitioner’s Guide*. Hoboken, NJ: John Wiley & Sons, 2006.

- [19] Jim Gatheral and Antoine Jacquier. “Arbitrage-Free SVI Volatility Surfaces”. In: *Quantitative Finance* 14.1 (2014), pp. 59–71.
- [20] Julien Guyon and Jordan Lekeufack. *Volatility Is (Mostly) Path-Dependent*. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4174589. Paris, France and Berkeley, CA, USA, July 2022. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4174589.
- [21] Steven L. Heston. “A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options”. In: *The Review of Financial Studies* 6.2 (1993), pp. 327–343.
- [22] C. C. Heyde. “On a Property of the Lognormal Distribution”. In: *Journal of the Royal Statistical Society, Series B* 25.2 (1963), pp. 392–393.
- [23] Eduardo Abi Jaber, Louis-Amand Gérard, and Yuxing Huang. “Path-Dependent Processes from Signatures”. In: *arXiv preprint arXiv:2407.04956* (2024). URL: <https://arxiv.org/abs/2407.04956>.
- [24] Olav Kallenberg. *Foundations of Modern Probability*. 3rd ed. Cham, Switzerland: Springer, 2021. ISBN: 978-3030810807.
- [25] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*. 2nd ed. Vol. 113. Graduate Texts in Mathematics. New York: Springer, 1991.
- [26] Hui-Hsiung Kuo. *Introduction to Stochastic Integration*. Universitext. New York: Springer, 2006.
- [27] Roger W. Lee. “The Moment Formula for Implied Volatility at Extreme Strikes”. In: *Mathematical Finance* 14.3 (2004), pp. 469–480.
- [28] Daniel Levin, Terry Lyons, and Hao Ni. “Learning from the Past, Predicting the Statistics for the Future, Learning an Evolving System”. In: *arXiv preprint arXiv:1309.0260* (2013). URL: <https://arxiv.org/abs/1309.0260>.
- [29] Michael B. Marcus and Jay Rosen. *Markov Processes, Gaussian Processes, and Local Times*. Vol. 100. Cambridge Studies in Advanced Mathematics. Cambridge, UK: Cambridge University Press, 2006. ISBN: 9780521863001. DOI: 10 . 1017 / CB09780521863001.
- [30] Robert C. Merton. “Theory of Rational Option Pricing”. In: *The Bell Journal of Economics and Management Science* 4.1 (1973), pp. 141–183.
- [31] Giulia Di Nunno *et al.* “From constant to rough: A survey of continuous volatility modeling”. In: *arXiv preprint arXiv:2309.01033* (2023). Accessed: 2025-08-20. URL: <https://arxiv.org/abs/2309.01033>.
- [32] Bernt Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. 6th. Universitext. Berlin, Heidelberg: Springer, 2003.
- [33] Léo Parent. *Essays on Asset Price Dynamics Modeling*. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5002182. Paris, France, Sept. 2024. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5002182.
- [34] Goran Peskir and Albert Shiryaev. *Optimal Stopping and Free-Boundary Problems*. Lectures in Mathematics, ETH Zürich. Basel: Birkhäuser, 2006.
- [35] Philip E. Protter. *Stochastic Integration and Differential Equations*. 2nd ed. Vol. 21. Stochastic Modelling and Applied Probability. Berlin: Springer, 2005.



- [36] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations”. In: *Journal of Computational Physics* 378 (2019), pp. 686–707.
- [37] L. C. G. Rogers. “Arbitrage with Fractional Brownian Motion”. In: *Mathematical Finance* 7.1 (1997), pp. 95–105.
- [38] Halsey L. Royden and Patrick M. Fitzpatrick. *Real Analysis*. 4th. Upper Saddle River, NJ: Prentice–Hall, 2010.
- [39] Steven E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Vol. 11. Springer Finance. New York: Springer, 2004.
- [40] S. James Taylor. “The Hausdorff α -Dimensional Measure of Brownian Paths in n -Space”. In: *Mathematical Proceedings of the Cambridge Philosophical Society* 49.1 (1953), pp. 31–39.
- [41] Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives: A Student Introduction*. Cambridge, UK: Cambridge University Press, 1995.
- [42] Toshio Yamada and Shinzo Watanabe. “On the Uniqueness of Solutions of Stochastic Differential Equations”. In: *Journal of Mathematics of Kyoto University* 11.1 (1971), pp. 155–167.

Appendix A

Auxiliary Results

This appendix collects results which are used in the main text but whose statements would have interrupted the argument there. Nothing here is original, and nothing here is proved; references are given in each case. The organising principle is simple: a result belongs in the body if the argument cannot be followed without it, and belongs here if the argument merely *invokes* it. Results which we do prove, but whose proofs are too long to sit comfortably in the body, are deferred instead to Appendix B.

A.1 Complements on conditional expectation

Throughout, $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and all identities are understood to hold almost surely. Proofs may be found in [4] or [15]. Items 8 and 9, which are used constantly in the main text, are stated there as Proposition 1.1.4; the numbering below is consistent with it.

Proposition A.1.1 (Properties of conditional expectation). *Let $X, Y \in L^1(\Omega)$. Then:*

1. $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$; that is, $\mathbb{E}[X \mid \mathcal{G}]$ and X have the same expectation.
2. If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$.
3. If X and $\mathcal{G}$ are independent — that is, $\{X \in U\}$ and A are independent events for every Borel $U \subseteq \mathbb{R}$ and every $A \in \mathcal{G}$ — then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.
4. (**Taking out what is known**) If Y is $\mathcal{G}$ -measurable and $\mathbb{E}[|XY|] < \infty$, then $\mathbb{E}[XY \mid \mathcal{G}] = Y \mathbb{E}[X \mid \mathcal{G}]$.
5. (**Tower property**) If $\mathcal{H}$ is a sub- σ -field of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}]$.
6. If $X \leq Y$, then $\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$; consequently $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$.
7. For all $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY \mid \mathcal{G}] = a \mathbb{E}[X \mid \mathcal{G}] + b \mathbb{E}[Y \mid \mathcal{G}]$. In particular $\mathbb{E}[\cdot \mid \mathcal{G}]$ is a bounded linear operator from $L^1(\Omega, \mathcal{F})$ into $L^1(\Omega, \mathcal{G})$, a fact used to pass to limits in Theorem 1.2.6.

The convergence theorems also carry over to the conditional setting, in exactly the form one would expect.

Proposition A.1.2 (Conditional Fatou lemma). *Let $X_n \geq 0$ with $X_n \in L^1(\Omega)$ for every $n \geq 1$, and suppose $\liminf_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then*

$$\mathbb{E}\left[\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G}\right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}] \quad \text{almost surely.}$$

Proposition A.1.3 (Conditional monotone convergence). *Let $0 \leq X_1 \leq X_2 \leq \dots$ and suppose $X = \lim_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then $\mathbb{E}[X | \mathcal{G}] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n | \mathcal{G}]$ almost surely.*

Proposition A.1.4 (Conditional dominated convergence). *Suppose $X_n \rightarrow X$ almost surely and $|X_n| \leq Y$ for every n , with $Y \in L^1(\Omega)$. Then $\mathbb{E}[X_n | \mathcal{G}] \rightarrow \mathbb{E}[X | \mathcal{G}]$ almost surely and in $L^1(\Omega)$.*

A.2 Characteristic functions

Recall that the characteristic function of a real random variable X is $\varphi_X(u) = \mathbb{E}[e^{iuX}]$, $u \in \mathbb{R}$. The following elementary properties are used without comment in the main text.

Proposition A.2.1. *For any real random variable X :*

- i) $\varphi_X(0) = 1$ and $|\varphi_X(u)| \leq 1$ for all $u \in \mathbb{R}$;
- ii) φ_X is uniformly continuous on $\mathbb{R}$;
- iii) $\varphi_X(-u) = \overline{\varphi_X(u)}$ (Hermitian symmetry);
- iv) $\varphi_{aX+b}(u) = e^{iub} \varphi_X(au)$ for all $a, b \in \mathbb{R}$;
- v) if X and Y are independent, then $\varphi_{X+Y}(u) = \varphi_X(u) \varphi_Y(u)$;
- vi) if $\mathbb{E}[|X|^n] < \infty$, then φ_X is n times differentiable and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k], \quad 1 \leq k \leq n.$$

Conversely, the existence of $2n$ continuous derivatives of φ_X in a neighbourhood of the origin implies $\mathbb{E}[|X|^{2n}] < \infty$.

Property ii) deserves a word, since it is the reason the continuity hypothesis in the converse half of Lévy's continuity theorem (Theorem 1.1.15) is a genuine restriction rather than an automatic one: it constrains the *limit* function, which is not assumed in advance to be a characteristic function.

A.3 Moment indeterminacy of the log-normal law

The following construction, due to Heyde [22], is referred to in Subsection 1.1.3 and shows that a distribution possessing moments of all orders need not be determined by them.

Proposition A.3.1 (Heyde). *Let f_X denote the density of the standard log-normal law, that is, the density of e^Z with $Z \sim \mathcal{N}(0, 1)$. For $a \in [-1, 1]$ define*

$$f_a(x) = f_X(x) (1 + a \sin(2\pi \log x)), \quad x > 0. \quad (\text{A.1})$$

Then every f_a is a probability density on $(0, \infty)$, the laws are mutually distinct for distinct a , and all of them share the moment sequence of the log-normal distribution.

Non-negativity holds because $|a \sin(\cdot)| \leq 1$, and the moment identity reduces to showing that

$$\int_0^\infty x^k f_X(x) \sin(2\pi \log x) dx = 0, \quad k \in \mathbb{N}.$$

Substituting $\log x = t + k$ turns the integrand into an odd function of t multiplied by the Gaussian weight, and the integral vanishes. Taking $k = 0$ shows in particular that each f_a integrates to 1.

The underlying obstruction is that the log-normal moments grow like $e^{k^2/2}$, which violates Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$; when Carleman's condition holds, the moment problem is determinate. See [15] for the general theory.



A.4 Grönwall's inequality

The following is the standard comparison tool for integral inequalities. It is purely deterministic, and in the stochastic setting it is applied pathwise or to expectations.

Lemma A.4.1 (Grönwall's inequality). *Let $g, \varphi \in L^1[0, T]$ and $\lambda > 0$ satisfy*

$$\varphi(t) \leq g(t) + \lambda \int_0^t \varphi(s) ds, \quad 0 \leq t \leq T.$$

Then

$$\varphi(t) \leq g(t) + \lambda \int_0^t g(s) e^{\lambda(t-s)} ds, \quad 0 \leq t \leq T.$$

In particular, if $g \equiv \kappa$ is constant, then $\varphi(t) \leq \kappa e^{\lambda t}$ for all $0 \leq t \leq T$; and if $\kappa = 0$, then $\varphi \equiv 0$.

The final sentence is the one actually used. Uniqueness arguments for differential equations almost invariably reduce to showing that a suitable distance between two candidate solutions satisfies a Grönwall inequality with $\kappa = 0$, whence it vanishes identically.

A.5 Radon–Nikodym and the Jordan decomposition

For completeness we record the decomposition underlying Corollary 1.1.3.

Theorem A.5.1 (Jordan decomposition). *Every finite signed measure ν on a measurable space $(\Omega, \mathcal{F})$ admits a unique decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. The measure $|\nu| := \nu^+ + \nu^-$ is called the total variation of ν , and $\nu \ll \lambda$ if and only if $|\nu| \ll \lambda$.*

See [38] for a proof.

Appendix B

Deferred Proofs

This appendix collects those proofs which are given in full but whose length would have interrupted the argument in the main text. The organising principle is the complement of the one governing Appendix A: a result whose proof is short enough to be read in passing is proved where it is stated, a result we merely invoke is recorded without proof in Appendix A, and a result we genuinely use but whose proof runs to more than a few lines is stated in the body and proved here.

Each section below carries a pointer back to the statement it establishes, and a reference to [26], whose treatment the arguments follow.

B.1 Doob's submartingale inequality

We prove Proposition 1.1.27 of Chapter 1.

Proof: Let Q be an enumeration of the rationals in $[0, T]$. Right-continuity gives $\sup_{0 \leq t \leq T} Z_t = \sup_{r \in Q} Z_r$ almost surely, so it suffices to bound the maximum over a finite set $\{r_1 < \dots < r_k\}$ and let $k \rightarrow \infty$. On such a set, decompose $\{\max_{\nu} Z_{r_{\nu}} \geq \varepsilon\}$ into the disjoint events A_{ν} on which r_{ν} is the *first* index at which the level is reached. Each $A_{\nu} \in \mathcal{F}_{r_{\nu}}$, and on A_{ν} we have $Z_{r_{\nu}} \geq \varepsilon$, so the submartingale property gives

$$\varepsilon \mathbb{P}(A_{\nu}) \leq \mathbb{E}[Z_{r_{\nu}} \mathbb{1}_{A_{\nu}}] \leq \mathbb{E}[\mathbb{E}[Z_T | \mathcal{F}_{r_{\nu}}] \mathbb{1}_{A_{\nu}}] = \mathbb{E}[Z_T \mathbb{1}_{A_{\nu}}] \leq \mathbb{E}[\max\{Z_T, 0\} \mathbb{1}_{A_{\nu}}].$$

Summing over ν and using disjointness bounds $\varepsilon \mathbb{P}(\max_{\nu} Z_{r_{\nu}} \geq \varepsilon)$ by $\mathbb{E}[\max\{Z_T, 0\}]$, uniformly in k . Letting $k \rightarrow \infty$ gives the claim. The martingale case follows by applying this to the submartingale $|X_t|$, which is one by the conditional Jensen inequality. ■

B.2 Density of step processes in $L_{ad}^2([0, T] \times \Omega)$

We prove Lemma 1.2.3 of Chapter 1.

Proof: The argument proceeds by successive approximation in three steps, each enlarging the class of integrands. Write $\|f\|^2 := \int_0^T \mathbb{E}[f(t)^2] dt$ for the norm of $L_{ad}^2([0, T] \times \Omega)$, and recall that it suffices to approximate an arbitrary f within ε by a step process, since the step processes then form a dense subspace.

Step 1: f bounded with continuous sample paths. Let $|f| \leq M$ and let $\mathcal{P}_n = \{0 = t_0^n < \dots < t_{k_n}^n = T\}$ be partitions with $\|\mathcal{P}_n\| \rightarrow 0$. Put

$$f_n(t) := \sum_{i=1}^{k_n} f(t_{i-1}^n) \mathbb{1}_{[t_{i-1}^n, t_i^n)}(t).$$

Each f_n is a step process, and it is *adapted* because the coefficient $f(t_{i-1}^n)$ is $\mathcal{F}_{t_{i-1}^n}$ -measurable and multiplies the indicator of an interval beginning at t_{i-1}^n — this is the left-endpoint choice, and it is the only reason adaptedness survives. For almost every ω the path $t \mapsto f(t, \omega)$ is continuous on the compact $[0, T]$, hence uniformly continuous, so $f_n(t, \omega) \rightarrow f(t, \omega)$ for every t . Since $|f_n - f|^2 \leq 4M^2$, which is integrable on $[0, T] \times \Omega$ (a finite measure space), dominated convergence gives $\|f_n - f\| \rightarrow 0$.

Step 2: f bounded, sample paths not assumed continuous. Let $|f| \leq M$. Extend f by 0 outside $[0, T]$ and mollify in time: for $n \in \mathbb{N}$ set

$$g_n(t) := n \int_{t-1/n}^t f(s) ds = \int_0^1 f\left(t - \frac{u}{n}\right) du.$$

Each g_n has continuous sample paths, since it is an indefinite integral, and $|g_n| \leq M$. Adaptedness is preserved because $g_n(t)$ averages f over $[t - 1/n, t]$ only, using no information after t ; this is where averaging *backwards* rather than symmetrically matters. By the Lebesgue differentiation theorem, $g_n(t, \omega) \rightarrow f(t, \omega)$ for almost every t , for almost every ω , and dominated convergence again gives $\|g_n - f\| \rightarrow 0$. Step 1 now approximates each g_n by step processes, and a diagonal argument produces step processes converging to f .

Step 3: general $f \in L_{ad}^2([0, T] \times \Omega)$. Truncate at level n by setting $f^{(n)} := (f \wedge n) \vee (-n)$. Each $f^{(n)}$ is bounded and remains adapted, being a composition of f with a continuous deterministic function. Moreover $|f^{(n)} - f|^2 \leq f^2$ pointwise and $f^{(n)} \rightarrow f$ pointwise, so dominated convergence — with dominating function f^2 , integrable precisely because $f \in L_{ad}^2$ — yields $\|f^{(n)} - f\| \rightarrow 0$. Combining with Step 2 completes the proof.

Adaptedness is the only delicate point throughout, and it is worth noting that each of the three steps preserves it for a different reason: the left-endpoint evaluation in Step 1, the backward averaging window in Step 2, and the deterministic truncation in Step 3. ■

B.3 Continuity of the sample paths of an Itô integral

We prove Theorem 1.2.7 of Chapter 1.

Proof: Let $\{f_n\}$ be step processes converging to f in $L_{ad}^2([0, T] \times \Omega)$; the associated processes $X_t^n = \int_0^t f_n dW$ have continuous paths by construction, being finite sums of continuous terms. By Theorem 1.2.6 the difference $X^n - X^m$ is a martingale, so Doob's L^2 maximal inequality (Proposition 1.1.28) together with Itô's isometry gives

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^n - X_t^m|^2 \right] \leq 4 \mathbb{E} [|X_T^n - X_T^m|^2] = 4 \int_0^T \mathbb{E} [(f_n - f_m)^2] dt \rightarrow 0.$$

Passing to a subsequence for which the right-hand side is summable, the Borel–Cantelli lemma yields $\sup_{t \leq T} |X_t^{n_k} - X_t| \rightarrow 0$ almost surely. A uniform limit of continuous functions is continuous, so almost all paths of X are continuous on $[0, T]$. ■

B.4 The Itô integral as a limit of Riemann sums

We prove Theorem 1.2.8 of Chapter 1.

Proof: Let $f_{\mathcal{P}}(t) = \sum_i f(t_{i-1}) \mathbf{1}_{[t_{i-1}, t_i)}(t)$ be the left-endpoint step approximation, so that the Riemann sum is exactly $I(f_{\mathcal{P}})$. By Itô's isometry it suffices to show that $\int_0^T \mathbb{E} [(f - f_{\mathcal{P}})^2] dt \rightarrow 0$. Expanding the square, this integral is a sum of terms of the form $\mathbb{E}[f(t)^2] - 2\mathbb{E}[f(t)f(t_{i-1})] + \mathbb{E}[f(t_{i-1})^2]$, each built from the function $(s, t) \mapsto \mathbb{E}[f(t)f(s)]$ evaluated at points within $\|\mathcal{P}\|$ of the diagonal. That function is continuous by hypothesis, hence uniformly continuous on the compact $[0, T]^2$, so the whole expression tends to 0 uniformly as $\|\mathcal{P}\| \rightarrow 0$. ■



B.5 Lévy's characterisation theorem

We prove Theorem 1.2.27 of Chapter 1.

Proof: The one-dimensional case is proved by applying Itô's rule to $t \mapsto e^{iuX_t}$ for fixed $u \in \mathbb{R}$. Since X is a continuous local martingale with $\langle X \rangle_t = t$, the second-order term contributes $-\frac{1}{2}u^2 e^{iuX_t} dt$, and after localisation and taking conditional expectations one obtains, for $s < t$,

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = 1 - \frac{1}{2}u^2 \int_s^t \mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s] dr.$$

This is *not* yet a deterministic equation: for each r the quantity $\mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s]$ is an $\mathcal{F}_s$ -measurable random variable, so the unknown is a random function and no uniqueness theorem for integral equations applies to it as it stands. The standard device removes the randomness by testing. Fix $A \in \mathcal{F}_s$, multiply through by $\mathbf{1}_A$, take expectations, and interchange expectation and the dr integral, which is legitimate by Fubini since the integrand has modulus one. Writing

$$h_A(t) := \mathbb{E}[\mathbf{1}_A e^{iu(X_t - X_s)}],$$

we obtain the genuinely deterministic linear integral equation

$$h_A(t) = \mathbb{P}(A) - \frac{1}{2}u^2 \int_s^t h_A(r) dr, \quad t \geq s,$$

whose unique solution is $h_A(t) = \mathbb{P}(A) e^{-u^2(t-s)/2}$. Since this holds for every $A \in \mathcal{F}_s$ and the right-hand side is the integral over A of the constant $e^{-u^2(t-s)/2}$, we conclude

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = e^{-u^2(t-s)/2} \quad \text{almost surely.}$$

The conditional characteristic function of $X_t - X_s$ given $\mathcal{F}_s$ is therefore deterministic and equal to that of $\mathcal{N}(0, t - s)$, which is simultaneously independence of the increment from $\mathcal{F}_s$ and the correct Gaussian law; Proposition 1.1.32 then supplies independence of increments. The multidimensional case is identical, applied to $e^{iu^\top X_t}$ with the cross-variations $\langle X^i, X^j \rangle_t = \delta_{ij}t$ making the exponent $-\frac{1}{2}\|u\|^2(t - s)$. ■

B.6 Girsanov's theorem

We prove Theorem 1.2.33 of Chapter 1.

Proof: By Theorem 1.2.27 it suffices to show that, under $\mathbb{Q}$, the process Z is a continuous local martingale null at the origin with $\langle Z \rangle_t = t$. Continuity and $Z_0 = 0$ are clear.

For the martingale property we use Lemma 1.2.32, which reduces the claim to showing that $Z(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale. Itô's product formula gives

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t) dZ_t + Z_t d\mathcal{E}_\alpha(t) + dZ_t d\mathcal{E}_\alpha(t).$$

Now $dZ_t = dW_t - \alpha(t)dt$ and $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW_t$, so by the Itô table $dZ_t d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dt$. Substituting, the two dt terms cancel exactly:

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t)(dW_t - \alpha(t)dt) + Z_t \alpha(t)\mathcal{E}_\alpha(t)dW_t + \alpha(t)\mathcal{E}_\alpha(t)dt = (1 + Z_t \alpha(t))\mathcal{E}_\alpha(t)dW_t.$$

The right-hand side has no drift, so $Z\mathcal{E}_\alpha$ is a $\mathbb{P}$ -local martingale, and under the stated integrability it is a martingale; hence Z is a $\mathbb{Q}$ -martingale.

Finally, $\langle Z \rangle_t = \langle W \rangle_t = t$, because subtracting the finite-variation term $\int_0^t \alpha(s)ds$ does not change the quadratic variation, and because quadratic variation is a pathwise limit and is therefore unaffected by the passage to the equivalent measure $\mathbb{Q}$. The same argument applied to $Z_t^2 - t$ confirms the compensator. Lévy's theorem now identifies Z as a $\mathbb{Q}$ -Brownian motion. ■